Design

# Graphic designers' work and skillset

## What companies talk about in their job advertisements

Paulo Roberto Nicoletti Dziobczenski

Aalto University

DOCTORAL DISSERTATIONS

Aalto University publication series
**DOCTORAL DISSERTATIONS** 131/2021

# Graphic designers' work and skillset

## What companies talk about in their job advertisements

**Paulo Roberto Nicoletti Dziobczenski**

**Aalto University**
**School of Arts, Design and Architecture**
**Design**
**Strategic Design Integration and Management research group**

**Supervising professor**
Guy Julier, Aalto University, Finland

**Thesis advisor**
Oscar Person, Aalto University, Finland

**Preliminary examiners**
Robert George Harland, Loughborough University, United Kingdom
Pinar Kaygan, Middle East Technical University, Turkey

**Opponent**
Robert George Harland, Loughborough University, United Kingdom

Aalto University publication series
**DOCTORAL DISSERTATIONS** 131/2021

© 2021 Paulo Roberto Nicoletti Dziobczenski

ISBN 978-952-64-0522-3 (printed)
ISBN 978-952-64-0523-0 (pdf)
ISSN 1799-4934 (printed)
ISSN 1799-4942 (pdf)
http://urn.fi/URN:ISBN:978-952-64-0523-0

Images: Paulo Roberto Nicoletti Dziobczenski

Unigrafia Oy
Helsinki 2021

Finland

CAPES, Brazil

NORDIC SWAN ECOLABEL
Printed matter
4041-0619

**Author**
Paulo Roberto Nicoletti Dziobczenski

**Name of the doctoral dissertation**
Graphic designers' work and skillset - What companies talk about in their job advertisements

**Publisher** School of Arts, Design and Architecture

**Unit** Design

**Series** Aalto University publication series DOCTORAL DISSERTATIONS 131/2021

**Field of research** Design studies

**Date of the defence** 12 November 2021

**Language** English

☐ **Monograph** ☒ **Article dissertation** ☐ **Essay dissertation**

**Abstract**

This doctoral thesis investigates the work and skillset of graphic designers as described by companies in their job advertisements. The literature on design suggests that the role of designers is changing and they are now making a more strategic contribution to organisations. In the case of graphic design, the literature describes the work and skillset of graphic designers as being traditionally connected to delivering visual outcomes, such as designing posters, and being knowledgeable about typography and visual composition. In addition to that, and parallel to the expansion of the role of designers in general, graphic designers also work and have skills in areas such as business, research and technology (e.g. coding). The broad work and skillset of graphic designers pose challenges to both educators and practitioners in being able to keep themselves up to date on graphic designers' work.

The studies of this doctoral thesis unveil the work and skillset of graphic designers mentioned in job advertisements from the UK, Finland and Brazil. This doctoral thesis investigates the job market for graphic designers in these different countries and assesses the variations between geographical contexts and their distinct design cultures, economic and educational infrastructures, while also drawing attention to their similarities.

The results of the studies show that the work and skillset of graphic designers are broader than often typified. Graphic designers mainly deliver digital and print work. To deliver this type of work, however, graphic designers need to have not only visual design skills, but also skills commonly associated with other fields, such as 'business, 'project management' and 'research'. Another result of the studies is that they shed light on how the skillset is described when graphic designers move (1) from junior to senior level positions, (2) from in-house departments to a design consultancy (or vice versa) and (3) from traditional to digital graphic design functions (or vice versa).

Overall, the results of the reported studies suggest that job advertisements reflect the past while also guide future developments in the field. For design practitioners and educators, job advertisements provide a proxy for understanding the job market that can shape educational activities and self-development efforts. For design researchers, job advertisements provide information about the qualifications sought by companies in design professionals and also about how much (or little) organisations know about a profession. For example, an organisation that believes designers should code would present 'coding' as one of the requirements in an advertisement. This thesis then suggests that design researchers should take advantage of the availability and coverage provided by job advertisements for investigating professional developments.

**Keywords** graphic design, design recruitment, design adoption, design management, design education, job advertisements

| | |
|---|---|
| **ISBN (printed)** 978-952-64-0522-3 | **ISBN (pdf)** 978-952-64-0523-0 |
| **ISSN (printed)** 1799-4934 | **ISSN (pdf)** 1799-4942 |
| **Location of publisher** Helsinki | **Location of printing** Helsinki **Year** 2021 |
| **Pages** 252 | **urn** http://urn.fi/URN:ISBN:978-952-64-0523-0 |

# Acknowledgements

I could never have finished this PhD without the support of so many great people. Let me acknowledge them here (when multiple people mentioned, I ordered them alphabetically).

Both before moving to Finland and during all these years, I received incredible support, patience and love from my dear wife Maria Valéria Assis. Thank you for being together during this whole, and sometimes painful, journey. My lovely daughter Nina also helped in her own way: by being cute. Maria and Nina are the loves of my life and a constant reminder that there is much more in life than a PhD.

This PhD would perhaps never have happened without Oscar Person, who I met virtually back in 2012. I am super thankful to Oscar for his support during the application process, through the many e-mail exchanges and Skype meetings. I still remember the day that he showed me the snow in Helsinki and asked me, 'Are you sure you want to move from sunny Brazil to this?' I also want to thank Toni-Matti Karjalainen for introducing me to Oscar. Lastly, to my former master thesis advisors - Julio van der Linden and Mauricio Bernardes – and evaluator – Filipe Campelo – for their help when considering universities to pursue my PhD.

When I moved to Finland and throughout the initial years of my PhD, Oscar's support and presence was much more than I, or any other doctoral student, could have expected. I am thankful for all the hours we spent together writing the initial articles of this thesis. He was also the main responsible for me joining winter/summer schools in research methods in the UK (2014, 2015) and

Germany (2016), which elevated considerably my research skills. I am extremely thankful to Oscar for his support with my PhD.

This thesis was supervised by two experienced gentlemen - first Sampsa Hyysalo and later Guy Julier. I want to thank Sampsa for his sharp and straight-to-the-point comments during our meetings and winter schools to guide the development of this thesis. Guy came, as we say in Portuguese, *aos 45 do segundo tempo* ('at 45 minutes of the second half', obviously a football reference), during the hardest period of my PhD journey (I will come back to this topic on the next page) and managed to bring me back to the project and finish the PhD. Thank you both.

In Finland, I met many smart and kind people at Aalto, which made the PhD journey more enjoyable. I want to thank my dear friend Seungho Lee for all the companionship during these years, sometimes over a beer or two. I was also lucky to have met Julia Valle, Namkyu Chun and Patricia Naves here in Finland. Last, but definitely not least, I want to thank the other members of the SDIM research group and Department of Design: Anna Kholina, Jack Whalen, Louna Hakkarainen, Marina Noronha, Markus Joutsela, Miikka Lehtonen, Pekka Murto, Peter McGrory, Pia Tamminen, Ramia Mazé and Tuuli Mattelmäki. Lastly, thanks to Turkka Keinonen for helping me make the career transition from doctoral student to Designer in Residence at Aalto.

I also want to acknowledge those who collaborated to develop the studies in this thesis. Thanks to the designers interviewed in Study 1, who we have agreed to keep anonymous. Sonja Meriläinen and Tania Rodriguez-Kaarto for your collaboration on Study 2. Leandro Tonetto and Roberta Mandelli for your collaboration on Study 3. And thanks to all the hiring managers and designers interviewed in Study 4, who we have also agreed to keep anonymous. Also, I want to thank Robert Harland and Pinar Kaygan for their thoughtful comments during the pre-evaluation process of this thesis. Lastly, I want to thank Érico Assis for translating this thesis abstract to Portuguese. Although Portuguese is my native language, Érico demonstrated the value of professional translators by elevating my text to a higher level.

I want to thank all my friends in Brazil and Finland who helped me to forget the PhD sometimes, you are the best: André Marques, Andrea Bandoni,

Andrea Capra, Bernardo Clementino, Bruna Zeni, Bruno Bin, Daniela Szabluk, Elisa Beretta, Emma Luode, Essi Bin, Igor Casenote, Felipe Luz, Fernando Santos, Francisco Veloso, Jussi Nykänen, Lauri Apajalahti, Luiz Nunes, Maaria Apajalahti, Maura Della Flora, Marcelo Halpern, Natalia Capra, Paula Marques, Rafaela Tomasi, Sérgio Callai and Thiago Tomasi. A special thanks to Luisa Vendruscolo, who constantly cheered me on, reminding me to graduate ASAP.

I also want to thank Harri Kiljander, Kimmo Timonen and Vanessa Törnblöm for all the conversations we had throughout these years and your views on my research and career from a more practical angle. Having a less academic perspective on my work was always helpful and I appreciate your insights.

*Eu gostaria de agradecer e dedicar esse título a minha família no Brasil. Vocês sempre acreditaram em educação e fizeram o seu melhor para que eu tivesse a oportunidade de estudar. Obrigado mãe Marine, pai Nery e a meu irmão Marcelo e família: Marla e Leonardo. Também quero agradecer o suporte da minha 'nova' família pelo suporte nessa incomum ideia de vir morar na Finlândia: Francisco Assis, Vera Assis, Érico Assis, Marcela Fehrenbach, Olivia Fehrenbach de Assis e Martim Fehrenbach de Assis.*

There are many reports of people who have had to overcome psychological issues to complete their PhDs. I am one of those people. In March 2020, after having many (~10) rounds of reviews of the introductory chapters of my thesis, I was not able to see the 'end of the tunnel' and wanted to give up. This affected my mental health and made me a less pleasant person to be around. At that time, I had no energy or motivation to continue working on my research. I am thankful for the support I got from Aalto psychologists Hannu Pesonen and Seija Leppänen during those tough times. I really hope that Aalto continues providing this service for doctoral students.

Last, but definitely not least, I want to thank the Science without Borders programme of the Brazilian Ministry of Education for the scholarship I received.

Helsinki, Finland, 20 August 2021

Paulo Roberto Nicoletti Dziobczenski

# Contents

1. Introduction ..... 15

1.1 Research questions and contributions ..... 21

2. Literature review ..... 29

2.1 Defining graphic design ..... 30

2.2 Graphic designers' work ..... 46

2.3 Graphic designers' skills ..... 49

2.4 Section summary ..... 52

3. Methodology ..... 55

3.1 Document analysis of job advertisements (Studies 1, 2 and 3) ..... 58

3.1.1 Data collection and management ..... 62

3.1.2 Data analysis ..... 65

3.2 Thematic analysis of interviews (Studies 1, 3 and 4) ..... 70

3.3 Systematic literature review (Study 4) ..... 74

3.4 Improving the research quality ..... 77

3.5 Ethical concerns ..... 80

4. Summary of the articles ..... 83

5. Cross-cutting contributions .....93

5.1 Cross-contribution 1: Defining graphic design: the work and skillset of graphic designers ..... 95

5.1.1 The state and maturity of the job market for graphic designers in the UK, Finland and Brazil .....108

5.2 Cross-contribution 2: The different career paths for graphic designers: From junior to senior positions, from in-house to consultancy (or vice versa), and/or from traditional to digital graphic design (or vice versa). ..... 110

5.3 Cross-contribution 3: The good, the bad and the ugly about job advertisements as data for (design) research, education and practice..... 118

5.4 Cross-contribution 4: Making secondary data great again: how to use job advertisements in (design) research, education and practice. ..... 123

6. Final implications.....127

7. References. .....137

Appendix A. Thesis abstract in Portuguese.....157

Publications…..……161

# List of Publications

This doctoral thesis includes the following research publications.

**Publication 1**

Dziobczenski, P.R.N., and Person, O. 2017. "Graphic Designer Wanted: A Document Analysis of the Described Skill Set of Graphic Designers in Job Advertisements from the United Kingdom." *International Journal of Design* 11 (2), 41-55.

**Publication 2**

Dziobczenski, P.R.N., Person, O., and Meriläinen, S. 2018. "Designing Career Paths in Graphic Design: A Document Analysis of Job Advertisements for Graphic Design Positions in Finland." *The Design Journal*. 21 (3): 349–70.

**Publication 3**

Dziobczenski, P.R.N., Person, O., Tonetto, L.M., and Mandelli, R.R. 2018. "Requests from Companies and Requirements for Design Education in Brazil: Where Do They Meet?" in C. Storni, K. Leahy, M. McMahon, P. Lloyd, & E. Bohemia (Eds.), *Proceedings of DRS 2018: Catalyst* (Vol. 7, pp. 2823-2835). (DRS International Conference Series). London: Design Research Society.

**Publication 4**

Dziobczenski, P.R.N., and Person, O. "The Job of Job Advertisements in Design Recruitment and Research." *In review process.*

# Author's Contribution

**Publication 1:** Main author

I was responsible for the collection and initial coding of the 1,406 job advertisements for the study. The second author and I built the final coding scheme, did the analysis of the job advertisements and discussed the results together. We also shared responsibilities in the research design and writing of the final version of the study.

**Publication 2:** Main author

I was responsible for collecting the 230 job advertisements for this study. I held the main responsibility for coding the data in partnership with the third author. During analysis, emergent findings and the implications of findings were formed in collaboration with the second author. The second author and I were responsible for the research design. We also held the main responsibility for writing the paper. The third author reviewed the final version of the article before submitting it to the journal.

**Publication 3:** Main author

I was responsible for collecting the 371 job advertisements for this study. I also held the main responsibility for coding and analysing the data in partnership with the fourth author. During analysis, emergent findings and implications were formed through discussion between the four authors. I developed the research design in collaboration with the second and third authors. The third and fourth authors conducted in-loco interviews with educators in Brazil. I

contributed to writing the interview guide and analysing the results of the interviews together with the third and fourth authors. The paper was written in collaboration with the other authors.

**Publication 4:** Main author

I was responsible for conducting and analysing the 18 interviews and reviewing 209 academic articles for this study. The research design was developed in collaboration with the second author. The second author assisted in the formation of emergent themes and findings during analysis. He also participated in writing the paper by providing comments and suggestions throughout the writing process and, in joining the process at the end, making more extensive edits in the manuscript.

# List of figures and tables

Figure 01: Examples of Shapes, Colors, Typography, Illustration and Photography. ..... 39

Figure 02: Conceptual Learning Model by Voorhees (2001) ..... 68

Figure 03: Coding schemes used to analyse job advertisements on Studies 01, 02 and 03 ..... 69

Figure 04: Requirements made in job advertisements by companies in the UK ..... 97

Figure 05: Requirements made in job advertisements by companies in Finland ..... 98

Figure 06: Skill requirements that differ significantly among junior, mid-level, and senior positions ..... 112

Figure 07: Skill requirements that differ significantly between internal and external positions ..... 114

Figure 08: Foundation and specialization themes for traditional and digital graphic design positions ..... 116

Figure 09: How hiring managers and designers used the job advertisement in design recruitment processes. ..... 119

Figure 10: Senior Designer Visualiser job advertisement published in 1986. 129

Table 01: Research questions of this doctoral thesis ..... 22

Table 02: Definitions of graphic design, displayed in chronological order ..... 36

Table 03: The Anatomy of Graphic Design. .....40

Table 04: Search parameters of job advertisements on Studies 01, 02 and 03.64

Table 05: Interviews used on the studies of this doctoral thesis .....73

Table 06: Types of professionals analyzed in the papers .....76

Table 07: Requirements made in job advertisements by companies in Brazil..100

Table 08: Requests made in above half of the job advertisements on Studies 01, 02 and 03……..……..…….104

Table 09: Comparison between job advertisements from the UK, Finland and Brazil…….. 109

Table 10: Job titles in the job advertisements (Finnish titles translated to English)……..115

# List of abreviations

2D – Two dimensions.

3D – Three dimensions.

AIGA – American Institute of Graphic Arts.

RQ – Research question.

SEBRAE - Brazilian Service of Support for Micro and Small Enterprises (*Serviço Brasileiro de Apoio às Micro e Pequenas Empresas* in Portuguese).

UI – User Interface.

UK – United Kingdom.

UX – User Experience.

# 1. Introduction

*#1 'You are experienced in design, typography, colour and animation'*

*#2 'Strong understanding of usability and user-centred design practices'*

*#3 'You have an eye for design'*

*#4 'Ability to write great copy'*

*#5 'A general understanding of HTML'*

*#6 'Desirable: A fondness of cats (we have two resident office kitties!)'*

*When graphic designers look for a job, they read phrases like those above, extracted from different job advertisements. Phrases 1 and 2 suggest specific skills sought by the organisations. Phrase 3 is more vague: does it refer to aesthetic sense? Or to the ability to spot trends in design? Phrases 4 and 5 show that graphic designers not only design visual outcomes, but are also involved in work traditionally delivered by copywriters and developers. And what can we tell from phrase 6? Obviously that the organisation has cats in the office. But more importantly, it gives a glimpse of the culture of the company: a more relaxed environment and less formal work style.*

*In short, one can learn quite a lot about a profession and a company by reading job advertisements.*

*Perhaps even write a doctoral thesis based on job advertisements.*

The classic narrative in graphic design goes as follows: graphic designers are knowledgeable in building and manipulating visual elements (shapes, colours, typography, illustration, photography) to design visual objects (e.g. a website) (e.g. Frascara, 2004). Nevertheless, the work and skillset of graphic designers are evolving. According to Harland (2011), the traditional domains of graphic design – typography, illustration, photography and print – are inadequate to describe what graphic design is and what graphic designers do. Graphic designers are demanded to expand their skills, as design problems are more complex and require a broader range of skills (Davis, 2006, 2008). In this sense, graphic design education and practice are in a 'state of flux' (McDonald, 2006, p. 354) or comprise a 'fluid practice' (Harland, 2016, p. 24), as changes in technology, the economy and a global marketplace have a direct effect on the work done by these professionals. The technological development of digital media was the starting point of multiple fields of practice and research in design – such as interaction design (e.g. Preece et al., 2004), user experience (e.g. Law et al., 2009) and web usability (e.g. Nielsen, 1999) – with a direct impact on the work and skillset of graphic designers.

As graphic design is changing, practitioners are challenged to review their skillsets constantly. Similarly, design educators face the challenge of preparing their graduates to get a position in the market. Debates on how design education can cope with the constantly changing design profession can be found in academic literature. For example, a recent special issue of *She-Ji: The Journal of Design, Economics, and Innovation* (2020) have focused on design education. Another example is Jorge Frascara's article 'Design, and design education: How can they get together?' (Frascara, 2017), which pointed out the need for designers to carry skills in a range of different areas. This broadening of the skills of designers is also identified in surveys of design professionals (AIGA, 2019; Roberts et al., 2015) and the conceptual articles by Valtonen (2016) and Norman (2010).

However, as will be demonstrated in the studies of this doctoral thesis, the ability to build visual outcomes through building and manipulating visual elements is still relevant for graphic design professionals. On top of that, new fields of design practice (e.g. UX and UI design) can potentially shape the development of graphic designers' skillset towards the digital realm. While scholars point to different directions in relation to what the 'right' skillset for

graphic designers consists of, these professionals are in need of a way to better define their career paths in terms of which skills they must develop further. In this sense, design educators have the task of informing their students about 'available career paths, which is informed by the real-life experiences of the designers in the job market' (Kaygan et al., 2020, p. 836).

Graphic design is not the only design discipline that is evolving. Overall, scholars have acknowledged that design is changing, assuming a more strategic role within organisations (Micheli et al., 2018; Perks et al., 2005; Ravasi & Lojacono, 2005; Valencia et al., 2013; Valtonen, 2005). In this more strategic role, graphic designers contribute by designing visual recognition for a brand (e.g. Karjalainen & Snelders, 2010). Specifically, by designing visual elements (e.g. logo design), graphic designers deliver visual touchpoints (e.g. Wheeler, 2009) that form a critical aspect of the branding strategy of an organisation (Borja de Mozota, 2003; Henderson et al., 2003). Yet, not all organisations elevate (graphic) design into a more strategic role. For example, the Danish Design Centre (Ramlau & Melander, 2004) pointed out in its 'design ladder' model that organisations initially use design as a means of 'form-giving', where designers' main task involves styling a product (for more on styling in design, see Person, 2011). Next, organisations see design as a process that is integrated early on into the development of products and services. Last, design is seen as a strategy, where design is a recurrent key informant for reaching organisational goals. Different organisations assign different roles for the design function and the skillset needed to work as a design professional. Thus, the skillset and work of design professionals change according to the role design plays in organisations.

The role played by designers in organisations is a topic recurrently debated in the design management literature. In short, design management[1] is a combination of short- and long-term activities for achieving organisational goals. In the short term, managing the design function refers to managing design projects in terms of budget, schedule and personnel (e.g. Farr, 1965). In the long term, managing design refers to the contribution provided by designers and the role designers have within an organisation's design capabilities (Borja

1 For historical reviews on the origins of design management, see Cooper et al. (2013) and Borja de Mozota (2003)

de Mozota, 2003; Gorb, 1990). For managing design, organisations have the options of recruiting a design professional and/or outsourcing design work to a design agency (Bruce & Morris, 1998; Bruce & Morris, 1994; Oakley, 2011). In design management literature, both the 'framework for design management' by Bruce et al. (1999) and the 'design management absorption model' by Acklin (2013) suggest that at the initial stages of managing the design function, organisations need to 'source' (Bruce et al., 1999) or 'acquire' (Acklin, 2013) design skills and professionals. Similarly, Cooper and Press (1995) proposed that design and human resources (HR) management can collaborate on 'preparing the organization to use design effectively' (p. 146). Thus, in this doctoral thesis, I define the recruitment of a design professional as an early-stage decision taken by an organisation for adopting and managing design. This reveals something about the expectations of the client and the ways in which skills are anticipated within the graphic design profession and how these skills shape the work done by graphic designers.

In this doctoral thesis, I investigate the skillset of graphic designers described in the initial stages of the recruitment process, during which organisations state the requirements for a designer position in a job advertisement. The studies of this doctoral thesis cover the skillset mentioned in job advertisements for graphic designers in the UK[2] (Study 1), Finland[3] (Study 2) and Brazil[4] (Study 3). They

[2] According to 'Design Economy 2018' (Design Council, 2018), there were over 1.6 million design professionals in the UK (1,693,200), 39% working with 'design (digital)', 22% with 'architecture and build environment', 12% with 'design (product and industrial)', 9% with 'design (multidisciplinary)', 7% with 'design (graphic)' and 'design (craft)', 2% with 'design (advertising)' and 1% with 'design (clothing)'.

[3] According to research by the Nordic Design Resource published in 2018 (nordicdesignresource.com) there were almost 55 thousand design professionals in Finland (54,985), 21% working with 'graphic and visual design', 20% with 'digital design', 18% with 'strategic design', 16% with 'service and experience design', 15% with 'product development' and 10% with 'formgiving'.

[4] According to 'Mapping of the Creative Industry in Brazil' (Mapeamento da Indústria Criativa no Brasil in Portuguese) (FIRJAN, 2019), there were over 76 thousand registered design professionals in Brazil. As found by previous reports published in the country (CBD, 2014), many designers in the country work either under different job titles or in an informal way. Thus, the number of designers in the country might be higher, especially considering that in January 2020, the Brazilian Ministry of Education website (emec.mec.gov.br) reported over 400 undergraduate programmes in the country.

also address how the skillset of graphic designers changes at different levels of seniority (Study 1), when designers work internally or externally to companies (Study 1) and in transitioning to more digital-focused positions (Study 2), and how the skills listed by organisations are similar/different to the skills outlined in documentation guiding educational institutions (Study 3). By looking at the skillset sought by organisations in their job advertisements, I set out to understand the type of work that is done by graphic designers. In other words, the skillset requested demonstrates the type of role the designer will have in the organisation.

In investigating the skillset of graphic designers, this doctoral thesis relies mainly on job advertisements as data. Following a growing interest in design and the great number of organisations seeking to recruit designers (e.g. Maeda, 2017; Sheppard et al., 2018), there are more job opportunities for (graphic) designers and, consequently, more job advertisements available. Job advertisements describe the qualifications sought by an organisation in a new professional (e.g. Todd et al., 1995). Overall, job advertisements represent a proxy for studying the qualifications sought in the job market (e.g. Harper, 2012).

To date, job advertisements have seldom been used in design research. As exceptions, not including the studies in this doctoral thesis, I highlight previous inquiries into the qualifications sought in designers in Taiwan (Yang et al., 2005), Turkey (Erkarslan, 2013) and the United States (Ramírez, 2012). In other fields, such as library and information science (e.g. Harper, 2012; Kim & Angnakoon, 2016), business (e.g. Bennett, 2002), education (e.g. Irons & Buskist, 2008) and health (e.g. Dosani et al., 2003), job advertisements are frequently used to unveil the qualifications organisations seek in a specific professional. Similar to job advertisements, other employment data (e.g. job titles) have been used by design researchers to investigate the industrial design profession in Ireland (Kiernan & Ledwith, 2014) and Turkey (Kaygan et al., 2020).

By revealing the skillset of a professional, scholars have contributed to professional development in terms of showing what career trends and paths are available based on the qualifications sought by companies (e.g. da Cunha, 2009). In addition, scholars have also pointed to the educational aspect of studying the requests by companies, as this contributes to graduates' employability (e.g. Choi

& Rasmussen, 2009). Scholars have acknowledged the benefits of using job advertisements, as they are a source of naturally occurring data (e.g. Todd et al., 1995) that is not created for research purposes (e.g. Kim & Lee, 2016). In addition, researchers have been able to cover a large share of the job market when using job advertisements, as these documents are increasingly available online (e.g. Chipulu et al., 2013). Yet, there is scant literature on the use of job advertisements in research, with little guidance in the literature on the type of questions that can be addressed through studies on job advertisements and what methodological challenges researchers may encounter in their studies. In Study 4 of this doctoral thesis, I investigate the relevance of job advertisements as data for research purposes and their contribution to design research, practice and education.

The job advertisements in this doctoral thesis come from countries with different histories and levels of maturity in terms of design. The UK and Finland are commonly considered leading design nations. On the other hand, Brazil is an 'emerging player' in the field, as argued by the International Design Scoreboard (Moultrie & Livesey, 2009). The initial assumption of this thesis is that design practice is contextually situated: designers in different countries – especially those with different design traditions, history and economic drivers – might have different responsibilities to fulfil within organisations. In looking at previous literature on design practice across different countries, mainly in terms of design promotion and policy strategies (e.g. Choi et al., 2012; Raulik-Murphy et al., 2016; Nil Gulari et al., 2017; Raulik-Murphy, 2010; Qian Sun, 2010) I found arguments both in favour of and against this idea. Similarities between different contexts exist. For instance, Candi and Saemundsson (2008) have argued that designers fulfil almost the same role in technology-based firms in the US and Iceland. In other words, designers working in technology-based firms have similar activities and responsibilities regardless of the country where they work. However, this technology emphasis is not necessarily representative of all roles undertaken within graphic design. To give a counterexample, Raulik-Murphy et al. (2010) report on the mistakes committed by the South African Design Institute when trying to copy the model of the British Design Council. In investigating design promotion across different countries, Raulik-Murphy (2010) claims elsewhere that social, political and economic contexts are crucial for building design policies in countries. Overall, across the studies of this

doctoral thesis, I found that the overall skillset portrayed in job advertisements in the UK and Finland more often mentioned 'traditional' graphic design skills (e.g. layout, typography) while also expanding towards areas such as 'business', 'research' and 'coding', while job advertisements from Brazil more often mentioned 'traditional' graphic design skills.

## 1.1 Research questions and contributions

This thesis explores the following overall research question: *What is the skillset organisations articulate in job advertisements when recruiting graphic designers?* Unveiling the skillset sought in graphic designers is important to understand the type of work done by graphic designers, and the role of design in general. The skillset of a (design) professional defines the work the professional does in an organisation. In addition, the work done shapes the role of the profession. Bringing this argument to the design field, when designers carry, for example, research skills (skillset), they will be responsible for conducting research activities (work), which therefore will affect the role design plays in the organisation (in this case, the role of design is research). Also, investigating the skillset of graphic designers is relevant in understanding the initial steps taken by organisations when adopting and managing the design function (e.g. Borja de Mozota, 2003; Valtonen, 2007). To answer this question, I report on how organisations articulate their requests in job advertisements for graphic design positions.

To answer the overall research question, this doctoral thesis investigates six research sub-questions. The initial four research sub-questions are relevant to understanding what skillset is necessary to operate as a graphic designer: specifically, how the skillset of graphic designers changes according to countries' different design maturity levels (RQ1, addressed across Studies 1, 2 and 3), to when design is located internally or externally to organisations (RQ2, addressed in Study 1), to the seniority of the position (RQ3, addressed in Study 1) and when facing technological advancements in the field (RQ4, addressed in Study 2). The fifth research sub-question (addressed in Study 3) is relevant for design education in addressing the 'education gap' (Todd et al., 1995, p. 20), where

there is an imbalance between the skills requested by companies and the skills provided by educational institutions. Lastly, the sixth research sub-question (addressed in Study 4) is relevant in pointing out the possibilities of using job advertisements as a (new) source of data for design research, practice and education. Looking across the results of different studies, I engage in discussions about the work and education of graphic designers. Also, I provide a methodological contribution to how to study the job market faced by (design) professionals. Table 1 presents the research questions in this doctoral thesis.

| **Research questions** | **Study/studies answering the RQ** |
|---|---|
| What is the skillset organisations articulate in job advertisements when recruiting graphic designers? | All studies |
| RQ1 – How does the skillset required from graphic designers change in countries with different design maturity levels? | Studies 1, 2 and 3 |
| RQ2 – How does the skillset required from graphic designers change from in-house design agencies (internal design) to design consultancies (external design)? | Study 1 |
| RQ3 – How does the skillset required from graphic designers change at different levels of seniority (junior, middle and senior)? | Study 1 |
| RQ4 – How does the skillset required change according to the type of graphic design position: traditional graphic design positions versus digital graphic design positions (i.e. UX designer, UI designer)? | Study 02 |
| RQ5 – How does the skillset required from graphic designers compare to the skillset defined in educational policy documents? | Study 3 |
| RQ6 – How are job advertisements in design constructed and how can they benefit design research, practice and education? | Study 4 |

**Table 1.** Research questions of this doctoral thesis

The first contribution of this doctoral thesis is in adding to the growing body of graphic design research, specifically by providing empirical studies on graphic design practice. Design research in general is 'seeking to establish itself as a discipline, and eventually as a science' (Horváth, 2007, p. 10). According to the author, there is a need for constant growth of knowledge in design research to differentiate designers from other professionals. More recently, Cash (2018) also claimed that there is a need to increase rigour in design research 'to avoid stagnation, low impact and potential obsolescence' (p. 84). Specifically, with respect to graphic design research, Frascara (1988) argued that graphic design has evolved without much theoretical reflection. Nowadays, even with a significant increase in the number of institutions providing graphic design education and researchers doing graphic design research (e.g. Harland, 2017; Harland et al., 2018), Walker (2017) argues that research focused on graphic design still lags behind design research in general. For example, in the UK, the 2014 Research Excellence Framework reported that research in graphic design showed 'little, if any, significant signs of improvement since the 2008 Research Assessment Exercise' (Harland et al., 2018, p. 2812). Overall, this doctoral thesis contributes to recent emerging debates (Corazzo et al., 2019; Davis, 2016; Griffin, 2017; Harland et al., 2018; Harland, 2020; Ross, 2018; Townsend & Armstrong, 2017; Walker, 2017) on the importance of establishing a graphic design research tradition. As pointed out by Corazzo et al. (2019), establishing a research culture in graphic design has the potential to demonstrate the value of graphic design practice, to reduce the perceptions of graphic design as a simply 'intuition-based' activity and, lastly, to establish clear pathways for career progression in graphic design.

Current research in graphic design is found mainly in textbooks and in a few empirical studies on the topics of graphic design education and practice. As pointed out by Harland (2020), there are few academic journals recognised for their focus on graphic design research. In relation to education, Heller's book *The Education of a Graphic Designer* (1998, 2005, 2015) presents short essays by multiple design educators reflecting on their experiences in teaching graphic design. I also highlight the special issue 'Territories of Graphic Design Education', published by the *Art, Design & Communication in Higher Education* journal in 2017. Following the editorial of this special issue (Harland, 2017), the nine articles published in that edition cover '*what* graphic design education

is today, and *how* it is taught.' (p. 4). With respect to this topic, I also highlight the work of Giloi and du Toit (2013) on how educators can assess graphic design work.

On the topic of graphic design practice, there are a number of books that guide students and practitioners on how to develop their careers and projects in graphic design. I highlight examples such as *How to be a Graphic Designer, Without Losing Your Soul* (Shaughnessy, 2012) and *The AIGA Guide to Careers in Graphic and Communication Design* (Cezzar, 2017). I also highlight the book *Graphic Designers Surveyed* (Roberts et al., 2015), whose authors surveyed more than two thousand graphic designers in the US and the UK about 'what they do and how they feel about the profession' (p. 13). Although relevant for graphic design practice, these books have been of limited significance in building theoretical contributions to the field, which are mainly based on systematic and peer-reviewed studies. In addition to that, developments in graphic design practice happen at a fast pace, shaped in part by technology, but also changing market conditions. Thus, the current relevance of the recommendations provided in these books needs to be assessed.

In academic journals, I highlight a few empirical studies published by *Design Studies*, based on interviews with practitioners, covering 'how the everyday work of graphic design happens in action' (Martin, 2012), specifically approaching topics such as ideation (Laing et al., 2017; Laing & Masoodian, 2016; Stones & Cassidy, 2010), conceptual design (Tan & Melles, 2010), design of interactive artifacts (Jung et al., 2017) and client-designer communication (Cornish et al., 2015). These empirical studies focus on how graphic design practice happens in a few studied cases; however, the question of how the whole breadth of activities performed by graphic designers occur in different contexts (e.g. countries) remains unanswered. In the studies of this doctoral thesis, I show the diverse activities performed by graphic designers in their professional routine, as described in job advertisements from the UK, Finland and Brazil.

Although graphic design is constantly evolving, as noted earlier, previous research on graphic design education and practice has not clearly specified the overall skillset needed to operate as a graphic designer. As the profession and the work of graphic designers change, the skillset needs to follow suit. Thus, this doctoral thesis contributes by providing a framework for studying the graphic

designer skillset by constantly monitoring the job market through the analysis of job advertisements. The studies in this thesis present the requirements set for the skillset of graphic designers in the studied countries according to seniority (Study 1, based on data from the UK), to when designers are located inside or outside companies (Study 1, based on data from the UK), to the increase in demand for digital design in graphic design (Study 2, based on data from Finland) and in relation to the skillset provided by educational institutions (Study 3, based on data from Brazil).

This doctoral thesis adds to the current design industry reports published by design associations in the UK, Finland and Brazil. The Design Council in the UK published 'The Design Economy' reports (Design Council, 2015, 2018) where they surveyed companies to understand the state of design in the UK and how design adds value for organisations. In Finland, Ornamo (Design Professional Association) and Grafia (Association of Visual Communication Designers in Finland) published annual member surveys to unveil the number of professionals working in the field in Finland, their job titles, salary levels and other such information (Grafia, 2019; Ornamo, 2019). In Brazil, the Brazilian Service of Support for Micro and Small Enterprises (SEBRAE) published an overview of the country's design sector, including a list of the design schools and organisations promoting design in the country (SEBRAE, 2014). While these reports present the big picture of the design industry, based on surveys in each of these countries, the studies in this doctoral thesis provide fine-grained information about the work and skillset of graphic designers in these countries.

This doctoral thesis also contributes to debates on how organisations adopt and manage the design function. For example, this thesis in part builds on the work of Perks et al. (2005), who defined an empirically derived taxonomy on the work done by designers in new product development. An underlying argument for much research on design management is that organisations need to formalise capabilities in design and learn how to manage it (Bessant & Bruce, 2002; Best et al., 2010). The recruitment of a design professional is the initial step for the long-term building of design capabilities and managing the design function within organisations, as pointed out in previous case studies (Bailey, 2012; Body, 2008; Valtonen, 2007) and in the design management literature (e.g. Cooper & Press, 1995; Farr, 1965; Sun et al., 2011). In design management literature, I also highlight the design management models by Acklin (2013) and Bruce et al.

(1999), who suggest that hiring designers is part of the initial steps for managing the design function.

In summary, the recruitment of a design professional is a step taken during the early stages of design adoption and management. In relation to the focus of this doctoral thesis on graphic design, the work provided by graphic designers (e.g. logo design) is one of the initial services that organisations seek when adopting design (Borja de Mozota, 2003). The studies of this doctoral thesis contribute to providing empirical knowledge about the early stages of the design adoption and management processes, during which organisations publish the requirements they seek of graphic design professionals in three different countries.

By studying these requirements, this doctoral thesis also makes a contribution to studies on the role of design in organisations (e.g. Micheli et al., 2018; Valencia et al., 2013). In their study on expanding the role of design in organisations, Micheli et al. (2018) concluded that designers need to develop a broad skillset in order to join cross-functional teams and elevate design to a strategic level. The skillsets sought by organisations serve as a proxy for understanding the work done by designers and, therefore, the role designers fulfil at companies. For example, an organisation hiring an in-house designer with 'business skills' is signalling that designers in this organisation have a role closer to 'process leadership' than 'design as [a] functional specialism' (Perks et al., 2005, p. 119). As job advertisements are located at the beginning of the design adoption and management processes, they show what organisations see as important in the work and skillsets of graphic designers.

Lastly, this doctoral thesis provides perspectives that may be important to educators in designing educational activities and also to practitioners in supporting their career planning and development. The need for constant reassessment in design education is suggested by Davis (2015, p. 31): 'There is little disagreement that the context for design practice has changed over the last decades and that design education is long overdue for rethinking curricular and pedagogical strategies.' Scholars in design (Lewis & Bonollo, 2002; Yang et al., 2005) and other fields (Bennett, 2002; Bettinger et al., 2016; Brumberger & Lauer, 2017; Chen & Zhang, 2017; Kennan et al., 2006; Laura Roberts & Mancuso, 2014) suggest that educators should continuously monitor the job

market their graduates are facing. Also, the main barrier for adopting design found by Liu and de Bont (2017), who conducted an interview study on design-led companies in China, is that graduate designers 'lack professional skills' (p. 140). By studying the job market, educators can plan educational assets to improve students' employability upon graduation. Similarly, this doctoral thesis provides information about the skillset sought in graphic designers, which can help practitioners in guiding their career development. According to Rafaeli (2006), who surveyed professionals on the reasons why they read job advertisements, many of them said that job advertisements provide a proxy for understanding the job market in terms of which companies are hiring and the qualifications sought. To this end, the skillsets sought in the UK, Finland, and Brazil can guide students and practitioners to define their career paths and development. In addition, the method for analysing job advertisements in the studies of this doctoral thesis can serve as a reference for practitioners interested in the job market in other countries.

In the next section, I present a literature review on definitions of graphic design and the work and skills of graphic designer

# 2. Literature review

In tandem with discussions on graphic design research, graphic design history and how it has evolved have been documented in many books and articles (see Triggs, 2011), such as in the books by Meggs & Purvis (2012, 2016) and Hollis (2001). Graphic design historians have acknowledged the development of the field by looking at important events such as the development of alphabets[5] and manuscripts, the invention of printing, the industrial revolution and the digital revolution (Harland, 2016). Yet, identifying the starting point for the history of graphic design is challenging (Margolin, 1994; Triggs, 2011). Margolin (1994) points out a narrative problem in graphic design, where some authors use 'graphic design' synonymously with any kind of visual communication. He then specifically points to the work of Hollis (2001), who made a clear distinction between graphic design and other types of visual communication in his account on graphic design history. Margolin (1994) then suggests that this separation of graphic design from other visual communication is helpful in understanding and tracing the history and development of the profession.

Technological developments over the last decades had a significant impact on the history and work of (graphic) designers. Specifically, Julier (2017) and Davis (2012) discussed how technological developments in the late 1980s affected how designers work. The most significant change was in terms of time: layouts became faster to create and manipulate than in the traditional way. Thanks to computers and software, graphic designers gained speed and agility in generating ideas and developing their visual work. In addition, computers became a *medium*, and the introduction of hypertext in 1987 allowed designers

5 For an account of the history of graphic design from the perspective of typography, see 'Modern Typography: an essay in critical history' (Kinross, 2004).

to design *using* screens and *for* screens (Julier, 2017). In short, as presented by Harland (2011) and Julier (2017), in the late 1980s and early 1990s, graphic design practice was transformed by technological advancements that affected how graphic designers work, which at the same time removed barriers to entry for new graphic design practitioners. As summarised by Bonsiepe (1997), the expansion of 'new media' (p. 6) – online media, hypertext and interactivity – transformed the graphic design profession.

This doctoral thesis aims to contribute to ongoing debates on strengthening graphic design research and practice in the face of the technological changes affecting the graphic design profession. The objective of this section is to review previous literature in design on the skillset and work of graphic design professionals. I start by reviewing previous literature on the importance of defining graphic design and how early scholars have defined the discipline. Specifically, I report on how graphic design is defined in terms of visual elements, visual outcomes and the functions performed by the work of graphic designers. Next, I review literature on the work of graphic designers. Scholars have depicted the practice of graphic design from opposite views, namely 'creative' and 'mundane' practices. Lastly, I present the skills mentioned as being necessary to graphic design practice.

## 2.1 Defining graphic design

Defining design is a long-standing topic in design literature. Definitions are crucial in determining boundaries and areas of interests in a profession. In their conceptual article, Poggenpohl et al. (2004) suggest that design lacks agreement on how it should be defined. Similarly, Julier (2013, p. 50) claims that definitions of design are constantly 'being constructed, but also decentred, dispersed and disorganized'. At the same time, defining the design profession is challenging: Buchanan (2001) suggests that a range of contrasting and contradictory definitions of design are found in the literature. In the following paragraphs, I focus on the relevance of defining graphic design and how graphic design was defined in literature (see Table 2 for definitions on graphic design).

Looking back in history, it has been argued that the term 'graphic design' was created by the type designer William Addison Dwiggins in 1922 (Meggs & Purvis, 2016; Triggs & Atzmon, 2019). Since then, graphic design started to encompass a range of activities such as printing art, commercial art, graphic art and advertising design (Heller, 1999). Graphic design as a 'named professional activity' started to grow and expand, mainly in America and Europe (Harland, 2016, p. 18). Yet, as noted by Barnard (2006), definitions of graphic design did not appear in mainstream dictionaries until the end of the 1990s.

Defining graphic design and the work of graphic designers used to be a simple task (Harland, 2007). In a conceptual paper, Harland presents the idea that 'typography', 'illustration', 'print' and 'photography' are the traditional domains of graphic design, but fail to fully capture the complexity and effect of graphic design in society. In another conceptual paper (Harland, 2011), he emphasised the relevance of discussing the language used to describe graphic design as it affects the understanding of the discipline. Language is a key part of graphic design practice, as graphic designers work towards translating language (e.g. a written briefing document, a request made verbally by the client) into visuals (Tomes et al., 1998). Exploring ways by which language operates to explain practices allows us to move beyond straightforward, received categories of the work and skillset of graphic designers to a more nuanced and dynamic approach for understanding the profession. Nonetheless, even the categorisations of the graphic designer's work are contested.

Defining the limits of graphic design is challenging, as the discipline 'concerns itself with issues of the humanities and social sciences, of objects and experiences, of people and messages' (Davis, 2012, p. 234). Davis claims that graphic design, as both a discipline of research and professional practice, is in constant relationship with so many other areas of practice and research that defining its boundaries is problematic. From an educational perspective, Giloi and Belluigi (2017, p. 8) suggest that the definition of graphic design affects 'what the designer does, the types of thinking that design involves, the processes of designing, and the nature of design products'. From both the educational and practice perspectives, Van der Waarde (2009) proposes that graphic design needs a more precise definition for improving practice and education. Overall, scholars recognise the challenges and benefits of defining graphic design as a field of practice and research.

An example of how difficult it is to define graphic design is that authors disagree on how to name the discipline or the professionals working in the field. In accounting for a narrative problem in graphic design history, Margolin (1994) criticises the historical narrative that graphic design accounts 'for all attempts to communicate with graphic devices since the beginning of human settlements' (p. 236). For Margolin, there is a need to make a distinction between graphic design and other visual communication practices. In research, the lack of consensus in naming the profession (graphic design, visual communication or communication design) has been argued as one of the reasons for poor performance of graphic design research in the Research Excellence Framework in the UK (Corazzo et al., 2019; Harland et al., 2018). In addition to that, Meron (2021), after reviewing online forums and interviewing graphic designers, pointed out that even graphic designers think that the term graphic design limits their professional contribution. Here, I present how authors have named the discipline in conceptual articles or practice textbooks in chronological order: Bonsiepe (1994) suggests that *graphic design* is associated directly with printing, then suggests that the profession should be called *information designer*. Buchanan (2001) pointed out an evolution from *graphic design* to *communication design* in attempting to emphasise the communication role of the profession. Similarly, Frascara (2004) acknowledged that *visual communication design* is a better term than *graphic design*, as *graphic design* emphasises the visual aspect of the profession, while *visual communication design* 'includes three essential elements of the profession: a method (design); an objective (communication); and a medium (vision)' (p. 3). Cezzar (2017) and Walker (2017) suggest that the term *communication design* is a synonym for *graphic design*.

Scholars like Buchanan (2001) and Frascara (2004) highlight the use of *communication design* as a way to reinforce the emphasis on the 'creation of effective communications' (Frascara, 2004, p. 4) instead of graphic elements. Similarly, Triggs and Atzmon (2019) suggest that by using *communication design*, graphic designers are blurring the boundaries of practice by collaborating with other disciplines (e.g. business, social sciences). Also, the authors suggest that the term 'communication design (…) reflects a reach beyond the traditional "subareas" of the design of posters, books, typography, etc.' (p. 11). A shift from *graphic designer* to *communication designer* suggests that the authors are 'claiming more ground' for the profession by expanding the contribution given by graphic

designers. However, one should not forget that graphic design is still built upon graphic elements such as illustration, typography, photography and print (Harland, 2011). Although I recognise the abovementioned limitations of the term *graphic design*, I use *graphic design* in this doctoral thesis as it is the most widely accepted term (Frascara, 2004). Also, I follow the reasoning presented by Triggs and Atzmon that the term *graphic design* has a clear connection to print materials that are key to understanding the past, present and future of the field. The authors then end the debate on nomenclature by saying that 'we have decided that using "graphic design" allows us to focus on other significant topics' (p. 11).

Following the goal of this doctoral thesis to contribute to the development of graphic design research and practice, I reviewed textbooks and academic articles on the graphic design field to locate definitions of graphic design. Despite the challenges in defining graphic design, previous authors have conceptualised what graphic design is in terms of what a graphic designer does and the functions performed by the work of graphic designers. Table 2 on the next page presents a compilation of definitions of graphic design published in both academic articles and textbooks.

| Source | Definitions of Graphic Design |
|---|---|
| Book *Thoughts on Design* by Rand (1970, p. 9) | Visual communications of any kind, whether persuasive or informative, from billboards to birth announcements, should be seen as the embodiment of form and function: the integration of the beautiful and the useful. |
| Article 'Graphic Design: Fine art or social science' by Frascara (1988, p. 20) | Graphic design is the activity that organizes visual communication in society. It is concerned with the efficiency of communication, the technology used for its implementation, and the social impact it effects. |
| Article 'A step towards the reinvention of graphic design' by Bonsiepe (1994, p. 9) | It is traditionally understood that the graphic designer is mainly a visualiser, one who organises visual components that are then reproduced with the aid of printing technology. Consequently, among the list of objects of professional action, we find: logotypes, stationery, book and magazine layout, advertisements and campaigns, posters, packages (labels), exhibitions, displays, corporate identity systems, signage systems. |
| Article 'Talking design: Negotiating the verbal-visual translation' by Tomes, Oates and Armstrong (1998, p. 131) | We are viewing graphic design as a process of translation from the verbal to the visual. |
| Article 'Design Research and the New Learning' by Buchanan (2001, p. 10) | Graphic design grew out of a concern for visual symbols, the communication of information in words and images. That the name of this profession or area of study has changed over the years only serves to emphasize the focus: it has evolved from graphic design, to visual communication, to communication design. Initially named by the medium of print or graphical representation, the introduction of new media and tools, such as photography, film, television, sound, motion, and digital expression, has gradually helped us to recognize that |

| | |
|---|---|
| | communication is the essence of this branch of design, independent of the medium in which communication is presented. |
| Book *Graphic Design: a Concise History* by Hollis (2001, p. 7) | Graphic design is the business of making marks and arranging them on a surface in order to convey an idea. |
| Book *Graphic Design as Communication* by Barnard (2005, p. 6) | Graphic design is considered as communication. Expressed in terms of functions, or the tasks that graphics performs, the definition covers magical, illustrative, persuasive and informative graphics. |
| Book *What is Graphic Design* by Newark (2008, p. 6) | Graphic design is the most universal of all the arts. It is all around us, explaining, decorating, identifying: imposing meaning on the world. |
| Book *The Fundamentals of Graphic Design* by Ambrose and Harris (2009, p. 10) | Graphic design takes ideas, concepts, text and images and presents them in a visually engaging form through print, electronic or other media. It imposes an order and structure to the content in order to facilitate and ease the communication process, while optimising the likelihood that the message will be received and understood by the target audience. A designer achieves this goal through the conscious manipulation of elements; a design may be philosophical, aesthetic, sensory, emotional or political in nature. |
| Article 'On graphic design: Listening to the reader?' by Van der Waarde (2009, p. 30) | Graphic design is a commercial service that develops visual means to support the relationship between clients and their contacts. In their designs, graphic designers accomplish this by balancing the desired effects, the visual goals and the visual elements and developing them coherently. |
| Book *The Thames & Hudson Dictionary of Graphic Design and Designers* by Livingston and Livingston (2012, p. 114 - 115) | Graphic design is the generic term for the activity of combining typography, illustration, photography and printing for purposes of persuasion, information or instruction. |

| | |
|---|---|
| Book *The AIGA Guide to Careers in Graphic and Communication Design* by Cezzar (2017, p. 15) | Graphic design, also known as communication design, is the art and practice of planning and projecting ideas and experiences with visual and textual content. The form it takes can be physical or virtual and can include images, words, or graphics. The experience can take place in an instant or over a long period of time. The work can happen at any scale, from the design of a single postage stamp to a national postal signage system. It can be intended for a small number of people, such as a one-off or limited-edition book or exhibition design, or can be seen by millions, as with the interlinked digital and physical content of an international news organization. It can also be for any purpose, whether commercial, educational, cultural, or political. |
| Article 'Research in Graphic Design' by Walker (2017, p. 549) | Graphic designers try to illuminate or explain things using graphic language, which may be verbal, pictorial, or schematic, and presented on paper or screen. Graphic designers work with illustrators, photographers, coders, editors, writers, and others involved in the graphic presentation of language. And, as with all designing, it is a planning and problem-solving discipline, concerned with purpose, process, visual judgement, engagement with users, and circumstances of use. |

**Table 2.** Definitions of graphic design, displayed in chronological order.

These definitions presented in Table 2 represent the authors' conceptualisations and ideas about what graphic design is. The authors do not expand on their grounds for coming to their definitions on graphic design. Most likely, these broad and overlapping definitions reflect an established narrative about what graphic design is over the years, which is based on the authors' professional and academic experience.

Past conceptualisations on graphic design have used different categories to summarise and explain what graphic design is. Newark (2008) in his textbook divides it into five categories: (1) alphabets, (2) typography, (3) images (e.g. illustration, photography), (4) tools (e.g. pencil, computer) and (5) disciplines (e.g. logos, posters). Van der Waarde (2009) in his work, based on a survey of more than 200 graphic designers in the Netherlands, defines graphic design under three categories: (1) visual elements (e.g. illustration, photography), (2) visual goals (e.g. film production, website design) and the (3) effects caused by the work of graphic designers (e.g. marketing, usability). Lastly, Harland (2011) in his conceptual article presents the dimensions of graphic design in terms of (1) actors (history, theory, criticism, research, education, craft), (2) functions (information, persuasion, interpretation, decoration, transformation, relation) and (3) context (social, economic, cultural, political, environmental, technological). While Newark's (2008) categories remain close to how graphic designers work (tools) and what they deliver (disciplines), Van der Waarde (2009) expands his categories to encompass the effects caused by this work. Harland (2011) highlights the 'human dimension' when referring to graphic design practitioners, but also takes a broader view of the field in referring to the functions performed by the work of graphic designers and the contexts where the work happens.

When reading the definitions presented in Table 2, and in line with the goal of the literature review of understanding the definitions of graphic design presented in earlier literature, I used thematic analysis (Braun & Clarke, 2006) to extract the main themes discussed in the definitions. By breaking down the definitions and extracting the topics discussed, I found the 'building blocks' that together form the definitions of graphic design. How scholars define graphic design and the overarching topics discussed in the definitions provided guidance for this thesis in understanding the current narratives present in graphic design. Thus, the studies of this doctoral thesis can provide empirical information to complement and discuss these definitions of graphic design.

Overall, I found four distinct themes in how authors defined graphic design: (1) *visual elements*, (2) *visual outcomes*, (3) *functions* and (4) *context*. *Visual elements* are specific elements used by graphic designers (e.g. typography, images) to produce *visual outcomes* (e.g. poster, website). These two themes resemble earlier conceptualisations by Newark (2008) and Van der Waarde (2009). These *visual*

*outcomes* perform a number of *functions*. For example, a poster has the *function* of informing about a product/service and also of persuading a person to buy a product. In this doctoral thesis, I cover seven functions for graphic designers' work: information, decoration, communication, persuasion, magic, metalinguistic and phatic. The *functions* given to a specific *visual outcome* are then positioned in a broader *context*, where I present how graphic designers have a social, cultural and economic (Barnard, 2005) impact on society through problem-solving practices (e.g. Frascara, 2004). These two last themes are more aligned with the dimensions of graphic design presented by Harland (2011). The next items in this section of the doctoral thesis elaborate further on these four topics. I also present some of the definitions used in Table 2 and highlight the parts that are directly connected to the topic.

### Visual elements

> 'Graphic design grew out of a concern for visual symbols, the communication of information in **words and images**.' Buchanan (2001, p. 10)

> 'Graphic design takes ideas, concepts, **text and images** and presents them in a visually engaging form through print, electronic or other media.' Ambrose and Harris (2009, p.10)

> 'Graphic design is the generic term for the activity of combining **typography, illustration, photography and printing** for purposes of persuasion, information or instruction.' Livingston and Livingston (2012, p. 114 - 115)

The visual aspect and visual elements of graphic design are recurrent topics approached in literature. As succinctly put by Tomes et al. (1998), graphic design uses visual elements to translate written requirements into visual outcomes. In this vein, graphic designers are described as being mainly responsible for *building*, *selecting* and *organising* visual elements for conveying a message (e.g. Ambrose & Harris, 2009). As succinctly put by Davis and Hunt

(2017), in their textbook on visual communication design, visual 'elements are the physical signs and symbols (for example, images, words, colours, and graphic devices such as lines and shapes) used to communicate the subject matter of a message' (p. 21).

These visual elements used by graphic designers are the 'building blocks' for creating visual outcomes. Some examples of visual elements that graphic designers build and select are shapes, colours, typography, illustrations and photography, as pointed out by design textbooks (see e.g. Cezzar, 2017; Dabner et al., 2014; Davis & Hunt, 2017; Lupton & Phillips, 2015; Newark, 2008). These visual elements overlap with what Harland (2011) calls the four traditional domains of graphic design: typography, illustration, photography and print. In Figure 1, I show examples of shapes, colours, typography, illustrations and photography, which are the basic visual elements that form all kinds of graphic design communication that people see in their everyday lives.

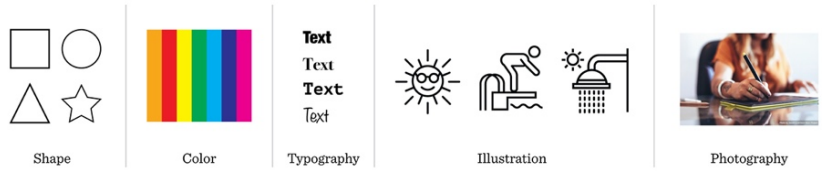



**Figure 1.** Examples of shapes, colours, typography, illustration and photography. Sources for the image: Typography: Fonts used: Helvetica Neue Condensed Black, Bodoni 72 Bold, PT Mono Bold and Noteworthy Light. Illustrations: 'Summer' collection, created by Creative Mania from the Noun Project (thenounproject.com). Photography: 'Close up of the hand of an illustrator making a design using a drawing pad' by Jacob Lund from The Noun Project (thenounproject.com)

The visual elements presented above are also found in the 'anatomy of graphic design', presented by Newark (2008) in his textbook *What Is Graphic Design?* (see Table 3 on the next page). The author complements the aforementioned shape, colour, typography, illustration and photography by adding *tools* and *disciplines*. *Tools* are defined as the means used to transfer one idea to a surface. For example, a computer is used to design a poster to communicate a certain idea. *Disciplines* are more similarly related to the visual outcomes, presented next

in this section. Contributing to a broader view of the visual elements in graphic design, Van der Waarde (2009) also suggests that animation, audio and video are part of the visual elements used by graphic designers in their work.

| Elements | Examples |
|---|---|
| Alphabets | Modules; Typefaces; Digital type; Full character set; Languages. |
| Typography | The grid; Hierarchy; Rules and other devices. |
| Images | Illustration; Photography; Using photography; Words and images. |
| Tools | Pencil; Materials; Paper; Computer. |
| Disciplines | Logos; Identity; Print – publicity, information; Packaging; Books; Magazines; Exhibitions; Signs; Web and film. |

**Table 3.** The anatomy of graphic design. Based on Newark (2008).

## Visual outcomes

> 'Consequently, among the list of objects of professional action, we find: **logotypes, stationery, book and magazine layout, advertisements and campaigns, posters, packages (labels), exhibitions, displays, corporate identity systems, signage systems.**' Bonsiepe (1994, p. 9).

> 'The form it takes can be **physical or virtual** and can include images, words, or graphics. The experience can take place in an instant or over a long period of time. The work can happen at any scale, from the design of a **single postage stamp to a national postal signage system.**' Cezzar (2017, p. 15)

> 'Graphic designers try to illuminate or explain things using graphic language, which may be verbal, pictorial, or schematic, and presented on **paper or screen.**' Walker (2017, p. 549)

The visual elements that graphic designers employ are then used to build visual design outcomes. In doing so, graphic designers arrange the elements on a specific surface. As stated by David and Hunt (2017), 'the meaning of a message depends not only on the choice of particular element, but also on their organization within a visual field' (p. 24). Visual concepts such as hierarchy, contrast, direction and grid are then relevant in organising visual elements into visual outcomes (e.g. Newark, 2008; Davis & Hunt, 2017).

Similar to design in general, where the work of designers is represented in terms of solving open and complex problems (e.g. Dorst, 2011; Dorst & Cross, 2001; Lawson, 2006), the visual outcomes designed by graphic designers are linked to problem-solving (e.g. Ambrose & Harris, 2009; Barnard, 2005; Harland, 2011; Walker, 2017). In an empirical study on the work of graphic designers, Tan and Melles (2010) found that graphic designers engage in problem-solving activities when moving from interpreting a briefing document to generating ideas for a visual outcome, even though the designers are not aware of their own reflection-in-action (Schön, 1984). In textbooks, Barnard (2005) differentiated graphic design from art by saying that graphic design 'can be characterised as "problem-solving", while art is "creative"' (p. 169). He also defines that the briefs and commissions from clients are the problems to be solved by graphic designers. By designing a visual outcome, Frascara (2004) mentioned that graphic designers are attempting to solve a problem proposed by an organisation. He claims that graphic designers need to be able to measure whether the problem was solved or not. For example, by designing a poster that invites people to buy organic products, graphic designers need to evaluate how the poster was able to change people's opinions and buying behaviours.

In his classic work on graphic design, Rand (1970) mentions billboards and birth announcements as examples of visual design outcomes. In their textbook, Ambrose and Harris (2009) classify the visual outcomes that graphic designers deliver into *physical* and *digital* graphic design outcomes. By printing a message on paper, graphic designers create physical outcomes of their work (for the history of print design, see Harland, 2016; Meggs & Purvis, 2016). Multiple examples of physical graphic design work have been pointed out in literature (Ambrose & Harris 2009; Cezzar, 2017; Frascara, 2004): from small-scale physical objects such as business cards, flyers, magazines and books to big-scale objects such as environmental graphic design/signage and outdoors. Some other

examples can be found in both small and big scales, such as packaging, posters, advertising, maps, among others.

Digital design outcomes are a result of technological advancements that have shaped the graphic design profession (e.g. Frascara, 2004). Graphic designers delivering digital outcomes then have the possibility of adding sound, motion and interactivity to their work. Over the past years, the rise of digital design has been pointed out by industry reports. For example, Maeda (2017) reviewed a survey made by the American Institute of Graphic Arts (AIGA) and suggested that 'the future of design is digital' (p. 25). Also, the *New Design Frontier* (Buley et al., 2019) suggests that for many organisations 'design is what happens on screens' (p. 15), as designers deliver wireframes and interfaces. Following that, graphic designers are suggested to be working with websites, animation, film and interfaces, among others (Ambrose & Harris, 2009; Cezzar, 2017; Frascara, 2004; Nini, 1996), which broadens the types of outcomes delivered by graphic designers and suggests that graphic designers need to carry a new set of digital skills.

### Functions: information, decoration, communication, persuasion, magic, metalinguistic and phatic

> 'Graphic design is considered as communication. Expressed in terms of functions, or the tasks that graphics performs, the definition covers **magical, illustrative, persuasive and informative graphics.**' Barnard (2005, p. 6)

> 'Graphic design is the most universal of all the arts. It is all around us, **explaining, decorating, identifying**: imposing meaning on the world.' Newark (2008, p. 6)

> 'Graphic design is the generic term for the activity of combining typography, illustration, photography and printing for purposes of **persuasion, information or instruction.**' Livingston and Livingston (2012, p. 114 - 115).

The visual design outcomes designed by graphic designers are described in textbooks as having the goal of informing (e.g. Barnard, 2005; Harland, 2011; Hollis, 2001; Newark, 2008; Livingston & Livingston, 2012) and/or decorating (e.g. Barnard, 2005; Harland, 2011; Newark, 2008). As they design visual materials, graphic designers are depicted as responsible for signalling valuable information about a product/company that distinguishes it from others (Newark, 2008). For example, by designing a packaging, graphic designers are able to identify the brand and the product owner at the same time that they inform people about the features of the product and ways of using it. Informing as a function happens in simple forms, such as telling that a specific place is a shop, and not a private house (Barnard, 2005). However, informing also happens in more complex ways, such as in environmental graphic design, where signs help people to navigate across a space. The decorative, or aesthetic, function refers to visual design outcomes having a purely ornamental role (Barnard, 2005). In other words, the visual design outcome exists as a 'source of enjoyment and pleasure' (p. 15). In contrast to the function of informing, this function has a primary emphasis on the visual appeal of a visual outcome. That said, in a number of cases, the informative and decorative function are intertwined. For example, signage has the goal of informing people how to move around a space, while at the same time it decorates the space. Also, a book cover has the function of informing – mainly, the title of the book and who wrote it – whilst also having a decorative function when displayed on a shelf.

By fulfilling identification or decorative functions, visual design outcomes also engage in the process of communication and persuasion. As discussed earlier in this section, a few authors (Buchanan, 2001; Cezzar, 2017; Harland, 2011; Frascara, 2004; Walker, 2017) recognise the communication aspect of graphic design to the extent that the word 'communication' should be used in the name of the profession. Buchanan (2001) claims that communication is the main essence of graphic design (for more on graphic design as communication and communication theories, see Barnard, 2005). Similar to the communication function, visual design outcomes are also described as having the role of persuading users towards a particular behaviour. As stated by Barnard (2005), the job of visual outcomes is 'to persuade, to convince or merely affect a change in thought or behaviour' (p. 15). For Frascara (2004), persuasion in graphic design happens in three different areas: advertising (commercial and non-

commercial), propaganda (political and ideological) and social interest communications (health, hygiene, etc.). In all these cases, graphic designers can deliver a range of physical and digital outcomes for persuasion interests. A specific example is a poster with the goal of selling (persuading people to consume) a product. Overall, in the words of Frascara (2004, p. 12), 'Every piece of communication design arises from the need to communicate a specific message, and to obtain a desired response'.

Following up on the graphic design functions, Barnard (2005) suggests magic as the least obvious function of graphic design work. The author defines the magic function as representing something in a different way than it really is. To clarify his point, he gives an example: 'Thirty graphics students asked to draw a garden will produce thirty different gardens, thus transforming one thing into another thing' (p. 16). Thus, he justifies the use of the word magic by pointing out that what magicians do is exactly 'turning one thing into another' (p. 16).

Lastly, Barnard (2005) presents together the metalinguistic and phatic functions. The former refers to a type of communication that explains or clarifies another type of communication. He then suggests quotation marks as an example of a visual element performing a metalinguistic function: these marks indicate that a specific sentence was said by someone else. The latter refers to elements that 'initiate, continue or conclude' (p. 16) a communication. He refers to comic books as an example, specifically the frames where the story happens: 'The framing, or the move from close-up to landscape, may have nothing to do with the narrative, but it keeps the story going, it tells the reader where to look next' (p. 16).

Overall, a visual outcome can perform many functions at the same time, while some are more obvious to a person than others. A poster can have, at the same time, the function of informing about a product, persuading a person to take action and also decorating an environment. The work done by graphic designers can fulfil a number of functions, which can be seen in the broader context of its impact on society, as discussed in the next section.

> 'Graphic design is the activity that organizes visual communication in society. It is concerned with the efficiency of communication, the technology used for its implementation, and the **social impact it effects.**' Frascara (1988, p. 20)

> 'And, as with all designing, it is a **planning and problem-solving discipline**, concerned with purpose, process, visual judgement, engagement with users, and circumstances of use.' Walker (2017, p. 549).

> 'It can also be for any purpose, whether **commercial, educational, cultural, or political.**' Cezzar (2017, p. 15)

Barnard (2005) defines the functions of graphic design on two levels: the individual functions of a graphic design outcome (discussed in the last section) and the 'social, cultural and economic functions of graphic design "as a whole"' (p. 13). By pointing to the functions 'as a whole', Barnard (2005) exemplifies the context where the work of graphic designers happens in line with the contexts suggested by Harland (2011). Harland then complements the social, cultural and economic contexts by adding the political, environmental and technological contexts. Overall, the context where the work of graphic designers happens has a connection with the impact this work has on society (e.g. Frascara, 2004).

The broad impact on society and the multiple contexts where the work of graphic designers occurs is covered in the literature. As pointed out by Harland (2007, p. 10): 'Graphic design can be described as a social phenomenon. It is a global form of representation that can be used as a measure of how people communicate with each other on a local, national and international scale.' In addition to affecting how people communicate, graphic design outcomes are portrayed as shaping how people behave (Frascara, 2004). For instance, advertising design is expected to make people buy certain products and services, while political propaganda shapes people's beliefs. Thus, graphic designers' work is suggested to enhance the likelihood that a message will be received (Ambrose & Harris, 2009). Graphic designers are described as possessing the ability to

shape behaviour and the 'obligation to contribute to the public understanding of environmental and social issues' (Meggs & Purvis, 2012). Similarly, graphic designers' work is responsible for building, reproducing and challenging societal values (Barnard, 2005). In other words, by generating visual outcomes to deliver messages, graphic designers are entitled to build a visual culture (Harland, 2007) that spreads discourses and promotes values.

In summary, authors define graphic design as an activity in charge of building and organising visual elements into visual design outcomes (i.e. physical, digital). In doing so, graphic designers attain their goals for information, decoration, communication, persuasion, magic, metalinguistic and phatic functions. Lastly, the work of graphic designers happens in social, cultural, economic, political, environmental and technological contexts. These contexts are then grouped according to how the work of graphic designers has an impact on society. These definitions of graphic design – in terms of what a graphic designer does and the functions of graphic designers' work – are mainly published in conceptual articles and textbooks. Their arguments are largely built on secondary sources of data, while this doctoral thesis builds on primary data drawn from graphic design job advertisements. The next sections of this doctoral thesis develop a discussion on defining graphic design by focusing on how additional literature has described the work performed by graphic designers and the skills that graphic designers should carry for delivering their work.

## 2.2 Graphic designers' work

An important part of describing graphic design is looking at what graphic designers do in their work. A similar case of looking at what *industrial* designers do was reported by Lloyd (2002), who presented how the work of industrial designers was represented in television programmes in the UK. As presented earlier, graphic designers build and select visual elements in order to deliver physical or digital design outcomes. In this subsection, I report on how literature – mainly empirical studies – describes the work graphic designers do in practice.

The work of graphic designers is portrayed by authors in two complementary ways: first, as a creative practice, where creativity and innovation are key

ingredients in developing graphic design projects and, second, as a mundane practice, where the creative work is complemented by routine activities in terms of management, scheduling and note taking. This distinction between creative and management practice is noted by Julier (2010, p. 238), who points to the 'messy, *mundane* and humdrum world of design in contrast to more widespread conceptions of *creativity* as being "spontaneous" or resistant to ordering and management' (emphasis by the author of this doctoral thesis). I cover these two complementary descriptions of the work done by graphic designers in the next paragraphs.

The description of the work of graphic designers as a creative practice recurs in the literature. In the case of Martin (2012), the author defined the choices made by designers as part of the creative process. He followed two graphic designers during a packaging design project and reported on how both physical and digital materials were used throughout the process. Other cases are provided by the studies by Laing and Masoodian (2016) and Laing et al. (2017), who observed how the influence of images affected the 'creative design process' (2016, p. 189) of graphic designers. In both studies, the authors studied how design students engage in the design ideation process. Another case is a study reported by Tan et al. (2009) and Tan and Melles (2010), who followed the development of a storybook. The authors described how a graphic designer went through four phases during the project: briefing, interpretation, idea generation and presentation. They defined that the initial stages of the design process involve 'a range of personal decision-making and creative activities, together with the use of preferred tools and strategies' (2010, p. 462). As a conclusion, the authors argued that graphic designers engage in 'creative' tasks such as sketching and generating ideas in order to solve the design problem. Overall, scholars presented the work of graphic design with an emphasis on the creative process, making references to graphic designers as professionals who bring creative 'spark' to solving design problems.

However, for delivering creative work, graphic design work is surrounded by management practices (Dorland, 2009). In reviewing the literature on how the creative process in graphic design studios is managed, Jacobs (2017) concludes that: 'Management and economic concerns shape the creative process more than many graphic design educators are willing to acknowledge' (p. 174). According

to the author, studios need to introduce management activities (e.g. time management) in order to operate efficiently from an economical perspective.

This dichotomy between creative practice and routine and management work is also noted by Julier (2010). In his conceptual article, the author points out that 'the public face or ethos of a design firm may be that of unrestrained creativity while the actual everyday practice may be a lot more humdrum' (Julier, 2010, p. 254). Similar to that, Dorland (2009) highlights how, despite the stereotypical image of 'playful practice', the creative process of graphic designers is surrounded by management, auditing and measurement activities. She points out that design studios are often described as a space of creativity and 'play': 'In the promotional material created by design studios, graphic designers (and creative workers in general) are often positioned as playful subjects or adolescent adults playing at work' (p. 106). However, after conducting interviews and observational research for over four years at a design consultancy in Canada, she found that graphic design work contains several management practices. For example, she mentions a project planning meeting that she followed where elements of time and budget shaped how the visual outcome was going to be constructed. In sum, designers position themselves closer to the idea of 'creative workers', as this is the image mainly associated with design practice. However, at the same time, designers must engage in lots of 'mundane' (Julier, 2010, p. 238) activities.

Overall, authors describe the work of graphic designers in a variety of ways; they present a stereotypical image of 'creative' professionals working in a creative environment, but also show how the work of graphic designers involves 'mundane' activities (e.g. filling spreadsheets). These different ways of working have implications for the skills (graphic) designers should have to operate within organisations. In other words, to perform 'creative' and 'mundane' activities, which skills should graphic designers possess? In this sense, graphic design professionals are in need of guidance on how to develop their skillsets for the job market in order to perform both 'creative' and 'mundane' activities. In the following subsection, I report on how the skills carried by graphic designers are described in literature.

## 2.3 Graphic designers' skills

In the literature, discussions on the skills held by graphic designers are found mainly in textbooks used in design education practices. The authors of these works present their ideas about the skillset a graphic designer should carry, mainly aiming to provide support for graphic design students and professionals in their career. I start this section by presenting the skills immediately related to graphic design – visual elements, tools, methods – and then move to presenting skills traditionally found outside the design field – communication, teamwork, research and business knowledge.

Knowledge of visual elements and how to organise them visually comprises one of the foundation skills of graphic designers. Starting from the basics, graphic designers select typography, colours, shapes, illustrations and photographs when building visual outcomes (see more about these graphic design elements in the following textbooks: Cezzar, 2017; Dabner et al., 2014; Lupton & Phillips, 2015; Newark, 2008). These are the elements that graphic designers use to convey a message in their work. For example, by selecting a handwritten typeface, a designer embeds the message of something exclusive or tailor-made to the work. After selecting the elements, the graphic designer needs to organise the elements in the visual workspace to deliver the message. Issues such as hierarchy, proportion, balance, proximity and grid, among others, have fundamental importance in organising the elements (e.g. Newark, 2008; Cezzar, 2017).

Knowledge about tools is another area of graphic design skill. As presented by Newark (2008, p. 108) in his textbook, 'design is a process of making, it is the transfer of ideas to a surface by means of tools'. Specifically, graphic designers use items such as pen/pencil, paper, computer and design software as tools for doing the job. Ambrose and Harris (2009, p. 150) in their textbook add that both digital and analogue tools enable designers to 'express themselves in any number of ways to experiment and develop a visual idea'. They also enumerate a number of tools such as cutting mats, brushes, drawing tablets and scanners, among others.

Knowledge of methods for solving a graphic design problem is an additional area of graphic design skills. A design method is a combination of repeated steps

to navigate across a design project. For example, Frascara (2004) suggests in his book that most communication design projects follow a planning, a visualisation and a production phase. Cezzar (2017) in her career guide for graphic designers adds that graphic design projects include research, ideation, prototyping, iteration and presentation. Regardless of the method used (or which the author followed), graphic designers acquire design method skills to be able to solve the problems they face.

In relation to a more expanded notion of the work of graphic designers, skills traditionally found outside the design field are also needed to operate as a graphic designer. Historically, these skills are considered *generic* and not necessarily linked only with graphic design work. Or, as said by Littlejohn (2017, p. 36) in her empirical study on the skills taught in design programmes in the US, these skills 'are noticeably different than the traditional conceptual, aesthetic, and form-giving skills that are the instructional focus of most design schools'. Following the argument that the work done by graphic designers is increasing in complexity (e.g. Frascara, 1988; Davis, 2006; Davis, 2008; Harland, 2016), professionals are asked to acquire new skills in order to solve design problems. In the following paragraphs, I highlight these skills suggested in literature.

Communication and teamwork are frequently mentioned as skills needed by graphic designers (Giloi & du Toit, 2012; Cezzar, 2017; Littlejohn; 2017; Frascara, 2017). In design management literature, specifically on the role of designers in organisations, the role of communication is also attributed to designers in terms of building integration and communication between departments (e.g. Bohemia, 2002; Perks et al., 2005; Valencia et al., 2013). One of the ways in which designers are suggested to promote communication is through idea visualisation, where designers represent an idea visually to share and discuss it with other professionals (Blaich & Blaich, 1993). The main reason for mentioning the need for communication and teamwork is that graphic designers are suggested to be involved in multidisciplinary teams, which requires collaboration and partnership for delivering the work. In the words of Frascara (2017, p. 129):

> The act of designing always involves people beyond the designer: clients, users, experts from other disciplines, and researchers. A

> designer must be educated in teamwork, and in the skill to speak different disciplinary languages. Design is not art, nor solitary activity. Teamwork involves recognizing clients and users as partners in the design process. One does not work for them, one works with them.

In discussing trends in how to assess graphic designers' work in education, Giloi and Du Toit (2012) also emphasise that communication and teamwork are required for graphic designers. Similarly, Littlejohn (2012, p. 36) argues for 'interdisciplinary competency' and 'collaborative competency' as two non-traditional (outside the design sphere) competencies for graphic designers.

Another generic skill mentioned in more recent literature for graphic designers is the ability to conduct research. As pointed out by Littlejohn (2017), graphic designers are being asked to base their project decisions on empirical research. Similarly, Frascara (2017) claims that more knowledge is required design research methods, such as participatory design, co-creation, interviews, among others. In the introduction to his book *The Education of a Graphic Designer*, Heller (2015) also highlights that skills outside the design area, such as research and writing, are 'increasingly more integral to a well-rounded career' (p. 11). Following this idea, graphic designers will expand from the traditional visual aspect to collaborate in generating and interpreting content/texts. Regarding this point, Frascara (2017, p. 129) argues that: 'Exclusive concentration on "the look of things" is long passed'; therefore, graphic designers need to have the ability to produce and interpret texts for their visual materials.

Another frequently mentioned skill for graphic designers is business knowledge. Heller (2015) points out that graphic designers should carry a basic understanding of fundamental business. Similarly, Giloi and Du Toit (2013) suggest that entrepreneurship is a requirement for graphic designers. In the design management literature, designers who acquire business skills are described as being able to 'move up' in the company's organisational hierarchy (e.g. Blaich & Blaich, 1993; Borja de Mozota, 2003; Green et al., 2004).

All in all, as design problems become more complex, graphic designers are required to carry a broader range of skills (e.g. Davis, 2006, 2008). Literature suggests that graphic designers should carry skills ranging from visual elements

to graphic design tools and methods. Also, literature points to many *generic* skills that graphic designers should have. These generic skills are not necessarily related only to graphic design, but are shared by a number of other professions. Examples of these generic skills cited in the literature are communication, teamwork, research and business. The studies of this doctoral thesis aim to provide guidance for graphic designers in how to plan their career and skills acquisition. In addition to that, this doctoral thesis contributes to narrative discussions on defining graphic design and the work done by these professionals, specifically by looking at how language is used in job advertisements and how it shapes the profession.

## 2.4 Section summary

Authors present different directions in defining graphic design, the work done by graphic designers and the skills needed to operate within organisations. The language used to describe graphic design shapes the practice of the profession (Giloi & Belluigi, 2017). Thus, the language used to describe graphic design is important in defining the value designers bring to organisations. In the studies of this doctoral thesis, the language used in the job advertisements plays an important role in defining the work of a graphic designer and the skillset of these professionals. On top of that, the language used in job advertisements constitutes part of the common narrative found in academia, practice and education about what graphic design is and what graphic designers do for organisations.

Definitions of graphic design have long been a topic in both textbooks and academia (see Table 2 for definitions of graphic design). Even the name of the profession is constantly discussed: while some authors prefer using *graphic design*, others would rather use broader terms such as *communication design* or *visual communication design*. Overall, a common narrative in design literature is that graphic design is a practice in charge of using visual elements (i.e. shapes, colours) to deliver visual outcomes (i.e. posters, interfaces). In doing so, graphic designers are represented as engaging in information, decoration, communication, persuasion, magic, metalinguistic and phatic functions. From a

more holistic view, the work provided by graphic designers is described as making an impact in multiple contexts: social, cultural, economic, political, environmental and technological contexts.

The work done by graphic designers is also highlighted in previous literature. Graphic design is commonly viewed as a creative practice, where designers are portrayed as creative professionals that work in an environment that encourages creative ideas. In graphic design literature, scholars present the creative practices happening in graphic design projects. For example, Martin (2012) described the decisions made by designers in a packaging design project, such as how they decided on the typeface. On the other hand, scholars also highlight more managerial and routine activities: Dorland (2009), Jacobs (2017) and Julier (2010) emphasise this mundane practice, where graphic designers also perform a number of activities involving management, auditing and measurement.

Literature suggests that graphic designers should have knowledge of visual elements, tools and methods. Also, considering the complexity of the problems faced by graphic designers and how collaboration is needed in order to solve those, graphic designers are portrayed as carrying skills outside the design realm in areas such as communication, teamwork, research and business.

Overall, descriptions on the work and skillset of graphic designers are located in conceptual articles and textbooks, forming a narrative that graphic designers are professionals concerned with delivering visual outcomes. The studies of this doctoral thesis present an extension to this narrative, looking at what happens in the job market: graphic design (similar to design in general) is expanding. Although knowledge on creating visual elements and outcomes is still highly relevant, graphic design practice is more complex and designers need to expand their contributions and skills towards other areas (e.g. business, technology). The challenge for design educators, practitioners and researchers is to consider how design is changing in local contexts and how technology affects graphic design nowadays. The studies of this doctoral thesis fill this gap by empirically studying the skills and work described by organisations in their job advertisements. Specifically, this doctoral thesis aims to help design practitioners in designing their career paths in terms of skills development. Also, the goal is to provide a framework for design educators in planning their

educational activities. Lastly, as an emergent field of research, the studies aim to contribute to current narratives found in literature on defining the work and skillset of graphic designers.

In the next section, I present the methodological approach for the studies in this doctoral thesis.

# 3. Methodology

Studying methods used in design research has become a relevant theme for design researchers in increasing the quality and relevance of the field. An increase in methodological rigour is noted as a way to improve quality in design research (Cash, 2018; Dorst, 2008). While scholars (Cash et al., 2013; Chai & Xiao, 2012) have mapped the themes and most cited papers in design research, an overall view on the methods selected by design researchers has not been identified earlier (as an exception, see Candi and Gemser (2010) and Liu and de Bont (2017) who classified the work of earlier scholars into 'quantitative' and 'qualitative').

Overall, design research is diverse in terms of both the topics studied and methodology (McMahon, 2012). At the same time, scholars note that design research has traditionally been conducted through experiments (Ball & Christensen, 2018; Cash et al., 2013) and/or in classrooms (Ball & Christensen, 2018). An example is research in design fixation, which Crilly (2019, p. 79) argues has been 'largely mono-methodological', mainly based on experiments. This lack of methodological variety is problematic as it 'limits the confidence that researchers can have in the recommendations they make for improving practice' (p. 79).

After reviewing the literature in the fields of graphic design and design management, it is evident that past research is not so extreme as to be considered 'mono-methodological' (Crilly, 2019, p. 79). Yet, I identified a predominance of empirical studies based on data generated for research (e.g. Flick, 2004), such as interviews and surveys. In this case, scholars based their findings on retrospective accounts of the phenomenon. As exceptions, I highlight ethnographic works in graphic design that have looked closely at how graphic

design happened in practice (e.g. Dorland, 2009; Martin, 2012; Tan & Melles, 2010) and the work done by Meron (2021), who looked at discussions happening in graphic design online forums. Empirical research in graphic design has mainly relied on interviews (e.g. Littlejohn, 2017) or experiments (Stones & Cassidy, 2007, 2010). In parallel to that, empirical research on design management and the value of design for organisations has been mainly conducted through interviews (e.g. Berends et al., 2011; Valencia et al., 2013) and surveys (e.g. Bohemia, 1998; Chiva & Alegre, 2007).

Multiple perspectives can potentially improve the understanding of graphic design research and practice. In this doctoral thesis, I selected *document analysis* of *job advertisements* as a method to investigate the skillset sought by organisations when recruiting graphic designers. According to methodology scholars (Bowen, 2009; Bryman, 2012; Rapley, 2007), document analysis is a systematic technique for analysing printed and electronic texts. Researchers can analyse the process of writing the document (e.g. authors involved, process) or the content (e.g. sentences, words used) a document has, according to Coffey (2014). Regarding the first option, researchers should focus on the reasoning behind building the document to unveil questions such as 'what was the purpose of the document? who was the author of the document?' (Coffey, 2014; Flick, 2004). In looking at the reasonings, the authors and audience of a document, researchers could better assess whether the document 'fits' the research study (Bowen, 2009). In sum, as pointed out by Coffey (2014), when looking at documents' construction and consumption, researchers develop an interpretive understanding of documents with a special interest in the language used to deliver the message (construction) to a specific audience (consumption). In relation to the second option, researchers concentrate on the messages embedded in documents by their producers (Coffey, 2014). Following the author, when focusing on the message carried by documents, the research approach can vary from quantitative measures (e.g. counting frequencies) to more qualitative theory building (e.g. grounded theory). In doing that, researchers are then responsible for interpreting and extracting the meaning of documents (Bowen, 2009).

Job advertisements are documents written and placed at the beginning of the recruitment process, in which companies communicate their needs in terms of the skillset of a professional (Barber, 1998; Dineen & Soltis, 2011; Walker &

Hinojosa, 2013). In other words, job advertisements are a communication tool between companies and candidates (e.g. Chipulu et al., 2013; Harper, 2012; Kim & Lee, 2016). In a job advertisement, an organisation typically describes (1) the ideal candidate (e.g. qualifications, educational requirements), (2) the recruiting company (e.g. size, location) and (3) the job terms (e.g. salary, benefits) (Todd et al., 1995). Similarly, Rafaeli and Oliver (1998) claim that job advertisements deliver information about the qualifications (recruitment message) sought by a company and also about the company's culture and values (impression management message).

In sum, *document analysis* of *job advertisements* represents a suitable method for answering the research questions of this doctoral thesis in terms of the skillset sought by organisations when recruiting graphic designers. First, job advertisements are recruitment documents, in which organisations describe their needs (skillset) when recruiting a new employee. Second, document analysis is a method designed to analyse texts. As presented earlier, researchers can analyse documents based on their written content and/or the process of writing the texts (Coffey, 2014). In the studies of this doctoral thesis, I take into account these two perspectives: Studies 1, 2 and 3 focused solely on the written content of job advertisements, while Study 4 looked at the process companies go through when writing a job advertisement. For studying the process of writing the job advertisements, I followed 12 recruitment processes in Finland, and interviewed the hiring managers and designers recruited for the positions. Also, I executed a systematic literature review on how previous scholars have used job advertisements in their studies.

In this section, I present the research approach followed throughout the studies of this doctoral thesis. In doing so, my goal is to provide an overview of the methodological decisions made in the process of investigating the skillset of graphic designers. I start by presenting how job advertisements were used throughout the studies in terms of data collection, management and analysis. Then, I cover how the interviews and a systematic literature review complemented the analysis of the job advertisements throughout the studies. Lastly, I discuss how the quality of the research was assured in terms of reliability, validity, generalisability and ethical concerns regarding the data analysed.

## 3.1 Document analysis of job advertisements (Studies 1, 2 and 3)

As stated earlier, document analysis was selected as the method for analysing job advertisements in this doctoral thesis. Document analysis involves reading and interpreting the text present in documents (Bowen, 2009). The author then suggests that document analysis 'combines elements of content analysis and thematic analysis' (p. 32) when reading methodology texts on content and thematic analysis. Similarly, Coffey (2014) suggests that document analysis can follow thematic analysis principles for a more qualitative approach or content analysis principles for a more quantitative approach. Based on the suggestions made by document analysis to follow thematic and content analysis, I reviewed methodology texts on how to conduct thematic and content analysis to execute document analysis of job advertisements in the thesis.

Thematic analysis (e.g. Boyatzis, 1998; Braun & Clarke, 2006; Fereday & Muir-Cochrane, 2006; Vaismoradi et al., 2013) is a method for finding patterns (themes) within data. Some authors (Braun & Clarke, 2006; Boyatzis, 1998) consider thematic analysis as a foundational method that provides skills for any researcher in using other qualitative techniques. Thematic analysis is 'a poorly demarcated and rarely acknowledged, yet widely used qualitative analytic method' (Braun & Clarke, 2006, p. 77) in fields such as social science (see Boyatzis, 1998), psychology (see Braun & Clarke, 2006) and health (see Vaismoradi et al., 2013; Fereday & Muir-Cochrane, 2006).

Content analysis (e.g. Krippendorff, 2013; Neuendorff, 2002; Vaismoradi et al., 2013) is a method for making inferences about a text, or a group of texts, by counting instances of a determined category. In other words, it is a 'systematic, objective, quantitative analysis of message characteristics' (Neuendorff, 2002, p. 1). Content analysis is considered one of the most important techniques in social science (Krippendorff, 2013), as it studies interactions in society that occur through talk and texts.

By comparison with thematic analysis, content analysis adds counting to the activities of coding and identifying themes in the data (Vairsmoradi et al., 2013). Following this more quantitative approach, content analysis provides a 'numerically based summary' (Neuendorff, 2002, p. 14) instead of a full report

about a dataset. Coding and counting instances of themes in a group of texts can follow two approaches: human coding or computer coding (Neuendorff, 2002). Briefly, in human coding, one or more persons establish a number of relevant topics and search for these in the data. Computer coding involves building a list of words or expressions and using software to automatically search and count the frequencies. Also, different from thematic analysis, scholars emphasise the possibility of assessing intercoder reliability. As stated by Krippendorff (2013, p. 18) 'researchers working at different points in time and perhaps under different circumstances should get the same results when applying the same technique to the same data'. Thus, content analysis puts more emphasis on the replicability of the results and assessing coders' 'quality'.

In Studies 1, 2 and 3 of this doctoral thesis, I executed document analysis of job advertisements by following thematic and content analysis recommendations in different parts of the analysis process. In sum, thematic analysis was used to build the coding schemes in the studies, while content analysis was used to code and count the frequencies of the requirements specified in the advertisements (the specific steps for analysing job advertisements will be further detailed in Section 3.1.2 Data analysis). I now present a review of how job advertisements and documents in general are used for research purposes.

Job advertisements provide primary data for understanding the skillset articulated by organisations when seeking to hire graphic designers. They are one example of *documents* that are used for research purposes. In design research, documents have been assigned a primary role in previous design research publications. For example, Lloyd and Snelders (2003) used interviews published in books and magazines for understanding the design process of the product Juicy Saliff – designed by Philippe Starck for the company Alessi. Another example is the study by Ryynänen (2009) who looked at how the design professional was described in the Finnish financial press (i.e. newspapers). Specifically in graphic design research, Meron (2021) reviewed discussions in graphic design online forums in order to assess the perceptions of graphic designers of their job titles. Documents have also been used by scholars as secondary or complementary types of data – for example Bruce et al. (1999) and Mutanen (2008) used company reports.

Job advertisements have the advantage of existing without the researcher's intervention and, therefore, provide information about the requirements specified by companies in their natural setting (e.g. Chipulu et al., 2013; Harper, 2012; Kim & Lee, 2016). As pointed out by Rapley (2007), job advertisements as a type of document represent naturally occurring data that were created for reasons other than for research inquiries. In this case, job advertisements are written by organisations for fulfilling a personnel need, not for answering a research question. The author emphasises that researchers should look at a phenomenon in its 'natural' setting, as people in a research setting, for example in interviews, behave differently than they would normally. He then gives an example of running a focus group: 'rather than only asking a focus group moderator about how they run focus groups, you can gain a good understanding of "how they run focus groups" through some form of recordings of them actually running focus groups.' (Rapley, 2007, p. 21). As pointed out by Potter (2002), researchers run into less interference in the data collection process – for example, not embedding their own agendas and ideas into questions – when using documents, which reinforces the idea of looking at a phenomenon in its natural form. Similar to the arguments made on behalf of documents, scholars who opted for job advertisements (Chipulu et al., 2013; Dörfler & van de Werfhorst, 2009; Marcusson & Lundqvist, 2015; Shou & Wang, 2017) claimed that this type of data overcomes challenges such as selection biases (e.g. self-selection and non-response) and response biases (e.g. social desirability) that are faced by researchers opting for other data collection methods. In addition, job advertisements are stable (Bowen, 2009; Lester et al., 2017) in the sense that they do not rely on retrospective accounts about a phenomenon that happened in the past, in which an informant might forget relevant details about how the event took place.

By studying job advertisements, I was able to cover a broader portion of the job market for graphic designers than would be possible with other data collection methods, such as interviews. As job advertisements are publicly and openly advertised documents (Chipulu et al., 2013; da Cunha, 2009; Kim & Lee, 2016), I collected a more significant number of recruitment cases in comparison to using other research techniques – 1,406 in Study 1, 230 in Study 2 and 371 in Study 3. As noted by Carnevale et al. (2014), between 80 to 90 per cent of

job openings requiring a bachelor degree are posted online, which is usually the case for graphic designers.

Yet, some limitations were faced by scholars while using job advertisements in research. First, as pointed out by other scholars, (Bettinger et al., 2016; Cloonan & Norcott, 1987; Dörfler & van de Werfhorst, 2009), not all positions are openly advertised. In this case, positions are filled through internal networks or direct recommendations. This is in line with the argument presented by previous scholars that some documents are not publicly available (Bowen, 2009; Lester et al., 2017; Yin, 2009). In trying to overcome this limitation, I collected job advertisements from multiple generic (e.g. LinkedIn) and design-specific job boards.

Second, in relation to the content of job advertisements, scholars recognised that they are written with the aim of convincing candidates to apply (e.g. Backhaus, 2004). In other words, job advertisements are part of self-representation communication (Atkinson & Coffey, 2004), where organisations control the content. Therefore, organisations might select words that attract qualified candidates, instead of reporting the 'real' description of the work and all the qualifications needed (e.g. Brumberger & Lauer, 2017; Chipulu et al., 2013; Ratledge & Sproles, 2017; Xu & Chen, 2000). However, organisations are incentivised to provide accurate descriptions in their job advertisements in order to attract qualified candidates for their open positions (García et al., 2010; Gatewood et al., 1993; Jackson, 2007).

The argument presented by scholars concerning the limitations of the content of job advertisements resembles discussions by methodology scholars in terms of limitations of documents as data. First, documents might not present the 'whole picture' (Bowen, 2009; Lester et al., 2017). In other words, documents might lack contextual information that is useful for understanding the phenomena involved. Also, documents provide a version of the reality. As noted by Rapley (2007) documents do not provide a single 'truth', 'but rather multiple and sometimes contradictory truths or versions' (p. 128). The term 'documentary realities' is then used by Coffey (2014) as a version of the reality which is covered in a document. The author uses the example of minutes of meetings to illustrate how this document is a 'partial and scripted' (p. 377) version of what happened in the meeting. Lastly, what is not written in a

document can be even more important than what is written. For example, Rapley (2007) discusses hidden information in documents using a case of a dating advertisement published in a local newspaper. In the document, a woman describes herself and what she is looking for in a gentleman. The author then reports the fact that her age was an expected, but not disclosed, piece of information: 'given the immediate context and our cultural knowledge about dating, the omission of her age is, potentially, a noticeably absent feature of the text' (Rapley, 2007, p. 112).

Overall, I acknowledge the fact that a job advertisement might not provide a complete description of all qualifications needed for a designer position, but still, as organisations describe the 'ideal' applicant, the job advertisement represents the qualifications for the 'ideal' professional (e.g. Todd et al., 1995).

### 3.1.1 Data collection and management

By studying job advertisements in three countries, some comparative sense of their similarities and differences can be explored. The UK was selected as a country with a strong history and tradition in design practice and education in Europe (e.g. Moultrie & Livesey, 2009; Harland et al., 2018). Similarly, despite its smaller size in terms of population and number of design schools, Finland is also recognised as a leading design nation (Moultrie & Livesey, 2009), where the role of design has expanded over time (Valtonen, 2005, 2007). For example, the Finnish capital Helsinki and its four neighbouring cities (Espoo, Vantaa, Kauniainen and Lahti) were granted the title of 'World Design Capital' in 2012 by the International Council of Societies of Industrial Design (WDC, 2012). In addition, the design government policy 'Design 2005!' is frequently cited as a reference in efforts to effectively introduce design into industry (Korvenmaa, 2011). Lastly, design education and practice in Brazil has a shorter history in comparison with the UK and Finland. For example, design education started formally in the 1960s. In regards to promoting design, a government design policy was only created in 1995 by the Ministry of Industry, Commerce and Tourism aiming to improve the competitiveness of Brazilian companies by using design (Raulik & Murphy, 2010).

Similar to previous studies using job advertisements in design research (e.g. Yang et al., 2005), I collected job advertisements available online. As seen in the literature review of studies using job advertisements (Study 4), the majority of studies used a sample size of less than 250 job advertisements and collected data for less than one year from online job boards. In this doctoral thesis, Study 1 (UK) had a sample size of 1,406 job advertisements, collected over five weeks from online job boards. Study 2 (Finland) had a sample size of 230 job advertisements collected over 18 months from online job boards. Study 3 (Brazil) had a sample size of 371 job advertisements collected over five weeks from online job boards (see Table 4).

The job advertisements were collected from both design (e.g. Design Week, Brand Republic) and generic (e.g. LinkedIn, Monster) recruitment websites. In Studies 1, 2 and 3, the process started by reaching out to educators and practitioners and asking them which websites were relevant in each of the studied countries. Together with other authors in the studies, we executed broad searches with the goal of encompassing a significant parcel of job advertisements for graphic designers. The terms we used in each study were as follows: in Study 1, I collected job advertisements with the words 'graphic designer' in the job title and with the word 'designer' in the job title and 'graphic design' in the job description. In Study 2, I collected job advertisements with the word 'designer' or *'suunnittelija'* (designer in Finnish) in the job title. In Study 3, I collected job advertisements with the words *'designer gráfico'* in the job title (graphic designer in Portuguese) and with the word 'designer' in the job title and *'design gráfico'* (graphic design in Portuguese) in the job description. (see Table 4 on the next page).

| | Job advertisements | Search terms |
|---|---|---|
| Study 1 | 1,406 job advertisements for graphic designers in the UK, collected over a five-week period (Jun/2015 – Jul/2015) from the following websites: Indeed (www.indeed.co.uk), LinkedIn (www.linkedin. co.uk), Monster (www.monster.co.uk), Brand Republic (www.brandrepublic.com) and Design Week (www.designweek.co.uk)<br><br>Total number of words: 325,847.<br><br>Language: English. | Search 1: 'Graphic designer' in job title<br><br>Search 2: 'Designer' in job title and 'graphic design' in job description |
| Study 2 | 230 job advertisements for graphic designers in Finland, collected over an 18-month period (Jun/2015 – Dec/2016) from the following websites: LinkedIn (www.linkedin.fi), the Ministry of Employment and Economy's job board (www.mol.fi) and Oikotie (www.oikotie.fi).<br><br>Total number of words: 85,655.<br><br>Language: English and Finnish. | Search 1: 'Designer' in job title<br><br>Search 2: 'Suunnitelija' (designer in Finnish) in job title |
| Study 3 | 371 job advertisements for graphic designers in Brazil, collected over a five-week period (Jun/2015 – Jul/2015) from the following websites: LinkedIn (www.linkedin.com.br), Indeed (www.indeed.com.br), Infojobs (www.infojobs.com.br), Trampos (www.trampos.co) and Adonline (www.adonline.com.br)<br><br>Total number of words: 50,838.<br><br>Language: Portuguese. | Search 1: 'Designer gráfico' (graphic designer in Portuguese) in job title.<br><br>Search 2: 'Designer' in job title and 'design gráfico' (graphic design in Portuguese) in job description |

**Table 4.** Search parameters of job advertisements in Studies 1, 2 and 3

The high number of job advertisements in this doctoral thesis required procedures to organise and handle the data for analysis. The process followed was similar in all three studies – collection, organising and cleaning – which I present below:

**Collection:** I performed the searches (following the terms defined in each study) in the selected websites manually. I saved a list containing the links of the job advertisements found in each search in a Microsoft Excel sheet. This list of links was then automatically (using an automatic script made on Automator for Mac) opened in a web browser. After the page was open, the script saved the content of the webpage using the software Evernote. Each job advertisement was then saved as a file into Evernote.

**Organising:** I added a code to each job advertisement file for controlling the data. For instance, job advertisements collected from LinkedIn were assigned the code 'L', plus a number. Following this, 'L400 – Graphic designer' was the name of the 400$^{th}$ job advertisement collected from LinkedIn. This list of files was then replicated in a Microsoft Excel sheet, and every job advertisement was then exported as a separate Microsoft Word file.

**Cleaning:** Organisations often publish their job advertisements on multiple websites. Also, organisations republish the same job advertisement to move their advertisement up on the website. As a result, some of the job advertisements collected initially were duplicates. I used the anti-plagiarism software Novus Scan to check for duplicates. The software compared each job advertisement to the whole dataset, pointing to the files that had repeated sentences. Following that, I removed duplicated job advertisements from the dataset.

### 3.1.2 Data analysis

For doing the analysis, I relied on coders (myself included) with professional experience and/or a degree in graphic design for performing a 'competent reading' (Atkinson & Coffey, 2004, p. 72). According to methodology scholars, documents are not neutral (Krippendorff, 2013; Rapley, 2007), as they are built for a particular purpose following social conventions (Coffey, 2014) and are part

of a system of relationships between people and society (Gorichanaz & Latham, 2016). In sum, documents carry a function (purpose) and are written in line with the standards expected by the audience. Furthermore, to better understand a document, a reader needs to have knowledge about the context. As pointed out by Atkinson and Coffey (2004), when reading documents, 'The reader brings to the text his or her stock of cultural knowledge, a knowledge (or ignorance) of similar texts, and his or her unique biography' (p. 72). Overall, the coders selected for the studies of this doctoral thesis were able to understand the language used in the job advertisements, based on their previous professional and educational backgrounds.

Thematic analysis and content analysis were used in different phases of the analysis process of Studies 1, 2 and 3 of this thesis. Due to the lack of classification of skills in the literature to serve as a coding scheme, thematic analysis was used for inductively building the coding scheme for the studies: a portion of the job advertisements was selected and every reference to a request (e.g. skills, competencies, personal traits) was coded in-vivo. Then, by discussing with the other authors of the studies, a coding scheme was built. Content analysis was used when deductively coding the whole dataset, using the coding scheme generated earlier. In the process, the frequencies of the themes in each job advertisement were assessed (presence or absence). In sum, thematic analysis was used to inductively build and refine the coding scheme while content analysis was used to deductively code and count frequencies of the requests made by organisations in the job advertisements.

The mixture of inductive and deductive approaches to coding job advertisements was relevant to maintaining the codes linked to the specific dataset (Braun & Clarke, 2006) and has been previously used by scholars in marketing (Gök & Hacioglu, 2010) and library and information science (Bronstein, 2015). In other words, by building the coding schemes for each study based on the data itself, we assured that our coding scheme would accommodate the varied range of requests made by organisations in their advertisements. In addition to that, in accordance with the different histories and traditions of the countries where the job advertisements were collected, the coding schemes should 'fit' the particularities and language of each country studied. In Study 1, we also discussed the coding scheme with design educators/practitioners and contrasted it with previous literature on design education to assess the completeness and

relevance of the coding scheme for design practice and education. Studies 2 and 3 also used the coding scheme of the first study as a basis for defining and naming the codes, while still being inductively built based on the job advertisements for each study.

The specific steps and actions taken in analysing the job advertisements were improved throughout the studies of this doctoral thesis. Specifically, (1) we changed the coding process and the number of coders as well as (2) revised how we mapped the skillset of designers based on learning theory from the field of education. In regards to the coding process and number of coders, Study 1 had a single coder (the author of this doctoral thesis) while Studies 2 and 3 had multiple coders (the author of this doctoral thesis and other researchers). The first reason for increasing the number of coders was to improve the understanding of the coding scheme and its reliability (e.g. Krippendorff, 2013, Harper, 2012). By having multiple coders in Studies 2 and 3, we were able to compare the coding process and improve it when needed (for more information, see 3.4 Improving the research quality). Another reason for having multiple coders in Study 2 was that a portion of the dataset was available only in Finnish. Then, we brought two other researchers that were fluent in the language to handle the coding in the job advertisements written in Finnish. Also, in Study 3 we had multiple rounds of coding in order to reduce errors in the process.

With no clear guidance from previous research on how to map the skillset of graphic designers, we organised the requirements specified in the advertisements inspired by the competency-based learning model by Voorhees (2001). In design education, Voorhees's model has been used earlier to describe the skillset and competencies of industrial designers (Hummels & Vinke, 2009). The author's pyramidal model (see Figure 2 on the next page) is divided into four levels: the first, lowest level refers to personal traits and characteristics that form the basis for learning; the second level refers to knowledge, skills and attitudes developed through educational activities; the third level refers to competencies developed through integrative learning practices; and the fourth and last level consists of demonstrations, where performance-based learning can be assessed.

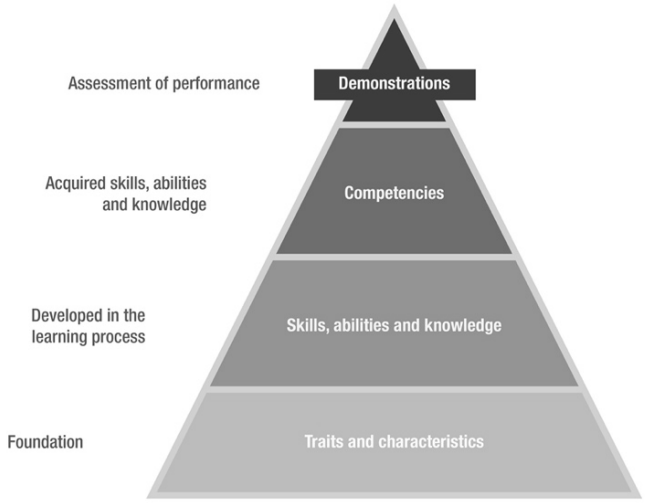
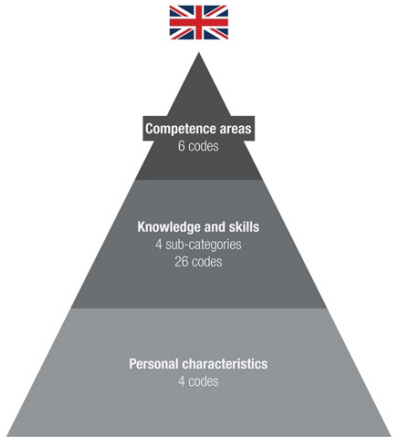



**Figure 2.** Conceptual learning model by Voorhees (2001)

As the model is developed for educational assessment, we used only the three initial levels (excluding demonstrations) to organise the requirements stated in the advertisements. We organised the requests made by companies in Studies 1, 2 and 3 inspired by Voorhees's model (2001). Specifically, we described the skillset of graphic designers in three main areas: (1) what graphic designers need to deliver (competencies in the Voorhees model); (2) what graphic designers are expected to know (skills, abilities and knowledge in the Voorhees model); and (3) what personal characteristics graphic designers are expected to have (traits and characteristics in the Voorhees model). In addition, we also made subcategories under the knowledge and skills categories, where we classified the requests under process management skills, conceptual design skills, technical design skills and software skills. Figure 3 on the next page demonstrates the codebooks used in each of the three studies of this doctoral thesis and how the categories are similar to Voorhees's (2001) model.




## Study 01 - UK - 36 codes

**Competence areas**
Brand visual identity
Digital design
Film and animation
Packaging and point of sale
Print and advertising
Retail and environmental design

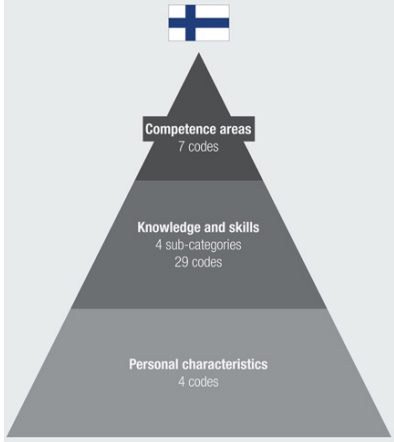
**Knowledge and skills - Process management skills**
Client relationship
Interpersonal (teamwork)
Presentation and communication
Project planning and administration
Team management

**Knowledge and skills - Conceptual design skills**
Briefing
Business orientation
Design research
Idea generation and concept development
Problem solving
Process understanding

**Knowledge and skills - Technical design skills**
3D modelling
Coding and platform management
Detailing and production
Digital photo manipulation
Illustration
Layout and composition
Motion design
Photography
Typography
Visual coordination

**Knowledge and skills Software skills**
2D software
3D software
Animation/video software
Office software
Web development software

**Personal characteristics**
Acumen
Aesthetic and creative sensitivi
Design passion
Self-driven




## Study 02 - Finland - 40 codes

**Competence areas**
Digital design work
Mobile/app design
Film and animation
Packaging and point of sales
Print and advertising
Retail and environmental design
Visual brand identity

**Knowledge and skills - Process management skills**
Client relationship
Interpersonal
Leadership
Presentation
Project management

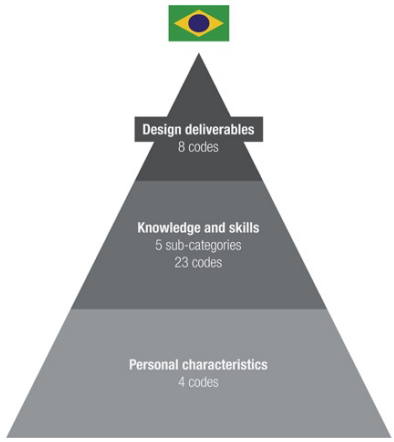
**Knowledge and skills Conceptual design skills**
Agile methods
Business orientation
Concept design
Data-driven design
Ideation
Holistic problem solving
Interaction design

**Knowledge and skills Conceptual design skills**
Prototyping
Service design
Trend analysis
User-centric design

**Knowledge and skills Technical design skills**
Coding Backend
Coding Frontend
Image retouching
Layout
Production process
Sketching
Typography
Visual coordination

**Knowledge and skills Software skills**
2D software
3D software
Animation/video software
Office software
Web development software

**Personal characteristics**
Acumen
Aesthetic and visual sense
Creativity/curiosity
Self-driven




## Study 03 - Brazil - 35 codes

**Design deliverables**
Brand design
Digital design
Packaging design
Print design
Editorial design
Signage and POS design
Service design
Video and animation design

**Knowledge and skills - Process management skills**
Client relationship
Teamwork
Project management

**Knowledge and skills - Conceptual design skills**
Business
Concept design
Design thinking
Problem solving
Research
User Experience (UX)

**Knowledge and skills - Technical design skills**
Coding
Design for web
Production and materials
Photography
Illustration
Layout

**Knowledge and skills - Content skills**
Content creation
Foreign language
Native language

**Knowledge and skills Software skills**
2D software
3D software
Animation/video software
Office software
Web development software

**Personal characteristics**
Acumen
Aesthetic sense
Creativity
Self-driven

**Figure 3.** Coding schemes used to analyse job advertisements in Studies 1, 2 and 3

Drawing on previous reports on the skills needed to deliver graphic design work (see Section 2 in this doctoral thesis), the skills suggested by scholars for graphic designers were included in the coding schemes for each study. For example, textbooks in graphic design emphasise the importance of visual elements and tools for graphic designers (e.g. Newark, 2008), which resonate with codes such as 'layout', 'aesthetic sense', '2D software' and 'illustration'. Also, a number of scholars suggest that graphic designers need to expand their skills to other fields (e.g., Frascara, 2017). These 'new' skills for graphic designers were embraced by the coding schemes, especially under the subcategories 'process management skills' and 'conceptual design skills'.

During Study 3, we changed our definition of design competence in comparison with earlier studies. During an interview with design educators, *we evolved the concept of design competencies from an outcome to a group of skills needed to perform a task*. In the interview, the design educators were asked to contrast the requirements specified in the job advertisements with an educational policy document that regulates design education in Brazil. The educators initially grouped the content to facilitate the task, which occurred unprompted. This group of skills established by the educators resembled the definition of competencies as 'a combination of skills, abilities, and knowledge needed to perform a specific task in a given context' (Jones & Voorhees, 2002). In other words, the new definition of design competence gives a broader view of the activities performed by designers, based on a combination of different skills. For instance, the educators defined 'representation' as a graphic design competence that encompasses knowledge in '2D software', 'illustration' and 'photography', among others (see Table 4 in Study 3 for the definitions of graphic design competencies).

## 3.2 Thematic analysis of interviews (Studies 1, 3 and 4)

Interviews provide an efficient route to accessing answers to a research question (Hox & Boeije, 2005; Rapley, 2007; Silverman, 2016). A researcher has control of the questions that will (ideally) lead to the answers needed. In comparison

with other naturally occurring data, where the researcher has no control over the creation of the data, in an interview, it is always possible to use 'probing questions' to explore or understand a concept better (Yeo et al., 2014, p. 195). For example, when there is the need for further clarification on an answer, a researcher can ask questions such as: 'Can you say a bit more about that?' or 'Can you give me an example?' Yet, there are challenges involved in conducting an interview. First, researchers need to have access to respondents and be able to convince them to contribute to the study. After that, researchers have to deal with the implied bias in the data. Respondents tend to present themselves in a positive or expected way (Fisher, 1993; Rapley, 2007), which might misrepresent the information provided. In addition, interviews provide a retrospective version of past actions (Rapley, 2004). In the words of Silverman (2015, p. 432), 'Interviews do not tell us directly about people's "experiences" but instead offer indirect "representations" of those experiences'.

The interviews conducted in Studies 1, 3 and 4 had different purposes for the studies in this doctoral thesis (see Table 5 on page 73). The interviews in Studies 1 and 3 resembled what the methodology scholar Flick (2004) calls *expert* interviews, which is a type of interview used to complement insights from applying other methods of research. These interviews were then used as a form of data triangulation (Silverman, 2016), where the findings from these interviews were then used to validate the previous analysis of job advertisements. In Study 4, *semi-structured* interviews (Silverman, 2016) were used to generate the study data that was then analysed and reported in the study. Although the purpose and means of generating data by interviews in the studies were different, I followed a similar rigorous approach as suggested by methodology scholars (e.g. Flick, 2004; Ritchie et al., 2013; Silverman, 2005) for collecting and analysing the data (Braun & Clarke, 2006). I describe how interviews were used in Studies 1, 3 and 4 over the following paragraphs.

In Study 1, I conducted five interviews with design practitioners and/or educators to assess the completeness and relevance of our coding scheme. During the interviews, the professionals had a sample of five examples of job advertisements and had to relate the requests with our initial coding scheme. Overall, the interviewees pointed out that the coding scheme covered the requests made in the advertisements, only suggesting small changes to the names and descriptions of the codes.

In Study 3, we[6] conducted two interviews with design educators in Brazil, where the interviewees had the task of commenting on and matching our coding scheme with the document *Guidelines for Undergraduate Programmes in Design* (*Diretrizes Curriculares Nacionais do Curso de Graduação em Design* in Portuguese), which regulates and defines the requirements for educational institutions offering design education in Brazil. The document specifies the skills and competencies that a design student should possess upon graduation. We used the latest version of the document (MEC, 2004), which is available at the Brazilian Ministry of Education website. The document was used as a proxy for the skills design graduates are being trained in at Brazilian universities and compared with the requests made in the advertisements in order to explore the possibility of an 'education gap' (Todd et al., 1995, p. 20), where the skills requested by organisations are not aligned with the skills provided by educational institutions.

Lastly, in Study 4, I conducted 12 interviews with hiring managers and six interviews with designers in Finland to understand how job advertisements were written and used throughout the recruitment process. Initially, I spoke with the hiring managers and used the published job advertisements to give context to the discussion. The interviews with the hiring managers covered topics such as how a specific job advertisement was written and used during recruitment, how the recruitment process happened in the company and which qualifications were sought in applicants. After the initial interviews, I contacted the hiring managers again and asked for the contact information of the designer who was hired to the position. I was then able to interview six designers on topics such as their reasons for applying for the position, the job search process and how the job application process happened. After providing a copy of the job advertisement, I asked them how well the description of the work in the document matched the work they were currently doing.

6 The interviews were conducted in-loco at a design school in the south of Brazil by Leandro Tonetto and Roberta Mandelli, co-authors of Study 3. I was involved in preparing the interview guide and analysing the results.

| | Interviews |
|---|---|
| Study 1 | Five design practitioners and/or educators: four of the interviewees were in Finland while one was in the UK (conducted via Skype); three interviewees had professional experience in the UK while two had experience in teaching graphic design.<br>Duration of interviews: Between 45 and 90 minutes.<br>Language: English.<br>Year: 2016. |
| Study 3 | Two interviews with three design educators in Brazil. The first interview was conducted with two design educators, and the second interview with one design educator.<br>Duration of the interviews: 90 minutes (first interview) and 60 minutes (second interview).<br>Language: Portuguese.<br>Year: 2017. |
| Study 4 | Twelve hiring managers and six designers in Finland.<br>Duration of the interviews: 33 minutes to 80 minutes.<br>Language: English.<br>Year: 2016 – 2017. |

**Table 5.** Interviews used in the studies of this doctoral thesis

All the interviews for this doctoral thesis were audio recorded. The interviews were conducted at the interviewee's work environment. As an exception, one interview in Study 1 was conducted over Skype while the interviewee was at home. The interviews in Studies 1 and 4 were conducted in English, while the interviews in Study 3 were conducted in Portuguese. I transcribed all interviews for this doctoral thesis, where the level of transcription (e.g. Flick, 2004) varied according to the role of interviews in each study: In Studies 1 and 3, the interviews were conducted to validate the coding scheme (Study 1) and highlight overlaps of the coding scheme with an external document (Study 3). Therefore, I listened to the interviews multiple times, transcribing relevant

parts and sentences. On the other hand, the 18 interviews in Study 4 were fully transcribed.

The interviews conducted for this doctoral thesis were analysed following thematic analysis principles (e.g. Braun & Clarke, 2006). In thematic analysis, coding and identifying themes are at the core of the method. A theme is a topic that repeatedly shows up in the data (Braun & Clarke, 2006). In other words, thematic analysis is a form of pattern recognition across multiple data points (Fereday & Muir-Cochrane, 2006). In Studies 1 and 3, I read the transcripts while highlighting relevant sections and writing initial themes. Later, the initial themes were discussed and revised together with the other co-authors of the studies. Similarly, in Study 4 I read the 18 transcripts and started building initial themes. I also mapped how the recruitment process was described by hiring managers and designers, with a special interest in how job advertisements were used throughout the process (see Figure 1 in Study 4). The description of the recruitment process and initial themes were then used as a basis to build the coding scheme in collaboration with the co-author of the study. The themes in the resulting coding scheme encompassed the design recruitment process and how job advertisements were used throughout the process. Examples of themes created were 'design recruitment process', 'role of job advertisements in the recruitment', 'job advertisement language' and 'process of writing the job advertisement'. Lastly, I revised the 18 transcripts, where relevant parts were coded following the generated coding scheme. The coding scheme and analysis of the interviews were later used as a framework to review the literature, which will be discussed in the next subsection.

## 3.3 Systematic literature review (Study 4)

A *systematic* literature review was conducted in Study 4 to acquire a more in-depth perspective of how job advertisements were used for research purposes. A literature review is an important step in understanding what has been done on a specific topic before (e.g. Hart, 1999). According to Potter (2004), a systematic literature review takes a more rigorous approach and aims to cover all studies within a certain timeframe. Two other literature reviews on how job

advertisements were used for research were conducted by Harper (2012) and Kim and Angnakon (2016). Yet, both studies only looked at studies published by library and information science scholars, not considering how job advertisements were used in other fields.

The steps taken for doing the review were as follows: First, I searched for 'job ad', 'job ads', 'job advertisement', 'job advertisements', 'job announcement', 'job announcements', 'job opening', 'job openings', 'job posting' and 'job postings' in the title, keywords and abstract sections in the following academic databases: Scopus, Academic Search Elite, Academic Search Ultimate and ProQuest. This initial search resulted in 1,199 academic articles published between 1987 and December 2017. Second, I reviewed all abstracts and removed papers that did not have job advertisements as primary or secondary data for analysis. Third, I then removed (1) duplicate papers, (2) papers written in languages other than English and (3) conference papers. The final number of articles found was 209, published in 166 different academic journals in fields such as library and information science, business and education. Table 6 on the next page presents an overview of the fields and job titles analysed in the studies.

| Field | Examples of job titles from the job advertisements analysed in the studies | # of papers* |
|---|---|---|
| Library and information management | Librarian, cataloguer, knowledge management professional. | 80 |
| Business | CEO, project manager, salesperson. | 45 |
| Education | Academic advisor, professor, teacher. | 17 |
| Health | Doctor, nurse, pharmacist. | 17 |
| IT | IT manager, software engineer, system analyst. | 14 |
| Biology | Conservation science professional, forestry professional, ranger. | 5 |
| Design & communication | Graphic designer, UX designer, UX researcher. | 5 |
| Writer | Technical writer. | 4 |
| Engineering | Civil engineer, engineering graduate. | 3 |
| Geography | Geography professional, geography academic. | 3 |

**Table 6.** Types of professionals analysed in the papers

* The table only presents fields that appeared in at least three papers

In exploring how job advertisements were used as data for research, in each paper I coded the purpose of the study (e.g. research question(s), hypothesis), advantages/disadvantages of job advertisements, the methodological choices made by scholars in terms of data collection (e.g. sample size, sources) and data analysis (e.g. type of coding, number of coders). When relevant, I also consulted other academic texts cited in the studies.

## 3.4 Improving the research quality

A number of actions were taken throughout the four studies of this doctoral thesis for improving its quality. Specifically, I relied on guidance of scholars discussing quality in qualitative research (e.g. Flick, 2004; Ritchie et al., 2013; Silverman, 2005), thematic analysis (Braun & Clarke, 2006; Fereday & Muir-Cochrance, 2006), content analysis (Neuendorff, 2002; Krippendorff, 2013), document analysis (Bowen, 2009) and job advertisement analysis (e.g. Harper, 2012; Kim & Angnakoon, 2016). Next, I discuss the quality of my research in terms of reliability, validity and generalisability.

Reliability refers to assuring that the research steps are appropriately communicated, thereby enabling other researchers to replicate them (e.g. Silverman, 2016). Content analysis scholars point out that reliability is improved when different researchers can repeat the research and obtain the same results (e.g. Neuendorff, 2002). As pointed out by the author, 'a measure is not valuable if it can be conducted only once or by one particular person' (Neuendorff, 2002, p.112). Similarly, Krippendorff (2013, p. 267) points out that reliable data 'mean the same thing for everyone who uses them'. For example, two (or more) researchers should be able to read, interpret and code a text in the same way. In other words, the instructions for coding a text should be clear enough that multiple researchers will arrive at the same conclusions, as 'replicability is the most important form of reliability' (Krippendorff, 2013, p. 24).

Two main actions were taken to improve the reliability of the studies in this doctoral thesis. First, I provide a thorough description of the steps taken for collecting, handling and analysing data. As pointed out by Flick (2004), reliability increases when a more detailed research process is communicated. Specifically, the method sections of the four studies present a detailed description of how the studies were conducted. However, due to the word limits of the journals and conferences where the studies were published, I had to summarise the description of the essential steps. To overcome this, the method section of this thesis provides a more detailed account of data collection and analysis.

Second, I involved multiple coders in the coding process. As a result, I employed intercoder agreement tests for validating the coding scheme when analysing job

advertisements (e.g. Neuendorff, 2002, Krippendorff, 2013). In sum, intercoder agreement measures the extent to which multiple coders agree when applying the same coding scheme to the same dataset (for more on intercoder agreement and how to calculate it, see Krippendorff, 2013). Despite the advantages of measuring the intercoder agreement for improving reliability, it is far from being a common practice in studies with job advertisements. In the literature review made in Study 4, we found that only 18 (out of 209) studies measured intercoder agreement. Similarly, Kim and Angnakoon (2016) pointed out that only six (out of 106) studies in the library and information science field adopted this procedure.

In the studies of this doctoral thesis, intercoder agreement tests were used as follows: In Study 1, the two authors of the study initially discussed and coded a portion of the job advertisements while building the coding scheme. Also, we assessed the completeness of our coding scheme during interviews with five educators/practitioners: Five job advertisements were given to the interviewees and they were asked to make notes on the requests made by companies. Next, we presented the coding scheme for the study and asked them to show how their notes were covered by our coding scheme. This procedure served to validate that our coding scheme encompassed the requests made in the job advertisements. Studies 2 and 3 had multiple coders and measured intercoder agreement. Krippendorff's alpha was selected to measure intercoder agreement as it is 'the most general agreement measure with appropriate reliability interpretations in content analysis' (Krippendorff, 2013, 221). To that end, we selected a random portion of job advertisements and compared the coding results. After measuring the intercoder agreement, the coders discussed the differences and made necessary corrections in the coding scheme. In doing that, we improved the reliability of the coding scheme and the coding process, as multiple coders have the same understanding of how to proceed in coding the job advertisements (Neuendorff, 2002; Krippendorff, 2013).

Validity (or internal validity) refers to ensuring that the research analyses the phenomena that it claims to analyse. Simply put, validity is 'a question of whether the researchers see what they think they see' (Flick, 2004, p. 387). The validity of the studies of my doctoral thesis was assessed in terms of (1) quality of data, (2) building the coding scheme and (3) applying the coding scheme.

First, the quality of job advertisements as a proxy for the qualifications sought by companies when recruiting design professionals is based on claims by previous scholars (see e.g. Todd et al., 1995), our systematic literature review and interviews with hiring managers and designers (Study 4). Content analysis scholars (Neuendorff, 2002; Krippendorff, 2013) suggest that researchers should question whether they are measuring what they want to measure. With respect to this point, job advertisements reflect the requests made by companies, and therefore analysing job advertisements provides me with relevant data for measuring what I want to measure (for more discussion on the quality of job advertisements for design research, see Study 4).

Second, the measurement instrument (coding scheme) needs to be able to capture accurately the requests made in the job advertisements. Therefore, in Studies 1, 2 and 3, we relied on inductive thematic analysis for building the coding scheme. In doing so, we assured that each coding scheme would 'fit' each study. For instance, I could have reused the coding scheme from Study 1 in Study 2 or 3. However, as the coding scheme was created based on the data from Study 1, it would embrace some of the requests made by companies in the other studies, but probably would lose some of the specific requirements. In addition, when building the coding scheme, the authors of the studies were involved in reviewing the codes and comparing them with how companies wrote their requests in job advertisements (Braun & Clarke, 2006). Lastly, the coding schemes were discussed with practitioners and educators in Studies 1 and 3 in order to assess their relevance and completeness.

Third, after verifying the validity of the coding scheme, I had to assure the validity of the process of applying the coding scheme. As discussed before (see Section 3.1.2. Data Analysis), Neuendorff (2002) suggests two different ways of coding a dataset: human coding and computer coding. In Studies 1, 2 and 3, human coding was preferred, as hiring managers write their requests in a multitude of forms, using varied words. For example, in Study 2, we coded the following sentences as a 'self-driven' personal characteristic: 'you need to be willing to learn', 'a never-ending will to learn more', 'grow and expand your skillset' and 'you are a self-starter'. Therefore, building a complete list of words *a priori* would not cover all the different variations and words used for each code. In addition, when building the coding scheme, the authors wrote detailed descriptions and examples of each code in order to improve the coding process.

A summary of the descriptions of the codes was reported in the results section of Study 1 and as appendixes to Studies 2 and 3.

Generalisability (or external validity) refers to how the research findings are valid in a different context from the one where the research was carried out. According to Flick (2004, p. 407) 'the generalizability of the results is often closely linked to the way the sampling is done'. The generalisability of this doctoral thesis was assessed in relation to the sampling strategy adopted. I started the sampling process by choosing to study the job market for designers in the UK, Finland and Brazil. The first two countries are recognised as leading design nations (e.g. Moultrie & Livesey, 2009), while the last is considered one of the 'emerging important players' (p. 24) in design. The diverse nature of these countries, in terms of culture, population numbers and economic situation, improves the generalisability of the findings of this doctoral thesis. In addition to selecting the countries, my sampling strategy involved selecting the websites for collecting job advertisements. In all studies, educators and practitioners in these countries were consulted when selecting the websites for data collection. In doing so, my goal was to cover the main online websites where designers in each of the countries look for design positions.

## 3.5 Ethical concerns

In the studies of this doctoral thesis, I dealt with ethical concerns in terms of collecting data and disclosing the names of companies and interviewees. As discussed before (see 3.1 Document analysis of job advertisements) job advertisements are publicly available documents. Similarly, the educational policy document is available on the Brazilian Ministry of Education's website, while academic articles are available for download using university access to scientific databases.

In addition to selecting publicly available documents, I decided not to reveal the names of the organisations behind the job advertisements. First, knowledge about which company published a specific job advertisement does not add relevant information to the studies in this doctoral thesis. As an exception, in Study 1 I had to know the hiring company in order to classify the position as

either in-house or external. Other than that, my focus was on the list of requirements defined by the organisations. Second, displaying the names of the companies would require a significant effort to contact and ask for permission from all the hiring managers. For example, Study 1 covered 1,406 job advertisements, which would require me to reach all the professionals who published the advertisements. I also decided not to report the names of the interviewees and their respective organisations in Studies 1, 3 and 4. Instead, I used the terms 'design educator' (Studies 1 and 3), 'hiring manager' (Study 4) and 'designer' (Study 4).

In sum, in this section I presented how I used job advertisements, interviews and academic articles to inform the studies of this doctoral thesis. In doing so, I presented the reasoning and limitations for each type of data. I also presented how I analysed the data following document, thematic and content analysis. Lastly, I discussed both how the quality of the research was assured and the ethical concerns regarding the analysed data. In the next section, I present the summary of the four articles that are part of this doctoral thesis.

# 4. Summary of the articles

This section presents a summary of each of the four articles in this doctoral thesis. For each article, I present an extended version of the published abstract, where I show the context of the study, the research question, the methods used and the main findings. The full version of each article is presented in Section 8 of this thesis.

The studies in this doctoral thesis are presented in the order they were executed. Studies 1, 2 and 3 present the investigations on the skillsets sought by companies in graphic designers in the UK, Finland and Brazil, respectively. Studies 2 and 3 are replications and extensions of Study 1. In other words, together with the other authors, I evolved the method for analysis in each of these three studies (see 3.1.2 Data analysis). Study 4 was the last one executed in this doctoral thesis, where I investigated the relevance of job advertisements in research and how they are used by practitioners in the recruitment of design professionals in Finland.

### Study 1: Graphic Designer Wanted: A Document Analysis of the Described Skill Set of Graphic Designers in Job Advertisements from the United Kingdom (Dziobczenski & Person, 2017)[7].

Graphic designers have traditionally been working with 'symbolic' and 'visual' communication (Buchanan, 2001, p. 8). Therefore, the realm of graphic designers' work has been in areas such as print and digital design (e.g. Frascara,

7 Dziobczenski, P. R. N., & Person, O. (2017). Graphic designer wanted: A document analysis of the described skill set of graphic designers in job advertisements from the United Kingdom. International Journal of Design, 11(2).

2004). However, more recently, the work of (graphic) designers has been transformed by new technology and societal changes (Davis, 2008; Harland, 2016). As a result, graphic designers are required to expand the type of work they offer to organisations, which results in the expansion of the skillset necessary to work as a graphic designer.

In Study 1, we address this change in the scope of graphic designers' work by investigating the requests made in job advertisements by organisations from the UK. The specific research question that we address in the study is: 'What is the expected skillset that employers reference in advertising for graphic design positions?' The contribution of this study for design educators is a structure for familiarising themselves with the requests made in the labour market facing design graduates. In the study, we recognise that design education seeks not only to fulfil current expectations, but also to prepare graduates for their long-term design career. Thus, our study provides a way for educators to assess and contrast their educational practices with the requests found in the job market and prepare design graduates accordingly. Similarly, for design practitioners, our study provides information to help them when deciding on what skills to develop in their careers.

To answer the research question, we collected 1,406 unique job advertisements for graphic designers that were publicly available on online job boards. The data collection occurred during a five-week period in June/July 2015 on both generic job boards (LinkedIn, Indeed, Monster) and specialised design websites (Brand Republic and Design Week). Data analysis started by inductively building the coding scheme based on a subset of job advertisements (40%). The coding scheme represented the requests made by companies. We organised the coding scheme inspired by a competence-based view of education (Voorhees, 2001). The coding scheme had 36 codes divided into three levels. The first of these levels referred to competence areas (what graphic designers are expected to deliver), the second to knowledge and skills (what graphic designers are expected to know) and the third to personal characteristics (what graphic designers are expected to have). Later, we deductively applied the coding scheme, assessing the presence/absence of each code across the whole dataset. We also compared the requests in the advertisements according to the level of seniority (junior, mid-level or senior) and the functional position (internal or external) within the hiring organisation.

The first finding of Study 1 concerns the frequency of the requests (theme) across the dataset. In other words, 'which requests are more and less frequent?' We assessed the relevance of each theme by comparing its frequency with other themes in the same category (Krippendorff, 2013). In doing so, we established a baseline for comparing the frequency of each theme versus the overall category (see Figure 1 in the article). Following that, the most frequent requests for graphic designers were: (1) the ability to deliver print and digital work; (2) knowledge and skills in teamwork, project planning, presentation and communication, idea generation, business orientation, detailing and production, coding, visual coordination, layout and 2D software; and (3) personal characteristics such as acumen, aesthetic and creative sensitivity and being self-driven. While previous literature suggests a more strategic role for designers (e.g. Perks et al., 2005), we found that employers predominantly recruit graphic designers for operational roles, where they need to deliver print and digital outcomes and be knowledgeable in 2D software.

The second finding points to the qualifications graphic designers should acquire for moving from junior to senior positions (see Figure 2 in the article). We performed independent chi-square tests to identify significant differences in the requests across junior, mid-level and senior positions. In sum, when moving towards senior positions, graphic designers are more frequently requested to have knowledge and skills in business orientation, process understanding, team management and illustration. These requests found in senior positions reinforce that senior designers are involved earlier in the development process.

The third, and last, finding concerns the differences in the skillsets of graphic designers working internally (in-house agencies) or externally (design consultancies) to organisations (see Figure 3 in the article). Once more, we performed independent chi-square tests to identify significant differences between internal and external positions. When working internally, graphic designers were requested (1) to deliver packaging and point of sales work; (2) to have knowledge and skills in presentation and communication, project planning, team management, teamwork, digital photo manipulation, coding, detailing and production, photography, visual coordination, Office software and Web software. This suggests that designers working in-house might have a more integrative/collaborative function, as they are in close contact with other departments inside an organisation. When working externally, graphic

designers were requested (1) to have knowledge and skills in idea generation, client relationships and motion design; and (2) to have design passion as a personal characteristic.

### Study 2: Designing Career Paths in Graphic Design: A Document Analysis of Job Advertisements for Graphic Design Positions in Finland (Dziobczenski et al., 2018)[8]

The graphic design profession is being transformed by technological changes (e.g. Frascara, 2004), which has created new types of work for designers under job titles such as interaction designer and user interface designer. As pointed out in Study 1, these changes in the profession force graphic designers to update their skillset.

Study 2 is a replication and extension of Study 1. In Study 2, we analysed job advertisements for design positions in Finland. In this study, we extended our data collection to all positions that graphic designers were eligible to apply for, and as a result we collected job advertisements for not only 'graphic designer', but also 'UX designer', 'UI designer', 'digital designer', among others.

The research questions tackled by the study are: 'What skillset is requested by employers in advertising for job openings in graphic design in Finland?' and 'How does the requested skillset compare for different career paths in terms of the requests made for traditional versus digital graphic design positions?' The study aims to contribute to design education by providing an overview of the requests made by employers of design graduates. Educators can potentially use the results of our study (and the method employed) to analyse their current educational practices and how they fit with the requests made in the job market. Similarly, design practitioners can also see our findings as a guide for career development and the types of design services companies are seeking to hire.

To answer the research questions, we collected 230 job advertisements over a period of 18 months (June 2015 – December 2016) from three websites:

[8] Dziobczenski, P. R. N., Person, O., & Meriläinen, S. (2018). Designing career paths in graphic design: A document analysis of job advertisements for graphic design positions in Finland. The Design Journal, 21(3), 349-370.

LinkedIn, the Ministry of Employment and Economy's job board and Oikotie. Similar to Study 1, we pursued a combination of thematic and content analysis. Initially, we thematically and inductively analysed the data in building a coding scheme to capture the types of requests made in the studied advertisements. The final coding scheme presented 40 codes divided into three levels, inspired by Voorhees's (2001) work on competency-based education: (1) competency areas in terms of the outcomes generated by graphic designers, (2) knowledge and skills and (3) personal characteristics. We improved our data analysis process from Study 1 by adding multiple coders to the study. The coding of the 230 job advertisements was done by three coders for two reasons. First, a portion of the job advertisements were written in Finnish. Second, multiple coders enable the measurement of intercoder agreement, which we did twice in order to improve the reliability and validity of the coding scheme. After measuring the intercoder agreement, we made changes to the coding scheme and the coded job advertisements.

In answering the first research question – What skillset is requested by employers in advertising for job openings in graphic design in Finland? – we classified the requests under 40 codes, where we followed Krippendorff's (2013) suggestion to distinguish the most/less frequent codes by comparing the frequency of each one in relation to the group of codes it belonged to. Following that, across both traditional and digital graphic design positions, designers were most frequently asked (1) to deliver mobile/app design work, digital design, visual brand identity and print design; (2) to have knowledge in teamwork, project management, user-centric design, business orientation, concept design, agile methods, coding frontend, layout, production process, 2D software; and (3) to be self-driven, have professional acumen and aesthetic and visual sense (see Figure 1 in the article). These results both replicate and extend the results of our prior study (Study 1 in this thesis). In short, while organisations in the UK highlighted the operational roles for graphic designers, in Finland we found elevated frequencies in requests emphasising the broader range of work done by designers in the country. For example, requests for 'user-centric design' and 'trend analysis' were more frequent in job advertisements from Finland.

In answering the second research question – How does the requested skillset compare for different career paths in terms of the requests made for traditional versus digital graphic design positions? – we initially classified the positions with

the word 'graphic' as a traditional graphic designer and the ones with one of the following words as a digital graphic designer: 'digital', 'interaction', 'UX', 'UI' and 'Web'. Then, we performed multiple logistic regression to uncover the most frequent themes in the traditional and digital graphic design groups. In doing so, we differentiated between the requests that are relevant for both groups (non-significant differences across groups) and those that are relevant for only one of the groups (significant differences across groups). In sum, traditional graphic design positions are portrayed as more generalist, being most frequently requested (1) to deliver digital design work, print and advertising, film and animation, retail and environmental design; (2) to have knowledge and skills in layout, 2D software, project management, trend analysis; and (3) to have professional acumen and be creative/curious. Digital graphic designers are described more as specialists, being most frequently requested (1) to deliver mobile/app design and (2) to have knowledge in agile methods, user-centric design, prototyping and interaction design.

### Study 3: Requests from companies and requirements for design education in Brazil: where do they meet? (Dziobczenski et al., 2018)[9]

As the professional realm for designers is changing in the industry (see e.g. Davis, 2006), the skillset required from design graduates and students is in constant flux. Following that, scholars (Todd et al., 1995, p. 20) have identified as an 'education gap' the imbalance between the skills demanded by organisations and the ones trained in universities. This educational gap is harmful to the employability of (design) graduates and hinders companies in their efforts to effectively recruit new personnel.

Study 3 is also a replication and extension of Study 1. Similar to Study 2, we added multiple coders to the analysis process in order to measure intercoder agreement. As a result, we improved the reliability and validity of the coding scheme. We also explored the possibilities of an educational gap in Brazil by analysing the requests from the industry and contrasting them with design

[9] Dziobczenski, P.R.N., Person, O., Tonetto, L., & Mandelli, R. (2018). Requests from Companies and Requirements for Design Education in Brazil: where do they meet? In Design Research Society International Conference: Catalyst. Design Research Society.

education guidelines. In doing so, we compared two sets of documents – job advertisements and design educational guidelines – which are written for different audiences and purposes. In the study, we recognised the challenges and limitations in comparing these different documents.

The specific research questions for this study were: 'What is the skillset requested by Brazilian companies in advertising for graphic designers? 'and 'How are the requests made by companies in job advertisements covered in a design education policy document?' The study contributes to design education and practice in Brazil and abroad. Regarding the former, we suggest that our findings – and method used – can be used by educators in reviewing the content of educational practices to potentially improve students' employability. Concerning the latter, as design graduates struggle to find positions in industry (Naveiro & Pereira, 2008), our results indicate the skills requested by organisations and therefore serve as a guide for career development.

In this study, we collected 371 job advertisements for graphic design positions in Brazil over a period of five weeks (June – July/2015) from five job boards: LinkedIn, Indeed, Infojobs, Trampos and Adonline. The coding scheme was built by thematically analysing a subset of the job advertisements and later applied by two of the authors. For improving the reliability of the coding scheme, we measured intercoder agreement and made the necessary corrections. The coding scheme contained 35 codes, divided into (1) design deliverables, (2) knowledge and skills and (3) personal traits. In addition to analysing the job advertisements, we interviewed three design educators in order to understand the educational relevance of our coding scheme. Specifically, the educators had to compare the requests made in the job advertisements with an educational policy document, published by the Brazilian Ministry of Education, which regulates and defines the requirements for institutions providing design education in the country. During the interviews, the design educators merged by similarities the content of the 35 codes into nine new groups. We then considered these nine groups as 'graphic design competencies', following the definition of competencies provided by Jones and Voorhees (2002).

To answer the first research question – What is the skillset requested by Brazilian companies in advertising for graphic designers? – we looked at the distribution of frequencies of the 35 codes in our coding scheme. Overall,

recruiting companies in Brazil are seeking for a more operational graphic designer. For example, 'representation' as a competency group merging the codes related to executing graphic design work was present in the majority (84.1%) of job advertisements. Also, the five most frequent codes also emphasised the operational and 'hands-on' work of graphic designers: 2D software (68.19%); print design (56.87%); digital design (45.01%); layout (39.08%) and photography (23.99%) (see Tables 1, 2 and 3 in the article). These findings demonstrate a challenge that faces design educators and practitioners in the country in their efforts to improve the role of (graphic) designers towards more strategic, rather than operational, work.

To answer the second research question – How are the requests made by companies in job advertisements covered in a design education policy document? – we relied on how the interviewees connected the content of the job advertisements with the educational policy document. The interviewees found that most of the requests made by organisations are covered in the guidelines defined by the Ministry of Education (see Table 4 in the article). As exceptions, the interviewees pointed out that the skill of 'coding' falls outside the scope of design education in the country, whereas 'acumen' and 'self-driven' are personal characteristics that are hard to regulate.

Overall, we discussed in the article the value of job advertisements as a source of knowledge for design education and practice. Our study suggests that by studying job advertisements, design educators and practitioners provide a snapshot of the requests sought in the market in a short period of time, as the recruitment process typically occurs during a period of a few weeks/months after the job advertisement is published. However, we recognise that design education should not only cover the immediate requests made by organisations. In supporting development in the field, design educators prepare their graduates for shaping how organisations and society perceive the role and responsibilities of design professionals.

### Study 4: The job of job advertisements in design recruitment and research (Dziobczenski & Person, In review process).

As job advertisements represent the main type of data in this doctoral thesis, Study 4 set out to investigate the relevance and use of job advertisements in the recruitment of design professionals and in research across different areas. Job advertisements are naturally occurring data (e.g. Silverman, 2016) that provides information about the requirements made by a company when seeking to hire professionals (e.g. Chipulu et al., 2013; Harper, 2012; Kim & Lee, 2016; Todd et al., 1995). This type of data is extensively used in fields such as business (e.g. Bennett, 2002) and education (e.g. Irons & Buskist, 2008), but predominantly in library and information science (e.g. Kennan et al., 2006). Study 4 adds to previous literature reviews on the use of job advertisements in library and information science (Harper, 2012; Kim & Angnakoon, 2016) by expanding the fields studied and also adding the perspective of hiring managers and designers. In this study, we set out to answer the following research questions: 'How do job advertisements function in the recruitment of design professionals?' and 'What are the opportunities provided by job advertisements for design researchers?'

To answer the research questions, we followed 12 design recruitment cases in Finland. In each case, we interviewed the hiring manager and, when possible, the designer hired for the position. The goal of the interviews was to unveil how job advertisements were built and used in the recruitment process. In addition to the recruitment cases, we carried out a systematic literature review of academic articles that selected job advertisements as data. We reviewed 209 articles, unveiling each study's purpose, the advantages/disadvantages of using job advertisements for research purposes, and the methodological choices made by the authors.

In relation to the functions of job advertisements in design recruitment (research question 1) we found that hiring managers used job advertisements to communicate with qualified candidates, evaluate candidates and also to promote the company to candidates and other audiences (e.g. clients, competitors, society). Designers used the job advertisements to evaluate their fit with the position and the recruiting company as well as to acquire knowledge about the

job market (Who is hiring? Which skills are required?), thereby guiding their self-development efforts (see Figure 1 in the article).

In relation to the opportunities provided by job advertisements for design researchers (research question 2), we found both advantages and limitations of using this type of data for investigating the labour market. For example, as noted in our review of past studies, job advertisements allow scholars to cover a large part of the job market (in comparison with interviews, for example). Yet, scholars also note that job advertisements might not provide all the qualifications sought for a professional, as some requests are considered 'basic' or intrinsic to the professional and, therefore, not necessary to cite in the text. As organisations need to communicate their requirements to candidates in an efficient/concise manner, it is reasonable to assume that the more frequently a certain qualification appears in job advertisements, the more important it is for recruiting companies in general.

We finish the article by suggesting that future research could further investigate the decisions made during the initial phases of design recruitment, when companies set out to define the requirements for designer positions in job advertisements. Overall, the article reinforces the potential of using job advertisements for studying the labour market conditions faced by (design) professionals.

# 5. Cross-cutting contributions

This doctoral thesis contributes to the understanding of the work and skillset of graphic designers through studying job advertisements from the UK, Finland and Brazil. By looking at the descriptions that organisations provide in their job advertisements when recruiting graphic designers (Studies 1, 2 and 3) and the process of writing a job advertisement (Study 4), I make four cross-cutting contributions to research in graphic design and design management.

The first contribution of this thesis is that it adds to current narratives defining graphic design and the work and skillset of graphic designers. Following that, I also discuss the similarities and differences in the job market for graphic designers in the UK, Finland and Brazil. The second contribution points to the different career paths available for graphic designers. More specifically, I present how the skillset changes in terms of seniority (junior, middle, senior), positioning (internal or external to organisations) and scope of work (traditional or digital graphic design).

These two initial contributions are aimed at design practitioners, educators and graphic design researchers. For practitioners, knowledge about the work and skillset of graphic designers provides information about the skillset required by organisations and helps professionals to better prepare for the job market. For educators, knowledge about the requests posted by organisations can inform curricula and course planning that facilitate the transition of graduates from academia to industry. For graphic design researchers, the thesis contributes empirical knowledge about how the graphic design profession is developing and evolving in the studied countries.

The third contribution highlights the value of using job advertisements as a source of data for design research and practice. In particular, I argue that job advertisements present an opportunity to acquire a more practice-grounded foundation when addressing key questions of interest to design and design management research, which to date have typically only been addressed through secondary data that is generated *for* research. The last contribution points to practical insights on how researchers and practitioners can use and analyse job advertisements. From evolving the use document analysis of job advertisements over the initial three studies, some essential learnings were extracted and synthesised in the section. In addition to that, I bring information from the interviews and literature review executed in Study 4 to provide more of a grounding to future uses of job advertisements in research and practice.

These two last contributions are aimed at design practitioners, educators and design management researchers. For practitioners and educators, I highlight the value of using job advertisements to investigate the job market in their specific countries/cities. While in contributions 1 and 2 I highlighted the findings from the studies, in contributions 3 and 4 I highlight the method and suggest that practitioners and educators can use and adapt it for their needs. For design management research, I contribute with knowledge on what happens in the early stages of design adoption and management. As pointed out by previous design management models (Bruce et al., 1999; Acklin, 2013), the recruitment of design professionals is one of the initial stages of adopting design. Design management researchers can then use job advertisements as a proxy to investigate the early steps taken by an organisation when adopting and managing design.

## 5.1 Cross-contribution 1: Defining graphic design: the work and skillset of graphic designers

While definitions of design are constantly being (re)constructed (Julier, 2013), a close look at defining graphic design, and the language used for it, is important because 'language limits an understanding about how far a subject can be appreciated and developed to understand the discipline' (Harland, 2011, p. 24). Hence, this thesis contributes to defining the graphic design discipline, the work done by graphic designers and the skillset carried by these professionals.

Early definitions of graphic design reinforce the narrative that representing concepts and ideas visually is at the core of the discipline (e.g. Ambrose & Harris, 2009). Furthermore, scholars highlight how the work delivered by graphic designers carries functions such as informing (e.g. Newark, 2008) and decorating (e.g. Barnard, 2005). In performing these functions, scholars also describe graphic design in terms of communication (e.g. Frascara, 1988) and persuasion (e.g. Livingston & Livingston, 2012). Overall, graphic design is described as a problem-solving activity (e.g. Walker, 2017) that has an impact on society (e.g. Frascara, 1988).

However, in order to fulfil these functions in their work, graphic designers have to carry a number of skills in both design and other areas. Previous literature in textbooks and conceptual papers points to four different groups of skills required by graphic designers. The first one is related to visual elements, e.g. colours, shapes, typography and photography (Ambrose & Harris, 2009; Cezzar, 2017; Dabner et al., 2014; Lupton & Phillips, 2015; Newark, 2008). The second one is related to tools, both physical (e.g. pencil, paper and pens) and digital (e.g. software) (Ambrose & Harris, 2009; Newark, 2008). The third one is related to the methods used to solve graphic design problems (e.g. Cezzar, 2017; Frascara, 2004). The last, and most varied, group consists of skills usually found outside the design field. Due to the increasing complexity of graphic designers' work (Davis, 2006, 2008; Frascara, 1988; Harland, 2016), this group of skills is necessary to operate as a graphic designer; these skills include communication (Cezzar, 2017; Giloi & du Toit, 2013), teamwork (Giloi & du Toit, 2013; Littlejohn; 2017; Frascara, 2017), user research (Heller, 2005; Littlejohn, 2017; Frascara, 2017), business (Heller, 2005), entrepreneurship (Giloi & du Toit,

2013), learning capacities (Frascara, 2017) and ability to produce and interpret texts (Frascara, 2017). Overall, the literature suggests many elements in terms of the skills needed to operate as a graphic designer.

While the literature reflects the authors' conceptualisations about the skills needed for graphic designers, the studies of this doctoral thesis present empirical evidence of the qualifications sought by organisations when hiring graphic designers. In addition to that, the studies extend the common narrative that graphic design is mainly concerned about visual outcomes. Similar to what was seen in the literature review, the work and skillset described in job advertisements for graphic design positions in the UK (see Figure 4 on page 97), Finland (see Figure 5 on page 98) and Brazil (see Table 7 on page 99) is also broad. The number of requests in each study varied from 35 codes in Brazil (Study 3) to 36 codes in the UK (Study 1) to 40 codes in Finland (Study 2). Each code represented one type of request made by organisations in their job advertisements (e.g. 2D software, project management skills). Graphic designers are expected to provide work in multiple areas, ranging from printed materials to digital interfaces. In all three studies, requests for 'print design' and 'digital design' were respectively the first and second most sought deliverables mentioned in the job advertisements. Other requests were found in the three studies but in different proportions. For instance, delivering 'packaging design' was more frequent in the job advertisements from the UK (18.3%) than in those from Finland (8.4%) and Brazil (9.4%).

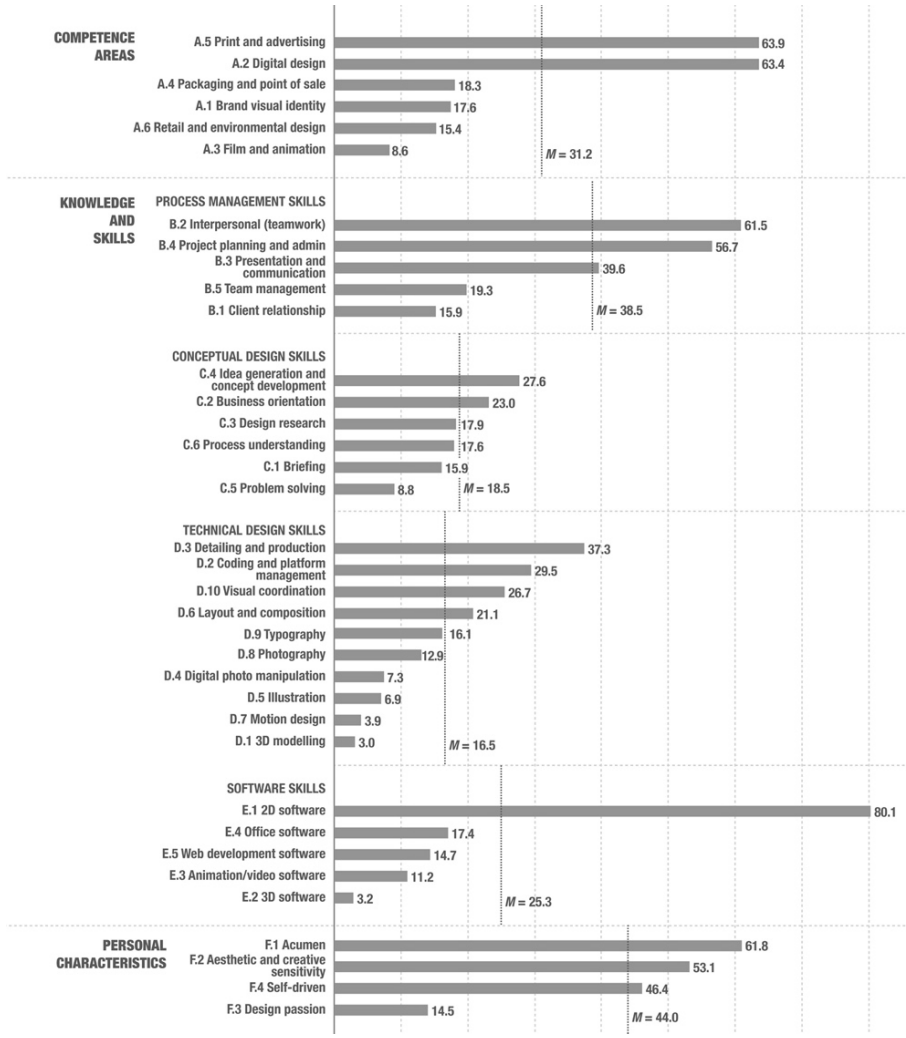



**Figure 4.** Requirements stated in job advertisements by companies in the UK.

Source: Study 1 (Figure 1).

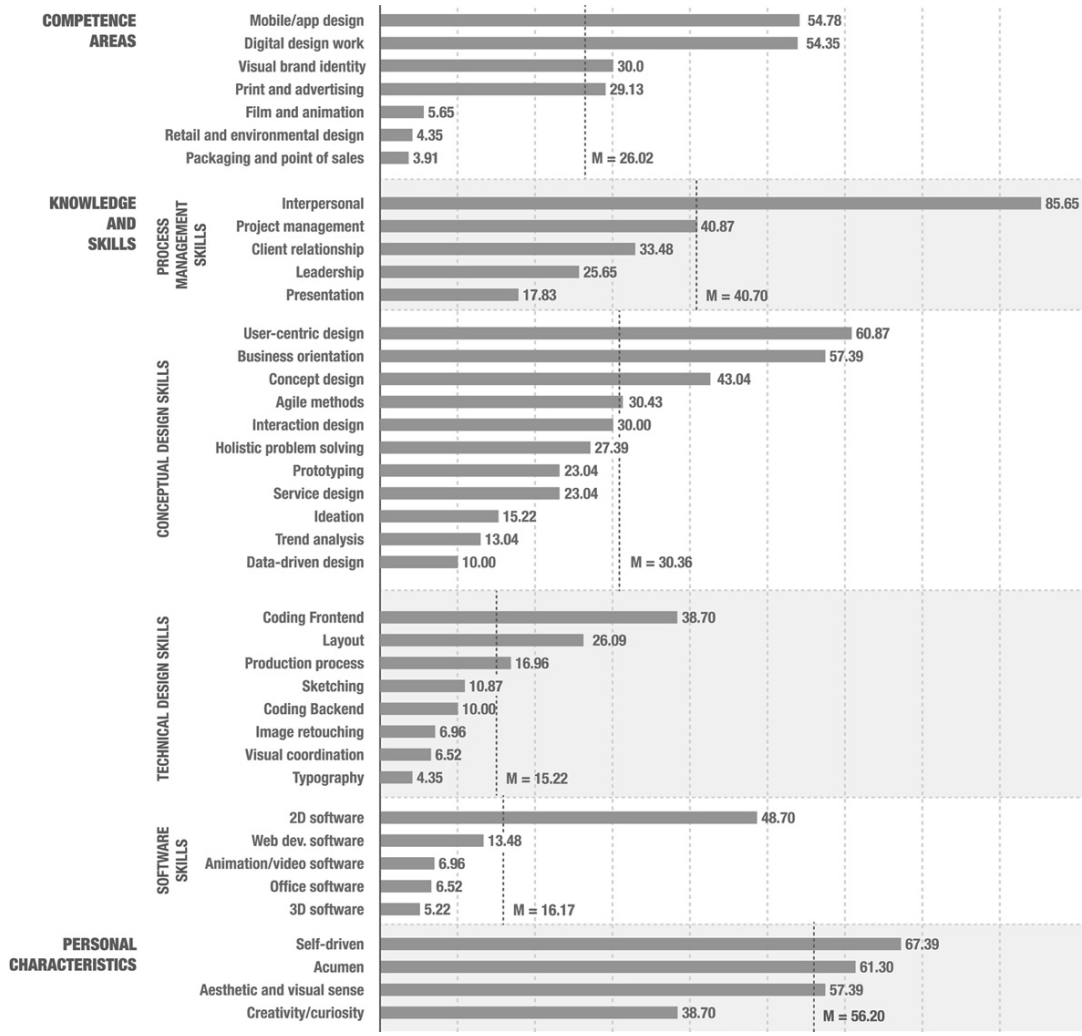



**Figure 5.** Requirements stated in job advertisements by companies in Finland.

Source: Study 2 (Figure 1).

| **Design deliverables** | **%** |
|---|---|
| D.1 Print design | 56.87 |
| D.2 Digital design | 45.01 |
| D.3 Brand design | 21.83 |
| D.4 Editorial design | 16.71 |
| D.5 Packaging design | 9.43 |
| D.6 Video and animation design | 7.28 |
| D.7 Signage and POS design | 5.39 |
| D.8 Service design | 0.27 |
| **Conceptual design skills** | **%** |
| K.1 Concept design | 6.74 |
| K.2 Business | 2.70 |
| K.3 Problem solving | 2.70 |
| K.4 User experience (UX) | 2.16 |
| K.5 Research | 1.62 |
| K.6 Design thinking | 0.54 |
| **Content skills** | **%** |
| K.7 Foreign language | 11.32 |
| K.8 Native language | 7.55 |
| K.9 Content creation | 6.20 |
| **Software skills** | **%** |
| K.10 2D software | 68.19 |

| | |
|---|---|
| K.11 Office software | 23.45 |
| K.12 Animation/video software | 11.32 |
| K.13 Web software | 7.28 |
| K.14 3D software | 2.70 |
| **Process management skills** | **%** |
| K.15 Teamwork | 13.75 |
| K.16 Project management | 7.28 |
| K.17 Client relationship | 1.62 |
| Technical design skills | % |
| K.18 Layout | 39.08 |
| K.19 Photography | 23.99 |
| K.20 Production and materials | 20.49 |
| K.21 Coding | 10.78 |
| K.22 Illustration | 7.28 |
| K.23 Design for web | 5.12 |
| **Personal traits** | **%** |
| P.1 Acumen | 12.40 |
| P.2 Self-driven | 10.24 |
| P.3 Creativity | 9.70 |
| P.4 Aesthetic sense | 1.35 |

**Table 7.** Requirements stated in job advertisements by companies in Brazil.
Source: Study 3 (Tables 1, 2 and 3 combined).

When looking at the requests made in job advertisements from the UK, Finland and Brazil, I noted a number of skills that overlapped with the skills described by earlier scholars. Knowledge on visual elements resembles 'layout', 'typography', 'illustration' and 'photography' skills, all part of the 'technical design skills' group in the three studies. In terms of tools, digital tools such as '2D software' were recurrently mentioned in the job advertisements. Knowledge in methods overlaps with organisations mentioning 'process understanding' or 'holistic problem solving'. Lastly, the skills mentioned by scholars outside the design field were covered in the job advertisements, as organisations seek for 'presentation', 'interpersonal', 'design research', 'business orientation' and 'content creation' skills.

Yet, I found a number of skills in the job advertisements that expand what earlier scholars suggest as the skills needed by graphic designers. In the job advertisements from the UK (Study 1), graphic designers are required to have, for example, 'client relationship', 'coding and platform management' and 'project planning and administration' skills. In the job advertisements from Finland (Study 2), organisations post requests for skills in 'agile methods', 'data-driven design' and 'trend analysis'. Lastly, job advertisements from Brazil (Study 3) mention requests for proficiency in a 'foreign language' (English or Spanish). In addition to that, the job advertisements from the three countries mention personal characteristics sought in graphic designers in terms of being 'self-driven' and having 'aesthetic and visual sense', 'acumen' and 'creativity'. Overall, considering the wide range of requests found in the advertisements (varying from 35 to 40), the graphic designer's skillset is described in a broader form in the advertisements than in the literature. This broader skillset sought in graphic designers suggests that, nowadays, organisations expect graphic designers to go beyond taking care of the visual aspect of products and services. For example, requests made in advertisements for 'business orientation', 'coding', 'service design' and 'user-centric design' signal how graphic designers have other responsibilities other than defining visuals.

Early literature described the work of graphic designers as 'mundane' (Dorland, 2009; Julier, 2010). The authors point out that graphic designers routinely carry out managerial activities, such as management of workload, planning and auditing. I found evidence for this administrative type of work in the job advertisements of the three studied countries in terms of 'project management',

'team management', 'leadership' and 'client relationship'. The three countries differed in terms of the frequency that the requests for administrative work were found in the advertisements: in the UK, 'project planning and administration' was mentioned in 56.7% of the advertisements, 'team management' in 19.3% (38.3% in senior positions) and 'client relationship' in 15.9%; in Finland, 'project management' was mentioned in 52.5% of the advertisements, 'leadership' in 10.2% and 'client relationship' in 25.4%. Lastly, in Brazil 'project management' was only mentioned in 7.3% of the advertisements and 'client relationship' in 1.6%, while no mentions of team leadership were found. Taken together with the overall list of requests made in the advertisements, I found evidence that the work of graphic designers is both 'creative' (e.g. building visual outcomes) and 'mundane' (e.g. managing the workload).

Literature in design management also pointed out to two different roles that designers fulfil in organisations: product-related and process-related roles (Valencia et al., 2013). In sum, the product-related role is at the 'core' of the profession, in terms of delivering visual work. The process-related role relates to designers being able to communicate and disseminate information across multiple departments and areas. I found that the requests made by companies in the UK and Finland emphasised both the product- and process-related roles, while in Brazil the focus was mainly on the product-related role. This claim is supported by looking at how often companies mentioned (1) process management skills, (2) conceptual design skills, (3) technical design skills, and (4) software skills in their job advertisements. Process management and conceptual design skills refer to a more strategic contribution provided by designers (e.g. Perks et al. 2005) in terms of client relationship, business and project management. Requests for skills that demonstrate this strategic role, such as 'client relationship', were much higher in Finland (25.4%) and the UK (15.4%) than in Brazil (1.62%). Other skills which had large differences were 'project management' (UK: 56.7%, Finland: 52.5%, Brazil: 7.3%), 'business' (UK: 23%, Finland: 62.7%, Brazil: 2.7%) and 'concept development' (UK: 27.6%, Finland: 32.2%, Brazil: 6.7%). On the other hand, technical design and software skills refer to a more 'hands-on' knowledge of developing and delivering design projects (e.g. Frascara, 2004). When looking at skills that demonstrate the traditional role, the requests across the three studies were similar. For example, requests for skills in 'illustration' (UK: 6.9%, Finland:

6.8%, Brazil: 7.3%) and '2D software' (UK: 80.1%, Finland: 72.8, Brazil: 68.2%) had smaller variations across the studies. In the case of '2D software', it was the most frequent request in job advertisements from the UK and Brazil, while being the second most frequent in job advertisements from Finland.

Overall, the work and skillset of graphic designers were broader in Finland than in the UK and Brazil. When it comes to the requests that were present in over half of the job advertisements (see Table 8 on the next page), in the Finnish study there were nine requests versus six in the British study and only two in the Brazilian study. In this sense, graphic designers in Finland are more often required to have knowledge in areas outside the design field, such as 'business'. In addition, all the requests present in over half of the advertisements in the UK and Brazil were also found in the requests in over half of the advertisements in Finland. In addition to the broader skillset requirements for graphic designers in Finland, the analysis of job advertisements showed how the job market in Brazil mainly focuses on the delivery of 'print design', while the requests in the UK and Finland were broader.

| | United Kingdom – Study 1 | | Finland – Study 2 | | Brazil – Study 3 | |
|---|---|---|---|---|---|---|
| | Request | % | Request | % | Request | % |
| #01 | 2D software | 80.1% | Interpersonal | 81.4% | 2D software | 68.2% |
| #02 | Print and advertising | 63.9% | 2D software | 72.9% | Print design | 56.9% |
| #03 | Digital design | 63.4% | Acumen | 72.9% | | |
| #04 | Acumen | 61.8% | Aesthetic and visual sense | 71.2% | | |
| #05 | Project planning and admin | 56.7% | Self-driven | 64.4% | | |
| #06 | Aesthetic and creative sensitivity | 53.1% | Print and advertising | 62.7% | | |
| #07 | | | Business orientation | 62.7% | | |
| #08 | | | Digital design work | 57.6% | | |
| #09 | | | Project management | 52.5% | | |

**Table 8.** Requests made in more than half of the job advertisements in Studies 1, 2 and 3.

All in all, the work and skillset of graphic designers are described broadly, both in the literature and in the job advertisements from the UK, Finland and Brazil. While some scholars prefer using the term *communication designer* (e.g. Frascara, 2004) to signal the broader contribution of graphic designers, others prefer to stick to the traditional term (e.g. Triggs & Atzmon, 2019). This broad description, in terms of the qualifications and deliverables required from graphic designers, is aligned with debates on how (graphic) design is expanding its contribution for organisations from product-oriented to process-oriented approaches (e.g. Valencia et al., 2013). In the job advertisements collected for this doctoral thesis, organisations request a wide skillset when hiring graphic

designers: in Studies 1, 2 and 3 of this doctoral thesis, I found that organisations seek between 35 (Brazil), 36 (UK) and 40 (Finland) different types of qualifications in graphic designers. This broad skillset and contribution expected from graphic designers brings both good and bad news for design educators and practitioners. Let me start with the latter.

Asking a graphic designer to be qualified and highly competent in such an extensive range of skills (35 to 40) is very demanding. The same applies to design educators: it is a big ask to require design graduates to develop such a varied skillset during their degree studies. On top of developing skills, (graphic) design graduates still need to develop their creative thinking, as creativity is frequently seen as the 'currency of designers' (Moor & Julier, 2009, p. 7). Back in 1998, Steven Heller claimed in his book *The Education of a Graphic Designer* that the greatest problem in graphic design education was 'not enough quality time' (p. 128). For the author, the solution was to expand the length of the degree programme from three to five years in order to cover a broad selection of necessary skills. Expanding the time allocated for design education would provide a temporary solution – but what happens in the future? For example, technological changes are currently pushing graphic design towards the digital realm. Should design educators then simply add more time to education to cover these new skills? Considering the constant transformation in the graphic design profession, this might work for the short term, but it is not sustainable in the long term. Furthermore, will design graduates still have enough time to build their creative practices? Before I address this point, I want to go back to the good news.

Graphic designers are traditionally recognised by their visual contributions for society. Yet, together with design in general, the contribution provided by designers is broadening towards more strategic and holistic contributions for society. As the contribution expected from graphic designers expands, their skillset should follow. The fact that organisations are requiring skills in 'business', 'research', 'problem solving' and 'user-centric design' demonstrates that the value brought by graphic designers goes beyond creativity and visual contributions. In addition to changes in the skillset, graphic designers are also working under new job titles such as 'UX designer' and 'UI designer', as reported by the Association of Visual Communication Designers in Finland (Grafia, 2019). As pointed out by Meron (2021), even graphic designers perceive that

the term 'graphic design' reduces their contribution to organisations to simply visual outcomes (Meron, 2021). Organisations do not limit themselves to asking only for graphic design skills or the use of the job title 'graphic designer' when hiring graphic designers: hiring managers are asking for skills that go beyond the traditional scope of graphic design as a visual activity.

While I celebrate the broader skillset, and therefore contribution, of graphic designers, I have concerns about how realistic it is for a graphic designer to master so many different skills. My overall recommendation for design educators and practitioners is to strategically and carefully consider which skills they want to develop further. For example, if a graphic designer is interested in working more closely with mobile development, developing skills in 'coding', 'user interface' and 'usability' would be relevant to their work.

Design educators might want to plan enough time in their educational programmes for designers to expand their skillsets outside the design realm. Similar to the main phases of the Double Diamond Model by the British Design Council, a well-known creativity tool in design practice and education, graphic design graduates can have divergent and convergent phases in their skills development. In the divergent phase, graphic design students can participate in courses and projects in other fields – such as business, engineering or social sciences – where they engage in creative problem solving together with students from other areas of expertise. By expanding their skillsets, graphic design students are able to improve their creativity practices, as creative design practices are strengthened by divergent thinking (e.g. Singh & Gu, 2012). In the convergent phase, graphic design students can start seeking an area of specialisation in which to build their career forward. For example, students interested in 'illustration' can therefore focus their time and effort towards this area. In design education, the convergent phase could then be organised during bachelor studies, while the divergent phase might be more relevant for master studies, where graduates can then establish their professional strengths and build their careers in one specific area.

Graphic designers have multiple options in developing their careers. While the job advertisements in this doctoral thesis presented between 35 to 40 different types of requests sought by companies, the average number per job advertisement ranged from five to eleven (see Table 9 in Section 5.1.1). That

said, graphic designers can tailor their choices on the skills they want to develop not only according to their professional interests, but also according to the organisations they want to work with. Organisations emphasise in their advertisements the skillset that is relevant for them, but might not be relevant for another organisation. For example, a graphic designer joining a team of hundreds of designers might be able to focus on a specific type of work (e.g. illustration), while a graphic designer joining an organisation as the first designer might need to 'wear multiple hats'. While I propose that design educators should help their graduates to expand their skillsets, graphic designers have multiple options on the career paths that they want to develop further and the skills they want to learn.

Graphic designers find work in different types of companies and collaborate with professionals from other areas (e.g. engineering, marketing, business). Proper collaboration happens through shared knowledge. In other words, graphic designers need to have a common language with other professionals. One way of building this common language is through expanding the skillset of graphic design professionals towards other areas. Luckily, other professionals are also striving to learn about design. For example, the *Harvard Business Review* routinely publishes articles on design thinking and how design can benefit businesses. However, design educators and practitioners should not forget their origins: visual design skills are at the core of the graphic design profession (as presented in the conceptualisations of graphic design presented in Section 2.1 Defining graphic design) and should be given enough attention.

Coming back to the suggestion presented earlier by Heller (1998) to expand the time allocated to graphic design education, my suggestion is that design educators should build a culture of life-long learning for their design graduates. As repeatedly seen in this thesis, graphic design has changed over history and there is little indication that it will remain stable. New technologies, developments in society and changes in economic conditions will require graphic designers to adapt to the situation at hand. To that end, a constant learning mentality will help graduates to adapt and pick up new requirements to work as a graphic designer in the future.

### 5.1.1 The state and maturity of the job market for graphic designers in the UK, Finland and Brazil

In the studies of this doctoral thesis, I used job advertisements from the UK, Finland and Brazil as a proxy for investigating the job market faced by design professionals. In other words, the job advertisements collected for each study were used to provide insight into the professional skillset and work delivered by graphic designers in each country. These three countries have different traditions and histories of design practice and education (Moultrie & Livesey, 2009). For example, there is a major difference in the number of schools offering design education in each country: 242 design schools in the UK[10], 9 design schools in Finland[11] and 480 design schools in Brazil[12]. In addition to the number of schools (and consequently, number of graduates), there is a significant difference in the population of these three countries: the UK has around 66 million; Finland around 5 million and Brazil around 210 million. Therefore, it is natural to expect that the job market for graphic designers in these countries would be different.

Nevertheless, when looking at the number of job advertisements collected for the studies in this doctoral thesis, the job market size in these countries is not aligned with the number of design schools and the population. In terms of the number of positions advertised, the biggest market studied was the British one, where we collected 1,406 job advertisements in five weeks (average of 281 positions per week). Next, in the same period, we collected 371 job advertisements from Brazil (average of 74 positions per week). Lastly, in Finland, even though we extended the period and the search keywords, we only collected 230 job advertisements throughout 18 months of data collection (average of three positions per week). As noted by previous scholars using job advertisements as data (e.g. Dörfler & van de Werfhorst, 2009; Okamoto & Polger, 2012; Surry & Robinson, 2001), not all positions are openly advertised,

[10] According to the website of the University and Colleges Admission Services (ucas.com), consulted in January 2020.

[11] According to the website of the Finnish Design Academy (fda.com), consulted in January 2020.

[12] According to the website of the Brazilian Ministry of Education (emec.mec.gov.br), consulted in January 2020.

meaning that organisations might use internal networks or other means to fulfil their personnel needs. Still, the number of job advertisements found throughout the studies demonstrates that the market for graphic designers in the UK is larger than in Finland and Brazil.

Even though I collected fewer advertisements from Finland, the requests were broader and more frequent than in the job advertisements from the UK and Brazil. First, the coding scheme built for the Finnish study contained more codes (40) than the study in the UK (36) and in Brazil (35). In other words, the range of requests was more varied in Finland than in the other two countries. Second, the number of requests per job advertisement was bigger in the job advertisements from Finland (11.5) than those from the UK (9.6) and those from Brazil (4.8). As the job advertisements from Finland contained more requests than those from the other countries, these advertisements were naturally also longer: Finnish advertisements had an average of 372 words, while the British ones had an average of 231 words and the Brazilian ones had an average of 137 words (see Table 9 below).

| | **United Kingdom** | **Finland** | **Brazil** |
|---|---|---|---|
| # of job advertisements (period) | 1,406 (5 weeks) | 230 (18 months) | 371 (5 weeks) |
| Number of requests per job advertisement (average) | 9.6 (std. deviation = 4.48) | 11.5 (std. deviation = 4.20) | 4.8 (std. deviation = 2.96) |
| Average length of job advertisements (in number of words) | 231 | 372 | 137 |

**Table 9.** Comparison between job advertisements from the UK, Finland and Brazil

In the introduction of this doctoral thesis, I pointed to the notion that design is contextually situated. In other words, design practice might differ according to the context in which it happens. Following this argument, design practice in

countries such as the UK and Finland should be more mature in comparison with design practice in Brazil, as design education in Brazil started in the 1960s and a national design policy was only introduced in 1995 (e.g. Raulik-Murphy, 2010). All in all, after conducting the studies and comparing the results, I argue that the requests found in the advertisements in the UK and Finland were broader, suggesting that graphic designers are asked to carry more skills than the traditional '2D software' and 'layout', which were prominent in the job advertisements from Brazil. At the same time, organisations recruiting graphic designers in the UK and Finland demonstrate a broader understanding of what is needed to operate as a graphic designer – and therefore set more extensive requirements.

## 5.2 Cross-contribution 2: The different career paths for graphic designers: From junior to senior positions, from in-house to consultancy (or vice versa), and/or from traditional to digital graphic design (or vice versa).

A second cross-contribution of this doctoral thesis is that it highlights the varied nature of career opportunities available to graphic designers. In this topic, I discuss three types of transitions available to graphic designers in terms of (1) seniority (from junior to senior positions), (2) location of the design position (internal or external to organisations) and (3) scope of work (traditional or digital graphic design). In analysing job advertisements representative of each type of transition, I show how the skillset requirements change when graphic designers face these transitions in their professional careers.

One of the contributions of this thesis is the presentation of seniority progression. The topic of seniority and design expertise is covered by previous scholars (e.g. Cross, 2004; Dorst & Reymen, 2004; Lawson & Dorst, 2015). For example, Dorst and Reymen (2004) mention (based on earlier lectures by Hubert Dreyfus) that designers have seven levels of seniority: (1) novice, (2) advanced beginner, (3) competent, (4) proficient, (5) expert, (6) master and (7)

visionary. In this doctoral thesis, I classified positions under three levels of seniority: (1) junior, (2) mid-level and (3) senior based on years of experience asked in the job advertisements and following the standards published by industry reports. In addition to that, the authors ask the question: 'How do the transitions between levels actually take place?' In other words, how do designers move from one level to another?

Study 1 in this doctoral thesis answers this question by providing specific information on what skills designers need to develop for growing their seniority. Specifically, I described how the skillset changes when moving from junior to senior in terms of the competencies, knowledge and skills and personal traits that a professional needs to acquire to accomplish such a transition (see Figure 6 on the next page). Four new skills need to be acquired when moving towards senior positions: (1) ability to manage teams, in terms of coaching and guiding other (junior) employees. Also, knowledge of (2) business, marketing and other commercial aspects becomes more common when moving from junior to senior level. Having (3) process understanding – in terms of being able to contribute to every step of the design process and knowledge – is important. Lastly, the ability to create (4) illustrations to be used in graphic design outcomes (e.g. drawing an illustration to explain a concept in a text) becomes more frequent in senior positions. As illustration is a complex skill – in fact, many professionals prefer to work as 'illustrators' and focus on that – it is seldom a skill possessed by novice designers. Two other skills present a non-linear development: (1) 'coding and platform management' is more frequent when moving from junior to mid-level positions, but then decreases when moving to senior positions. An explanation for that is that senior designers are more involved in more managerial or complex activities, which do not allow them to perform coding activities. (2) 'Self-driven' in terms of proactivity and willingness to learn was less frequent when moving towards senior levels. As professionals evolve in their career, the need to mention this personal characteristic comes up less consistently in job advertisements.

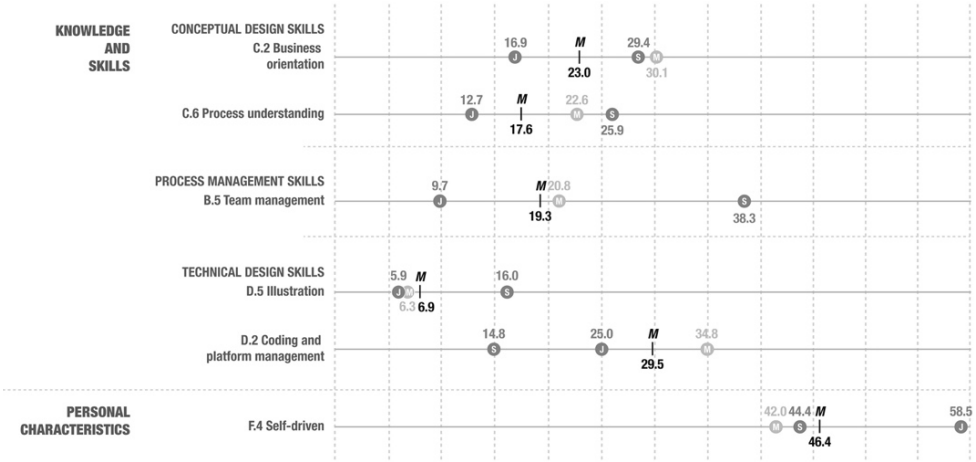



**Figure 6.** Skill requirements that differ significantly among junior, mid-level, and senior positions. Source: Study 1 (Figure 2).

Overall, graphic designers expand their contribution when moving towards senior positions. Specifically, graphic designers start managing the process and teams, taking responsibility for more complex design tasks and also demonstrating the business value of design. The work of senior graphic designers resembles the work of design managers as discussed in previous literature. In sum, design managers are designers who were elevated in a company's hierarchy and are responsible for implementing and managing the design function (see e.g. Blaich & Blaich, 1993; Borja de Mozota, 2003). As seen in Study 1, senior graphic designers are also responsible for managing the process and teams, which forces these professionals to expand their skillsets to areas outside the design realm.

Another cross-contribution of this thesis in terms of career transitions is related to how the positioning of the graphic design professional – internally or externally to an organisation – affects his or her skillset: In the first case, the organisation has an in-house design department, where designers work internally and mainly for the hiring organisation. In the second case, the organisation is a design consultancy or agency offering design services to one or multiple clients. The positioning of graphic designers is similar to the typical types of employment found for industrial designers: in-house designer or design consultant (Kaygan et al., 2020). Previous literature has focused on the advantages and disadvantages for organisations in adopting one approach or a combination of the two (e.g. Bruce & Morris, 1994; Bruce & Morris, 1998). I

also highlight the recent study by Kaygan et al. (2020), who historically mapped the job titles of design professionals in Turkey. I show how the skillset differs when transitioning from in-house to external (or vice versa) in Study 1.

By collecting job advertisements in the UK, Study 1 revealed that the market for internal designers is almost three times bigger than that for external designers. Specifically, in our data collection, we found 903 job advertisements for internal designers and 363 positions for external designers. Besides, we found multiple differences in the requests made by hiring organisations when seeking internal or external designers (see Figure 7 on the next page). Internal designers were more often requested to deliver packaging and point of sales material than external designers. In relation to knowledge and skills, internal designers receive more requests in terms of presentation and communication, project planning, team management, teamwork, digital photo manipulation, coding, detailing and production, photography, visual coordination, office and web software. External designers were more often requested to have knowledge and skills in idea generation, client relationships, motion design and design passion as a personal characteristic. These differences in the sought skillset also demonstrate the differences in the work of internal and external designers: internal designers need to focus not only on a broader number of design tasks than external designers, but also planning and management activities. On the other hand, as pointed out by previous literature (e.g. Bruce & Morris, 1994; Bruce & Morris, 1998; Von Stamm, 1998), external designers usually bring a fresh perspective and innovative solutions to a design problem. Thus, the requests for idea generation were most often found in the advertisements for external designers in comparison with internal positions.

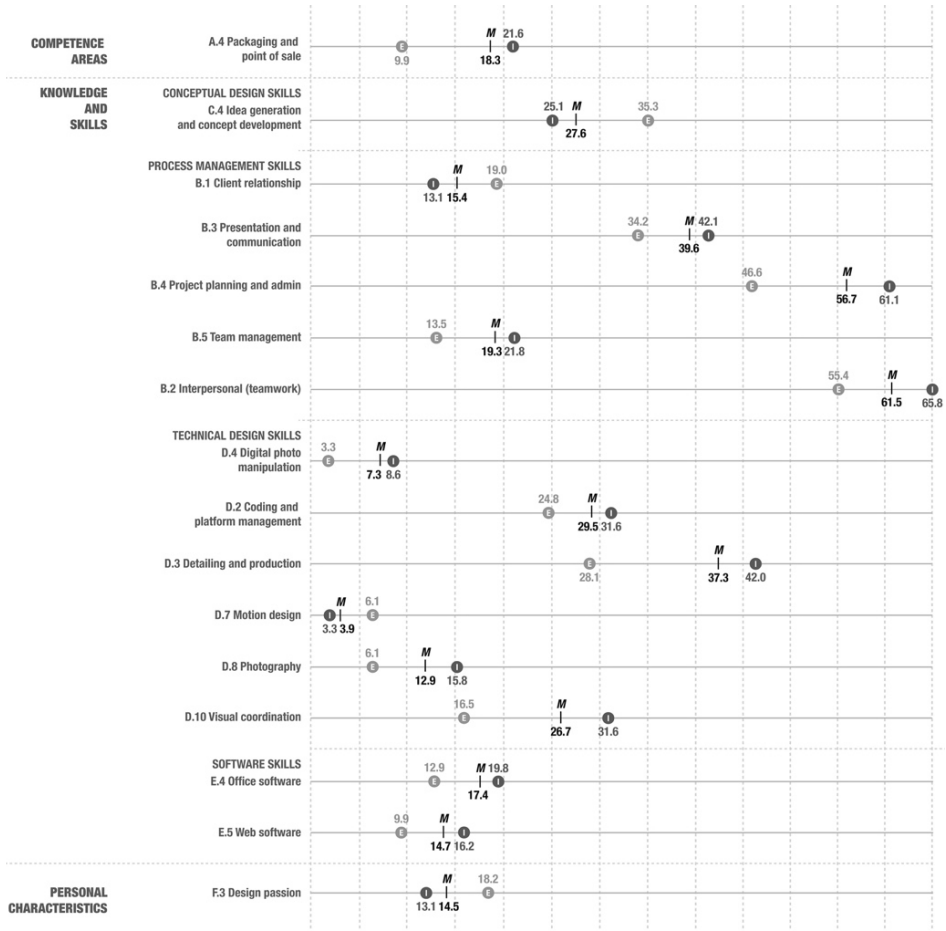



**Figure 7.** Skill requirements that differ significantly between internal and external positions. Source: Study 1 (Figure 3).

Lastly, this doctoral thesis contributes by pointing out how the skillset of graphic designers changes when the scope of work changes. In this case, I follow previous scholars (e.g. Frascara, 2004; Davis, 2006; Davis, 2008) who have claimed that the scope of the work of graphic designers expands due to technological and societal developments. The type of work done by graphic designers is in constant flux, being nowadays more oriented towards the overall user experience, instead of only the visual outcomes (e.g. Cezzar, 2017; Frascara 2017). The more traditional production of physical outcomes is still relevant nowadays, as people interact with (physical) books, signage and other elements. However, the rise of digital design has shifted the work of designers

towards digital platforms. This trend has been documented in internet publications, like for instance the 'In Tech Report' (Maeda, 2017) and internet blogs from internet companies such as Adobe and the Interaction Design Foundation. However, in academic outlets, this movement towards digital design has been less acknowledged.

In Study 2 of this doctoral thesis, I covered how the graphic design profession is expanding its work towards the digital realm. For example, graphic designers now find employment under new job titles, such as interaction designer, UX designer, UI designer and web designer. In the study, we collected 55 job advertisements for 'graphic' designers and 102 for 'digital', 'interaction', 'UX', 'UI' and 'web' designers. In other words, the number of open positions for digital graphic designers in Finland is almost double than that for 'traditional' graphic designers. In addition to that, in Finland designers have also found employment under titles such as 'marketing designer', 'service designer', 'design director' and 'head of design' (see Table 10 below).

| *Job title* | *No.* | *Job title* | *No.* | *Job title* | *No.* |
|---|---|---|---|---|---|
| Graphic Designer[a] | 54 | Marketing Designer | 3 | Design & Marketing Coordinator | 1 |
| UX Designer[b] | 38 | Digital Designer[b] | 2 | Design Director | 1 |
| Web Designer[b] | 24 | Lead Designer | 2 | Digital Graphic Designer | 1 |
| UI Designer[b] | 21 | Motion Graphic Designer | 2 | Graphic Maker[a] | 1 |
| Visual Designer | 21 | Graphic UI Designer | 2 | Head of Design | 1 |
| UI/UX Designer[b] | 11 | Brand Visual Designer | 1 | Lead Creative Strategist | 1 |
| Designer | 10 | Communication Coordinator | 1 | Marketing Artist | 1 |
| Service Designer | 8 | Content Designer | 1 | Marketing Content Designer | 1 |
| Interaction Designer[b] | 6 | Creative Design Manager | 1 | Marketing Manager | 1 |
| Art Director | 6 | Creative Director | 1 | Usability Expert | 1 |
| Packaging Designer | 3 | Creative Strategist | 1 | Web Graphic Designer | 1 |

[a]Traditional graphic design positions.
[b]Digital graphic design positions.

**Table 10.** Job titles in the job advertisements (Finnish titles translated to English). Source: Study 2 (Table 1).

The expansion of the work of graphic designers to digital spheres also affects the skillset needed to operate as a graphic designer. Besides the overall required qualifications (see primary and foundation themes in Figure 8), five qualifications appeared in our analysis as relevant only for the digital graphic design positions (see primary and secondary specialisation themes in Figure 8). Digital graphic designers are requested to deliver (1) mobile and app interface design as well as to have knowledge in (2) agile methods, (3) interaction design, (4) prototyping and (5) user-centric design. These five requests are then essential for graphic design students and professionals who plan to move their career towards the digital design arena. Overall, as shown in Figure 8, in order to build a career in digital design, professionals might want to prioritise a few skills to develop their career on this professional path.

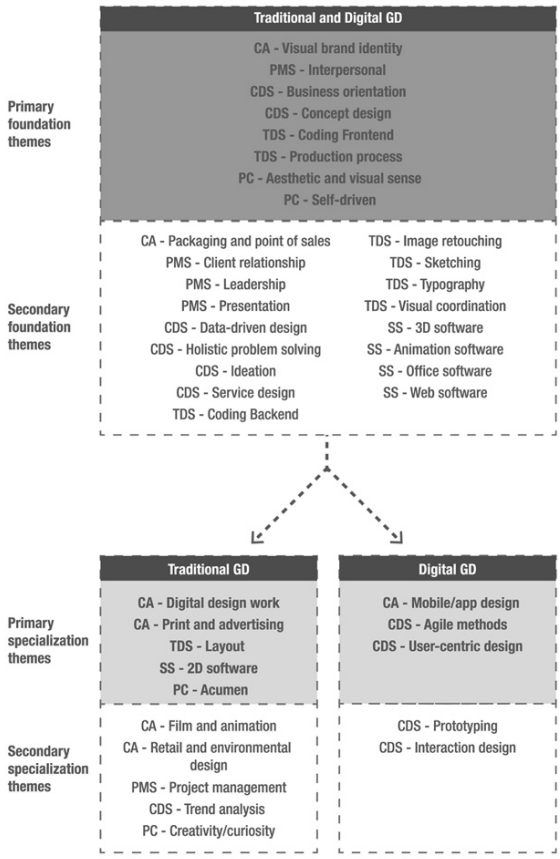



**Figure 8.** Foundation and specialisation themes for traditional and digital graphic design positions. CA = competence area; PMS = process management skills; CDS = conceptual design skills; TDS = technical design skills; SS = software skills; PC = personal characteristics. Source: Study 2 (Figure 3).

The findings of this doctoral thesis suggest that graphic designers have different options when planning their career paths. For example, they can develop their careers in in-house design departments or design consultancies. Also, by accumulating experience on the job, graphic designers move towards senior positions. Or, lastly, they can decide to move towards digital-focused roles (e.g. UX, UI). As seen in the job advertisements collected in the studies of this doctoral thesis, the work and skillset of graphic designers change according to the location, level of seniority and focus of the position. Hence, this suggests that graphic designers are required to constantly revisit and update their skillsets in line with their career goals.

These multiple career paths for designers are aligned with current discussions on the expanded role of designers. Designers have been delivering a broader contribution for organisations by joining the development process in earlier phases. In other words, designers are coming on board in the research or conceptual phases, instead of only in the technical and visual phases. This broader contribution has affected the profession in different ways. For examples, designers are located inside the organisation, signalling that design is an important competency that should be developed internally. Also, new job titles are being created, suggesting a higher degree of specialisation in the field. Nowadays, multiple job titles suggest that designers are moving towards specialising in specific subfields of the design profession. One example is the current popularity of 'UX/UI' designer as a job title, suggesting that designers are moving towards specialising in designing digital outcomes.

How does this movement towards digital design affect graphic designers? First, it suggests that graphic designers have more opportunities to deliver their work within companies. More importantly, it suggests that 'traditional' graphic designers need to pick up skills that they might not have learned through their formal education. Current design blogs have published several guides for professionals seeking to make the transition 'from graphic design to UX design' (e.g. uxdesign.cc). Also, Study 2 of this doctoral thesis presented a guide for practitioners on what skills are more/less relevant in the 'traditional' and 'digital' career paths for designers. In sum, the career options for graphic designers are almost endless, as long as the skillset follows suit.

## 5.3 Cross-contribution 3: The good, the bad and the ugly about job advertisements as data for (design) research, education and practice.

The third contribution in this doctoral thesis is to highlight the value offered by job advertisements as a source of data for research, education and practice. Studies using job advertisements are mainly found in the library information science field (see the literature reviews by Harper, 2012 and Kim & Angnakoon, 2016), but also in business, education and health. In design, a few researchers have used job advertisements to investigate the requirements posted by companies (e.g. Yang et al., 2005). By reviewing the literature in the field of graphic design and design management, I found a predominance of studies based on data generated for research such as interviews and surveys. In the studies of this doctoral thesis, I selected job advertisements as a type of document not generated for research.

This doctoral thesis contributes to understanding the relevance of job advertisements by studying how each such document was built and used by hiring managers and designers in the recruitment process (see Study 4). The interviewed hiring managers described how they used job advertisements to communicate their personnel needs to candidates, to advertise the company and the position to candidates and other audiences (e.g. clients, competitors) and to filter candidates who applied to the position. The designers described how they used job advertisements to get information about the job market (e.g. who is hiring and which skills are required?), to evaluate and filter positions and companies and lastly as a guide for self-development (see Figure 9 on the next page).

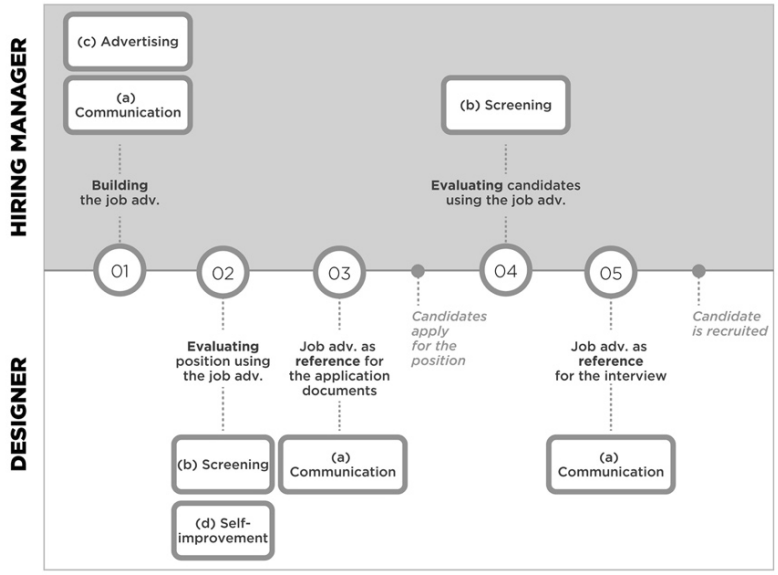



**Figure 9.** How hiring managers and designers used the job advertisement in design recruitment processes. Source: Study 4 (Figure 1)

From a research perspective, prior studies have pointed to the benefits of studying job advertisements in terms of the coverage of the job market provided (see e.g. Todd et al., 1995). By selecting job advertisements, scholars have managed to study a large share of the job market, which also provided the opportunity to compare cities and countries (e.g. Brumberger & Lauer, 2017). In addition to that, scholars have mentioned the benefits of using naturally occurring data (Silverman, 2016), in the sense that job advertisements are not generated for research and are found in their natural setting (Chipulu et al., 2013; Dörfler & van de Werfhorst, 2009; Marcusson & Lundqvist, 2015; Shou & Wang, 2017).

On the other hand, scholars have pointed to limitations, such as the fact that not all positions are openly advertised (Bettinger et al., 2016; Cloonan & Norcott, 1987; Dörfler & van de Werfhorst, 2009). In addition, challenges were found when interpreting and analysing the data, as companies might use technical jargon (Debortoli et al., 2014; Den Hartog et al., 2007; McArthur et al., 2017; Meyer, 2017), over-represent themselves and the position (Kuokkanen et al., 2013; Shi & Bennett, 2000; Todd et al., 1995), fail to mention all the requirements for the position (Chipulu et al., 2013) and/or reflect in the job

advertisement only the opinions of those involved in writing the document (Ahmed, 2012; Brumberger & Lauer, 2017; Cleary & Cochie, 2011; Kennan et al., 2006). As stated by Coffey (2014, p. 368), job advertisements, as a type of document, provide a 'documentary reality', in the sense that they show only a partial version of a phenomenon. In addition to that, as job advertisements are contextually situated – they are written with a specific audience and goal in mind – a 'competent reading' is needed to fully understand their content (Atkinson & Coffey, 2004, p. 72). They allow access into the shared, specialist understandings of language within the graphic design profession.

In response to the limitations pointed out by previous literature in using job advertisements for research, all the hiring managers interviewed for Study 4 mentioned that they had written the text in cooperation with other colleagues (designers, project managers, creative directors, human resources professionals) in order to bring multiple perspectives to the text. Also, in the 12 cases followed, the job advertisements were publicly available, as hiring managers wanted to ensure that more candidates would get access to the document. Hiring managers also mentioned that the quality of the text in the job advertisement affects the applicants. In the words of one hiring manager interviewed in Study 4: 'the better the (job) advertisement is written, the better the applications'.

Based on the arguments of previous scholars and the interviews with hiring managers and designers, this doctoral thesis suggests that job advertisements can serve (design) researchers, educators and practitioners as a proxy for the job market. These professionals can learn about the current situation of the job market by studying the requests organisations make in their advertisements. For researchers, job advertisements provide the opportunity to investigate how professions evolve in terms of the skillset required and the activities performed by these professionals. One example of this is the work by Choi and Rasmussen (2009), who tracked the changes that occurred in the requirements set for librarian professionals by analysing job advertisements over a period of nine years. In addition to that, job advertisements allow analysing this professional development by looking at what organisations say about their needs, without being limited in terms of sample size (as in the case of interviews, for example) (e.g. Todd et al., 1995).

For design management researchers, the use of job advertisements can provide a new layer of information to their inquiries on how companies adopt and manage design. Recruiting design professionals is noted as one of the initial steps in adopting design, according to design management models (Bruce et al., 1999; Acklin, 2013) and case studies on how organisations adopt design (Bailey, 2012; Body, 2008; Valtonen, 2007). In addition to that, design management research has mainly been executed based on interviews (e.g. Berends et al., 2011; Valencia et al., 2013) and/or surveys (e.g. Bohemia, 1998; Chiva & Alegre, 2007). Similar to the studies in this doctoral thesis, using job advertisements would provide design management researchers with the means to expand the breadth of their analysis by relying on a type of data that is openly available online.

Another topic of interest for design management researchers is the role of design in organisations. By using job advertisements as data, researchers can look at the descriptions provided by organisations in their advertisements as a proxy for assessing the role of design in a specific organisation. This connection between the activities performed and the skillset was previously presented in education literature by Voorhees (2001), who suggested that in order to execute an activity, a professional needs to have a specific group of skills, abilities and knowledge. Design management researchers can also assess the knowledge an organisation has about design based on the requirements posted in their advertisements. For example, if an organisation seeking to hire a designer only emphasises requests for technical skills (e.g. software, layout) in the advertisement, this suggests that a designer in this company would have a more tactical, instead of strategic, role.

For practitioners and educators, job advertisements serve as a proxy for monitoring the job market in one or multiple countries. Practitioners and educators face the challenge of constantly reviewing their skillset, as design as a profession is in constant change, which has a direct impact on the skillset of designers and the work they do. The description of the skillset and activities in a job advertisement informs practitioners and educators about what organisations seek in design professionals. However, organisations provide a description of the 'ideal' candidate, which does not always align with what a candidate thinks about the profession. This is in line with previous literature claiming that job advertisements are a reflection of how a company defines a profession's identity: Snyman (2001) concluded that companies lacked an

understanding of the information and knowledge management professional based on the requirements posted in their advertisements. For example, in design blogs (e.g. uxdesign.cc), discussions on whether designers should learn programming languages are recurrent. At the same time, a survey of over two thousand graphic designers in the US and in the UK showed that 'coding' was the most useful skill to develop in the future (Roberts et al., 2015). Thus, when an organisation mentions in a job advertisement that the candidate must know 'coding', this signals to applicants that designers in this specific organisation are required to work with coding and know programming languages. Similarly, the types of requests made in the job market can then benefit educators by informing educational practices that improve graduates' employability levels. In the words of Bennett (2002, p. 471), who analysed one thousand job advertisements for business graduates, 'higher education institutions which are serious about wanting their graduates to be more employable should study carefully current trends in employers' demands for particular personal skills (as evidenced by, for example, the contents of job advertisements) and then develop these skills in undergraduate programmes'. In this regard, in Study 3 we compared the requests made in the job advertisements with design education guidelines in order to check overlaps between what organisations are seeking and what educational institutions are teaching in Brazil.

Overall, job advertisements reflect the past of the profession, by showing how organisations describe their needs throughout time, at the same time they influence the future developments of the profession, such as by potentially shaping the educational and self-development practices of practitioners. In addition to that, the language used in job advertisements to describe the work and skillset of design professionals gives an indication of how the industry visualises the design profession identity.

## 5.4 Cross-contribution 4: Making secondary data great again: how to use job advertisements in (design) research, education and practice.

The fourth contribution of this doctoral thesis adds to the previous one by highlighting the practical aspects of adopting job advertisements for research and practice. As mentioned by Kim and Angnakoon (2016, p. 334), who performed a methodological assessment of over 100 studies using job advertisements in the field of library and information science, 'there remains considerable room for improvement in the methodological rigour of job ad studies'. Similarly, Harper (2012, p. 32) also pointed out that in library and information science, the main field using job advertisements as data, 'there has been little guidance on the collection and analysis of job advertisements'. Thus, after reviewing the relevance and the contribution provided by job advertisements (see Cross-contribution 3), this doctoral thesis offers guidance for researchers, educators and practitioners in planning their data collection, management and analysis phases.

Starting with data collection, Harper (2012) and Kim and Angnakoon (2016) pointed out that previous researchers have mainly relied on purposive sampling or relevance sampling (Krippendorff, 2013), selecting relevant websites or printed media (e.g. newspaper, magazines) to extract published job advertisements. The selection of the websites and media were mainly made based on the researchers' knowledge about the field or by consulting practitioners and experts (e.g. Dziobczenski & Person, 2017). Nowadays, online job advertisements are openly available on websites (e.g. Chipulu et al., 2013) such as LinkedIn and Indeed, which increases the access to a broader range of job advertisements in the market. At the same time, as seen in Studies 1, 2 and 3, organisations republish their advertisements in order to bring them to the attention of more job seekers. Therefore, researchers need to remove duplicate advertisements when collecting data. In the literature review conducted for Study 4, we found that the majority of studies selected online job boards as the main source to collect job advertisements. In the studies of this doctoral thesis, online job boards were used as rich sources for collecting data.

In addition to selecting the sources for collecting data, researchers also need to decide on the sample size for their studies. As pointed out by scholars (e.g. Todd et al., 1995), job advertisements have the advantage of not being restricted to a 'small' sample size. However, then the question is 'what is a "good" sample size to study using job advertisements?' and the answer is unclear (Kim & Agnakoon, 2016). However, Harper (2012) suggests that defining a minimum sample size is possible by looking at the sample size set in previous studies. In Study 4, we found that the sample size of the articles ranged from five to 250,000, where almost half of the studies had a sample size ranging from five to 250. Also, depending on the location and parameters for the data collection, reaching a specific number of job advertisements might be not possible. For example, in this doctoral thesis it took 18 months to collect 230 job advertisements from Finland (Study 2) while it took five weeks to collect 1,406 job advertisements from the UK (Study 1).

The decision made regarding the sample size will then directly affect data management. As discussed in the methodology section of this doctoral thesis, the time spent for collecting, organising, cleaning and analysing the data is affected by the number of job advertisements in the sample. Researchers should then balance the number of resources (e.g. time, people, equipment) available in their projects with the desired sample size accordingly.

For analysing job advertisements, researchers initially have to decide between adopting human or computer coding (see e.g. Neuendorff, 2002). Human, or manual, coding is when one or more coders analyse and make decisions about how to code the data. Human coding is more suitable for smaller sample sizes, when one or multiple professionals can manually read and interpret the dataset. On the other hand, computer, or automatic, coding is when a list of words is defined or extracted from the dataset, where (e.g.) frequencies and cooccurrences of terms are assessed. Computer coding is better suited to bigger samples. In Study 4, we found that a large share of studies (n = 189, 90.4%) used manual coding, with the number of coders ranging from one to five. In Studies 1, 2 and 3, we used human/manual coding to unveil the requests made by companies in their job advertisements. One of the reasons for that was to be able to identify the requests made by companies, which are described using a diverse range of different words. By reviewing the job advertisements manually, we could then interpret the meaning of the sentences and not rely on lists of words/sentences,

which are one way of analysing job advertisements automatically. Furthermore, throughout the development of Studies 1, 2 and 3, we improved the quality of our coding process by adopting a few practices:

- **Selection of coders**: An essential step at the beginning of the coding process is the selection of the coder(s) for the task. Krippendorff (2013) emphasises that coders need to have experience and familiarity with the phenomena under consideration. As pointed out by the author (2013, p. 128), 'Even the most detailed recording/coding instructions take for granted that coders and content analysts have similar backgrounds and so will interpret the written instructions alike'. In addition to that, job advertisements might contain professional jargon that complicates their interpretation (Debortoli et al., 2014; Den Hartog et al., 2007; McArthur et al., 2017; Meyer, 2017). In Studies 1, 2 and 3 of this doctoral thesis, all the coders had a combination of (1) formal training (degree) in graphic design and/or (2) practical experience working as a graphic designer. This educational and/or professional experience in graphic design was essential in the process of understanding and coding the job advertisements for graphic designers throughout the studies.

- **Training of coders**: In addition to a proper selection of coders for the project (see the previous item), an initial phase where the coders are trained to understand properly the coding scheme is recommended (e.g. Neuendorff, 2002). In addition to the initial understanding of the coding scheme, Neuendorff (2002) suggests that researchers undertake a 'pilot coding' (p. 133), where a portion of the data is coded by multiple coders, and the results are shared and discussed. In Studies 1, 2 and 3, we selected an initial share of advertisements and applied the coding scheme, and the results were discussed across the coders and other authors of the studies. The initial discussion and training of the coders, followed by a pilot coding, were essential to improve the coders' understanding of the coding scheme and, therefore, enhance the coding process and outcomes.

- **Multiple coders and intercoder agreement**: Manual coding of a significant amount of text is a demanding activity, which can lead to errors (e.g. Neuendorff, 2002; Krippendorff, 2013). One way of improving the coding process is to employ multiple coders and measure intercoder agreement. Yet, these two practices are far from common in studies analysing job advertisements:

in our literature review conducted for Study 4, we found that less than one-fifth of the studies had multiple coders (30 studies out of 209, 14.3%). In addition, less than one-tenth of the studies used intercoder agreement measurements (18 studies out of 209, 8.6%). These practices were used across Studies 1, 2 and 3 (see Subsection 3.4 Improving the research quality). In sum, in Study 1, the authors discussed and built the coding scheme together before the whole dataset was coded, while in Studies 2 and 3, we relied on multiple coders and measured intercoder agreement.

- **Break the coding process into multiple parts**: In Studies 1, 2 and 3, while reading the job advertisements, the coder(s) had to decide on between 35 to 40 codes to apply in each sentence. The use of multiple coders and intercoder agreement (previous item) helped minimise the number of errors in the coding scheme. Another action taken in Study 3 was to break the coding process into parts. As the coding scheme contained codes in three categories – design deliverables, knowledge and skills, and personal traits – we divided the coding process according to these three categories. First, each coder read the job advertisement looking for mentions of design deliverables, then the job advertisement was reread looking for knowledge and skills and lastly looking for personal traits. In other words, each job advertisement was read at least three times when coding. In doing that, we reduced the number of codes each coder had to consider when reading the documents. In addition to that, by reading each job advertisement multiple times, we reduced cases where a sentence was skipped or not coded.

In sum, I recommend that researchers should take advantage of the easy access to several job advertisements online to collect a larger sample size. Yet, the greater the sample size, the greater the resources needed to collect, organise, clean and analyse the data. Following that, I suggest that researchers should have multiple coders analysing the data. These multiple coders should be selected based on their professional and educational experience on the phenomena being studied. Furthermore, coders need to have a period of training and complete a pilot study to practice and agree on the coding scheme. Intercoder agreement should also be adopted to enhance the quality of the coding process. Lastly, in the case of a coding scheme with a large number of codes, I suggest that the coding process should be split into parts and/or repeated a few times to assure a reliable coding process and result.

# 6. Final implications

*Our company is hiring a graphic designer!*

*Apply if you meet these nine mandatory requirements:*

*#1 You have good interpersonal skills: you are good with people, collaborate and communicate properly with designers and other professionals;*

*#2 You master the following 2D software: Adobe Illustrator, Adobe InDesign, Adobe Photoshop, Figma and Sketch;*

*#3 You have professional acumen: you are careful about your work and pay attention to the minimum details;*

*#4 You have an 'eye for design';*

*#5 You are self-driven: you know when to take action and need very little supervision. You are also interested in learning and following trends, every single day;*

*#6 You can deliver print design work, such as brochures, books and posters;*

*#7 You can deliver digital design work, such as interfaces, websites and newsletters;*

*#8 You know that design is important for business and that every action you take has a financial impact on our clients' businesses;*

*#9 You know how to manage your schedule effectively, so we can give you several projects and you'll be able to work well under pressure.*

*It would be great if you would also fulfil these 31 optional requirements:*

*- Deliver work in areas such as film/animation, packaging/point of sales, retail/environmental design and visual brand identity;*

*- Are able to work with clients, lead teams and present your work;*

*- Have knowledge in 3D software, animation/video software, agile methods, coding back-end, coding front-end, concept design, data-driven design, ideation, image retouching, holistic problem solving, interaction design, layout, office software, production process, prototyping, service design, sketching, typography, trend analysis, user-centric design, visual coordination and web development software;*

*- Are creative and curious.*

*Send your application today!*

This is a fictitious job advertisement.

I wrote it including all the 40 requirements found in the job advertisements from Finland (Study 2). The Finnish study in this thesis was the one with the highest number of requests mentioned by companies in their job advertisements. In the first part of the fictitious advertisement (mandatory requirements) I presented the nine requirements mentioned in over half of the job advertisements in descending order of frequency (see Table 8 in Cross-contribution 1). In the second part (optional requirements) I included the 31 requirements mentioned in less than half of the advertisements. If a person who fulfils all these 40 requirements exists, he or she would be an extremely qualified candidate for this fictitious job advertisement. However, I do not believe in the existence of a single person able to master all these 40 requirements.

Looking back in time, Figure 10 shows an example of a job advertisement published in the first edition of Design Week magazine in the UK, back in 1986.



**SENIOR DESIGNER VISUALISER**

**Reading** **c.£15,000**

Our client is a well established graphic design consultancy with a prominent position in below the line work. The company now seeks to strengthen the design team with the appointment of a Senior Designer with exceptionally good visualising skills. The successful candidate (male or female) must also be able to demonstrate the ability to take and fully understand client briefs and then to run the job through to completion. Candidates must also have well developed skills in creative design, artwork, typography, print and art direction.

Candidates, probably aged up to 35, will be able to show a well balanced and well presented portfolio and attention to detail will be important. Suitable backgrounds could include the design/marketing department of a major blue chip plc or a graphic design group with an emphasis on creative design and a high level of service to their clients.

Please send full career history, in total confidence, to:

**Nicholas Potter, quoting reference 263/DW**
**Mainstay Management Services Limited**
**34 York Street, Twickenham, Middlesex TW1 3LJ.**
**Telephone: 01-891 3301.**

**MAINSTAY**
Management Services

**Figure 10.** Senior Designer Visualiser job advertisement published in 1986.
Source: Julier (2017, p. 23)

There are many ways in which this job advertisement tells us about what was happening in the design market (and society in general) over 30 years ago. First of all, there is no 'graphic' in the job title. Instead, visualisation is emphasised through the use of the word 'visualiser'. By asking that 'candidates must also have well developed skills in creative design, artwork, typography, print and art direction', the advertisement highlights the work of designers in terms of delivering visual outcomes. Following the same idea, the advertisement emphasises that 'attention to detail will be important', in a similar way as the 'eye for design' mentioned in the fictitious job advertisements; this idea was also repeated many times in the real job advertisements examined in the studies of this doctoral thesis. In addition to design requirements, the advertisement also describes the successful candidate as 'male or female', which tells about how the job market during the 1980s functioned. In this sense, job advertisements demonstrate dynamics of the job market that could interest researchers with different agendas (e.g. gender studies).

This job advertisement from the 1980s represents a different picture of the work and skillset of graphic designers, in comparison with what I found by looking at job advertisements from the UK, Finland and Brazil. For example, there are no mentions of the need for 'presentation', 'business' and/or 'project management' skills, which were recurrently found in the advertisements in this doctoral thesis. Does this mean that graphic designers in the 1980s did not present their work, engage in business discussions or manage their workload? Probably not. Still, as in many physical magazines, where the size of the advertisement affects the price paid to publish, it is fair to assume that organisations are financially incentivised to be careful about what they want to add to their advertisements. The fact that this organisation in the 1980s decided not to include these skills suggests that these were not central to the design position.

Job advertisements reflect the past while also guiding the future development of a profession. Organisations describe what their professional requirements are at a specific time. This idea was exemplified by looking at the earlier job advertisement from 1986. More importantly, job advertisements provide important information that shapes professions. A few designers in Study 4 and scholars (e.g. Rafaeli, 2006) mentioned that the qualifications described in job advertisements serve as a guide for future development. The alignment between the requirements posted in job advertisements and self-development is one that design graduates and practitioners should have in mind when looking for positions in the market.

In guiding the future developments of a profession, job advertisements also carry the role of contributing to the discourse about professional identity. In a study looking at how television programmes show the work done by industrial designers, Lloyd (2002) argues for an accurate representation of the profession to avoid false expectations from clients willing to buy design firms' work and from future design students. A similar argument can be made in the case of job advertisements: an inaccurate representation in the job advertisement might create false expectations from professionals applying to the position and professionals looking at the job advertisement for self-development purposes.

After spending the last seven years investigating the requirements stated in job advertisements for graphic design positions, and job advertisements per se, I reflect on what graphic design is, what the work done by graphic designers

involves and what skills are needed to operate as a graphic designer. Also, I address the relevance and value of job advertisements for practitioners, educators and researchers.

Graphic design nowadays is different from when Paul Rand described it in terms of designing billboards and birth announcements (Rand, 1970). On a similar note, Harland (2016, p. 23 - 24) in his book Graphic Design in Urban Environments makes his argument for how graphic design has changed over the years:

> Clearly, the idea of graphic design is continually evolving and the range of activities and intentions undertaken in its name diminish and increase. No longer is calligraphy significant in the way Dwiggins considered it to be, but typography remains central. No longer is printing the dominant media, but screen communication, is in the ascendant. No longer is graphic design undertaken only by individuals, but also by large teams who fulfil multiple junior and senior roles across a spectrum of expertise. No longer is it about the present and future, but also the past, with an established and continually evolving history.

The visual aspect of graphic design, strongly repeated in the common narrative of definitions of graphic design (see Table 2 in Section 2) is still present. Creativity and visual communication are at the core of the profession. When looking at what deliverables organisations mentioned in their job advertisements in the UK, Finland and Brazil, they expect graphic designers to provide physical (e.g. posters) and digital (e.g. interfaces) outcomes. In this sense, technology has transformed how graphic designers work since the late 1980s. The introduction of computers gave designers a better tool to work. Nowadays, it is impossible to disconnect graphic design from technology: '2D software' was the most frequent request in job advertisements from the UK and Brazil and the second most frequent request in job advertisements from Finland. This technological improvement also added a new medium for graphic designers as part of their work: delivering digital design work was frequently mentioned in over half of the advertisements from the UK (63.4%) and Finland (57.6%) and in slightly less than half (45%) of the job advertisements from Brazil.

Nowadays, graphic designers are located in multidisciplinary teams, both in design consultancies or in in-house departments, solving more complex problems (e.g. Davis, 2012). Ability to 'teamwork' was a recurrent request made in the advertisements collected in this doctoral thesis, especially in those from Finland (81.4%) and the UK (61.5%), while being less frequent in those from Brazil (13.7%). In other words, graphic designers are brought to collaborate at the beginning of the process, instead of being asked to add 'visual sparkle' at the end of the project.

This expansion of the scope of graphic design has a direct effect on how graphic designers work and the skills they need. For example, graphic designers face administrative tasks (Dorland, 2009; Julier, 2010), which lead them away from 'creative' work. At the same time, delivering visual outcomes is just part of what graphic designers do. Based on the job advertisements in this doctoral thesis, graphic designers are also involved in doing research, planning service journeys and writing content. Following the complexity of the problems and the multidisciplinarity of the teams in which graphic designers are positioned, a range of new skills become necessary in order to work, such as 'agile methods', 'business', 'coding', 'user-centred design', among many others.

In sum, the constant changes in the scope of the work of graphic designers requires a constant analysis of the relationship between the work done and the skills needed. Following the recognition of the value design offers for organisations, designers expand their activities and contributions. This expansion of the work forces an expansion of the skillset as well – for delivering digital design projects, a person should have digital design skills. The other way around is also true: by expanding the skillset, designers are able to expand the activities they perform and the contribution provided – after learning business skills, a person can now participate in business planning activities.

This constant updating of the work delivered and the skills carried by designers is a topic to which design educators, practitioners and researchers could pay attention. Educators have the power to shape design education and the skills their graduates will acquire before moving to the market. Practitioners should be aware that design is a profession that requires constant learning and, therefore, monitor developments in the profession on a regular basis. Researchers interested in design

practice and management can monitor the changes in the profession and how they affect the work, role and skills of designers.

Still, I identify opportunities to further develop the contributions made in this doctoral thesis and suggest a few future research opportunities.

Job advertisements tell us one facet of the story, where organisations have control of the message. As mentioned above, the work and skills of design professionals is a currently debated topic of interest to educators, practitioners and researchers. I suggest that future work should bring multiple perspectives about what designers do and the skillset they carry. In sum, job advertisements carry the point of view of recruiting organisations, but multiple perspectives – such as from practitioners, educators and researchers – can be added to yield a full understanding of the work and skillset of design professionals.

Similarly, by using job advertisements, we expanded considerably the breadth of the analysis by reviewing the language (e.g. words, sentences) used to describe graphic design professionals. Yet, we were not able to analyse each recruitment case deeply. Overall, throughout the studies of this doctoral thesis, we were able to cover a significant number of recruitment cases – ranging from 230 to 1,406. Such elevated numbers would be difficult to cover following other research methods. However, it was not possible to gain an in-depth understanding of each recruitment process and, specifically, how the company defines the work and skillset of graphic designers. Future scholars might want to aim to study a few cases deeply, instead of aiming for breadth. Similar to that, scholars might want to look at recruitment cases in more senior positions (e.g. Head of Design, Chief Design Officer) for which companies might not publish a job advertisement, but perhaps use headhunting companies and/or personal networks to find a qualified candidate.

Job advertisements can also be beneficial for design researchers interested in design policy, the job market for designers and design maturity across countries. Job advertisements provide information about what the organisations in a country think about the work and skillset of designers. For example, Brumberger and Lauer (2017) compared the requests for technical writers in the UK, Ireland and India using job advertisements. For the authors, a comparison study 'requires consistent and detailed sampling that job advertisements are especially

good at providing' (p. 313). In Section 5.1.1, I presented an overall view of the job market in the UK, Finland and Brazil, based on the number of advertisements published. In this case, I suggest that design researchers can look at the findings of this doctoral thesis, and the method employed here, as a starting point for their future inquiries. Yet, as said earlier, job advertisements provide the perspective of the hiring organisations only and, therefore, should be considered (if possible) together with other perspectives.

As this doctoral thesis moves towards its final paragraphs, I want to restate the importance of **monitoring**[13] the job market as an important practice for design educators in preparing their students for the market. Monitoring the job market can add a layer of information about what it means to be a graphic designer today. Graphic design educators are well positioned to critically assess the requirements posted by organisations hiring graphic designers and make changes to their educational practices accordingly. They can, for example, compare these requirements with academic literature in graphic design and also their experience in teaching graphic design. Yet, a clarification needs to be made in regards to how education and the job market operate. While design education prepares students for the future, meaning that students who start their design studies will graduate in a few years, the job market has a much more short-sighted view: organisations posting job advertisements expect to complete the recruitment after a few weeks/months. In other words, design education and the job market work on different time spans: while design education looks a few **years** ahead, the job market looks a few **months** ahead. In this sense, monitoring the market provides short-term direction on the developments happening in the graphic design profession.

Although design education and the job market operate on different timelines, I reinforce that design educators can monitor what happens in the market to identify trends in where the design profession is heading towards. For example,

[13] I made a similar argument when presenting Study 3 of this thesis at the DRS Conference, in Limerick/Ireland, in 2018. At that time, I was in a room full of design educators. I had to make sure that the argument presented in the study was not that the design education should **follow** the market, but instead that design education should **monitor** the market. Luckily (for me), the educators in that room understood my point and decided not to throw any heavy objects at me. Some of them even talked to me after the presentation, which shows that design educators at some level accept the argument presented in the article.

a retrospective account of job advertisements tells us the story of where the profession was in the past. For instance, Todd et al. (1995) sought to understand the profession by studying past job advertisements, looking at how the requests for information systems changed over a 20-year period. Another example is what was done in Study 3 of this thesis, where we compared the requests made by organisations with the guidelines defined by the Brazilian Ministry of Education for design schools in the country. Future researchers could explore further connections between these types of documents, such as the ones used in Study 3: job advertisements and educational guidelines. All in all, the results of the studies of this doctoral thesis, and the method followed, provide detailed information for responding to immediate requests posted by organisations recruiting designers. Thus, the results and methods presented here can be combined with the expertise and experience of design educators for planning future scenarios for what it means to be a designer and what is the role of design in society.

To conclude, job advertisements tell multiple stories in a single document. Organisations describe what they believe designers should do and the skills they need to have. This description provides a lens to understand how much (or little) the organisation knows about the design profession. Also, a job advertisement provides a description about the company and its values. For instance, a description of the workplace reveals how 'formal' or 'relaxed' the culture is at that company. Researchers, practitioners and educators just need to know what to look for and how to do it.

If you are interested in using job advertisements for your research, practice or education, this doctoral thesis can help you.

# 7. References


Acklin, C. (2013). Design Management Absorption Model – A Framework to Describe the Absorption Process of Design Knowledge by SMEs with Little or No Prior Design Experience. *Creativity and Innovation Management*, *22*(2), 1–14.

Ahmed, F. (2012). Software requirements engineer: An empirical study about non-technical skills. *Journal of Software*, *7*(2), 389–397.

AIGA. (2019). *Design Census 2019*. Retrieved from https://designcensus.org/data/2019DesignCensus.pdf

Ambrose, G., & Harris, P. (2009). *The fundamentals of graphic design* (1st ed.). AVA Publishing.

Atkinson, P., & Coffey, A. (2004). Analysing documentary realities. *Qualitative Research*, *3*, 77–92.

Bailey, S. G. (2012). Embedding service design: the long and the short of it. *ServDes. 2012. Third Nordic Conference on Service Design and Service Innovation.*, 31–41.

Ball, L. J., & Christensen, B. T. (2018). Designing in the wild. *Design Studies*, *57*(1995), 1–8.

Barber, A. E. (1998). *Recruiting employees: Individual and organizational perspectives*. Sage Publications.

Barnard, M. (2005). *Graphic Design as Communication*. Routledge.

Bennett, R. (2002). Employers' demands for personal transferable skills in graduates: A content analysis of 1000 job advertisements and an

associated empirical study. *Journal of Vocational Education and Training*, *54*(4), 457–476.

Berends, H., Reymen, I., Stultiëns, R. G. L., & Peutz, M. (2011). External designers in product design processes of small manufacturing firms. *Design Studies*, *32*(1), 86–108.

Bessant, B., & Bruce, M. (2002). *Design in Business: Strategic innovation through design*. Pearson Education.

Best, K., Kootstra, G., & Murphy, D. (2010). Design Management and Business in Europe: A Closer Look. *Design Management Review*, *21*(2), 26–35.

Bettinger, P., Merry, K., & Cieszewski, C. (2016). The importance of mapping technology knowledge and skills for students seeking entry-level forestry positions: evidence from job advertisements. *Mathematical & Computational Forestry & Natural Resource Sciences*, *8*(1), 14–24.

Blaich, R., & Blaich, J. (1993). *Product design and corporate strategy: managing the connection for competitive advantage*. McGraw-Hill.

Body, J. (2008). Design in the Australian Taxation Office. *Design Issues*, *24*(1), 55–67.

Bohemia, E. (2002). Designer as integrator: reality or rhetoric? *The Design Journal*, *5*(2), 23–34.

Bonsiepe, G. (1994). A step towards the reinvention of graphic design. *Design Issues*, *10*(1), 47–52.

Bonsiepe, G. (1997). Design - the blind spot of theory or Visuality | Discursivity or Theory - the blind spot of design. *The Sciences*, 1–12.

Borja de Mozota, B. (2003). *Design management: using design to build brand value and corporate innovation*. Allworth.

Bowen, G. A. (2009). Document analysis as a qualitative research method. *Qualitative Research Journal*, *9*(2), 27–40.

Boyatzis, R. E. (1998). *Transforming qualitative information: Thematic analysis and code development*. Sage.

Braun, V., & Clarke, V. (2006). Using thematic analysis in psychology. *Qualitative Research in Psychology*, *3*(2), 77–101.

Bronstein, J. (2015). An exploration of the library and information science professional skills and personal competencies: An Israeli perspective. *Library and Information Science Research*, *37*(2), 130–138.

Bruce, M., & Morris, B. (1998). In-house, outsourced or a mixed approach to design. In Margaret Bruce & B. Jevnaker (Eds.), *Management of Design Alliances - Sustaining Competitive Advantage*. Wiley & Sons Ltd.

Bruce, M., Cooper, R., & Vazquez, D. (1999). Effective design management for small businesses. *Design Studies*, *20*(3), 297–315.

Bruce, M., & Morris, B. (1994). Managing external design professionals in the product development process. *Technovation*, *14*(9), 622–623.

Brumberger, E., & Lauer, C. (2017). International Faces of Technical Communication: An Analysis of Job Postings in Three Markets. *Technical Communication*, *64*(4), 310–327.

Bryman, A. (2012). *Social research methods* (4th ed.). Oxford university press.

Buchanan, R. (2001). Design Research and the New Learning. *Design Issues*, *17*(4), 3–23.

Candi, M, & Saemundsson, R. (2008). How different? Comparing the use of design in service innovation in Nordic and American new technology-based firms. *Design Studies*, *29*(5), 478–499.

Candi, M. & Gemser, G. (2010). An agenda for research on the relationships between industrial design and performance. *International Journal of Design*, *4*(3), 67–77.

Carnevale, A., Jayasundera, T., & Repnikov, D. (2014). *Understanding Online Job Ads Data: A technical report* (Issue April).

Cash, P. J. (2018). Developing theory-driven design research. *Design Studies*, *56*, 84–119.

Cash, P. J., Hicks, B. J., & Culley, S. J. (2013). A comparison of designer activity using core design situations in the laboratory and practice. *Design Studies*, *34*(5), 575–611.

Cash, P., Škec, S., & Štorga, M. (2013). A bibliometric analysis of the design 2012 conference. *Proceedings of the 19th International Conference on Engineering Design (ICED13) : Design For Harmonies*, *2*, 43–52.

CBD. (2014). *Diagnostic Review of Design in Brazil*. Retrieved from https://www.cbd.org.br/materiais-cbd/diagnostic-review-of-design-in-brazil/

Cezzar, J. (2017). *The AIGA Guide to Careers in Graphic and Communication Design*. Bloomsbury Publishing USA.

Chai, K.-H., & Xiao, X. (2012). Understanding design research: A bibliometric analysis of Design Studies (1996–2010). *Design Studies*, *33*(1), 24–43.

Chen, H., & Zhang, Y. (2017). Educating Data Management Professionals: A Content Analysis of Job Descriptions. *Journal of Academic Librarianship*, *43*(1), 18–24.

Chipulu, M., Neoh, J. G., Ojiako, U., & Williams, T. (2013). A Multidimensional Analysis of Project Manager Competences. *IEEE Transactions on Engineering Management*, *60*(3), 506–517.

Chiva, R., & Alegre, J. (2007). Linking design management skills and design function organization: An empirical study of Spanish and Italian ceramic tile producers. *Technovation*, *27*(10), 616–627.

Choi, Y., Lim, S., & Evans, M. (2012). Supporting Design: National Business Support Programmes in the UK and South Korea. *Design Journal*, *15*(1), 79–104.

Choi, Y., & Rasmussen, E. (2009). What Qualifications and Skills are Important for Digital Librarian Positions in Academic Libraries? A Job Advertisement Analysis. *Journal of Academic Librarianship*, *35*(5), 457–467.

Cleary, J., & Cochie, M. (2011). Core Skill Set Remains Same In Newspaper Job Ads. *Newspaper Research Journal*, *32*(4), 68–82.

Cloonan, M. V., & Norcott, P. C. (1987). Evolution of preservation librarianship as reflected in job descriptions from 1975 through 1987. *College and Research Libraries*, *50*(6), 646–656.

Coffey, A. (2014). Analysing Documents. *The SAGE Handbook of Qualitative Data Analysis*, 367–379.

Cooper, R., Junginger, S., & Lockwood, T. (2013). *The handbook of design management*. A&C Black.

Cooper, R., & Press, M. (1995). *The design agenda : a guide to successful design management*. John Wiley and Sons.

Corazzo, J., Harland, R. G., Honnor, A., & Rigley, S. (2019). The Challenges for Graphic Design in Establishing an Academic Research Culture: Lessons from the Research Excellence Framework 2014. *Design Journal*, *23*(1), 7–29.

Cornish, K., Goodman-Deane, J., Ruggeri, K., & John Clarkson, P. (2015). Visual accessibility in graphic design: A client-designer communication failure. *Design Studies*, *40*, 176–195.

Crilly, N. (2019). Methodological diversity and theoretical integration: Research in design fixation as an example of fixation in research design? *Design Studies*, *65*, 78–106.

Cross, N. (2004). Expertise in design: An overview. *Design Studies*, *25*(5), 427–441.

da Cunha, M. V. (2009). The information professional's profile: An analysis of Brazilian job vacancies on the internet. *Information Research*, *14*(3).

Dabner, D., Stewart, S., & Zempol, E. (2014). *Graphic Design School: A Foundation Course for Graphic Designers Working in Print*. Thames & Hudson.

Davis, M. (2006). Raising the Bar for Higher Education. In S. Heller (Ed.), *The education of a graphic designer* (3rd ed.). Allworth Press.

Davis, M. (2008). Why do we need doctoral study in design? *International Journal of Design*, *2*(3), 71–79.

Davis, M. (2012). *Graphic design theory*. Thames & Hudson.

Davis, M. (2015). Interdisciplinarity and the Education of the Design Generalist. In S. Heller (Ed.), *The Education of a Graphic Designer* (3rd ed., pp. 31–54). Allworth.

Davis, M. (2016). Tenure and design research: A disappointingly familiar discussion. *Design and Culture*, *8*(1), 123–131.

Davis, M., & Hunt, J. (2017). *Visual communication design: an introduction to design concepts in everyday experience*. Bloomsbury Publishing.

Debortoli, S., Müller, O., & Vom Brocke, J. (2014). Comparing business intelligence and big data skills: A text mining study using job advertisements. *Business and Information Systems Engineering*, *6*(5), 289–300.

Den Hartog, D. N., Caley, A., & Dewe, P. (2007). Recruiting leaders: An analysis of leadership advertisements. *Human Resource Management Journal*, *17*(1), 58–75.

Design Council. (2015). *The Design Economy*. Retrieved from https://www.designcouncil.org.uk/sites/default/files/asset/document/Design Economy report web Final - 140217 Yea 1.pdf

Design Council. (2018). *The Design Economy 2018 - The state of design in the UK*. Retrieved from https://www.designcouncil.org.uk/sites/default/files/asset/document/Design_Economy_2018_exec_summary.pdf

Dineen, B. R., & Soltis, S. M. (2011). Recruitment: A review of research and emerging directions. In S. Zedeck (Ed.), *APA handbooks in psychology®. APA handbook of industrial and organizational psychology, Vol. 2. Selecting and developing members for the organization* (pp. 43–66). American Psychological Association.

Dörfler, L., & van de Werfhorst, H. G. (2009). Employers' demand for qualifications and skills: Increased merit selection in austria, 1985-2005. *European Societies*, *11*(5), 697–721.

Dorland, A. (2009). Routinized labour in the graphic design studio. In G. Julier & L. Moor (Eds.), *Design and creativity: Policy, Management and Practice* (pp. 105–121). BERG.

Dorst, K. (2008). Design research: a revolution-waiting-to-happen. *Design Studies*, *29*(1), 4–11.

Dorst, K. (2011). The core of "design thinking" and its application. *Design Studies*, *32*(6), 521–532.

Dorst, K., & Cross, N. (2001). Creativity in the design process: co-evolution of problem–solution. *Design Studies*, *22*(5), 425–437.

Dorst, K., & Reymen, I. (2004). Levels of Expertise in Design Education. *International Engineering and Product Design Education Conference*, 1–8.

Dosani, S., Schroter, S., MacDonald, R., & Connor, J. (2003). Recruitment of doctors to non-standard grades in the NHS: analysis of job advertisements and survey of advertisers. *BMJ: British Medical Journal (International Edition)*, *327*(7421), 961–964.

Dziobczenski, P. R. N., & Person, O. (2017). Graphic designer wanted: A document analysis of the described skill set of graphic designers in job advertisements from the United Kingdom. *International Journal of Design*, *11*(2).

Dziobczenski, P. R. N., Person, O., & Meriläinen, S. (2018). Designing Career Paths in Graphic Design: A Document Analysis of Job Advertisements for Graphic Design Positions in Finland. *The Design Journal*, *21*(3), 349–370.

Dziobczenski, P. R. N., Person, O., Tonetto, L. M., & Mandelli, R. R. (2018). Requests from Companies and Requirements for Design Education in Brazil: where do they meet? In C. Storni, K. Leahy, M. McMahon, P. Lloyd, & E. Bohemia (Eds.), *Design Research Society 2018: Catalyst* (pp. 2823–2835). Design Society.

Erkarslan, O. (2013). A systematic review of the relations between industrial design education and industry in Turkey through SWOT analysis. *The Design Journal*, *16*(1), 74–102.

Farr, M. (1965). Design management: Why is it needed now. *Design Journal*, *200*, 38–39.

Fereday, J., & Muir-Cochrane, E. (2006). Demonstrating Rigor Using Thematic Analysis : A Hybrid Approach of Inductive and Deductive Coding and Theme Development. *International Journal of Qualitative Methods*, *5*, 80–92.

FIRJAN. (2019). *Estudos e Pesquisas: Mapeamento da Indústria Criativa no Brasil*. Retrieved from https://bit.ly/2YjkngY

Fisher, R. J. (1993). Social Desirability Bias and the Validity of Indirect Questioning. *Journal of Consumer Research*, *20*(2), 303.

Flick, U. (2009). *An Introduction to Qualitative Research* (4th ed.). SAGE Publications.

Frascara, J. (1988). Graphic Design: Fine art or social science? *Design Issues*, 18–29.

Frascara, J. (2004). *Communication design: principles, methods, and practice*. Allworth.

Frascara, J. (2017). Design and design education- How can they get together? *Art, Design & Communication in Higher Education*, *16*(1), 125–131.

García, M. F., Posthuma, R. A., & Quiñones, M. (2010). How benefit information and demographics influence employee recruiting in Mexico. *Journal of Business and Psychology*, *25*(3), 523–531.

Gatewood, R. D., Gowan, M. A., & Lautenschlager, G. J. (1993). Corporate Image, Recruitment Image and Initial Job Choice Decisions. *Academy of Management Journal*, *36*(2), 414–427.

Giloi, S., & Belluigi, D. Z. (2017). Underlying Knowledge-knower structures in graphic design: Contributing to establishing a cohesive language for use in graphic design education. *Art, Design & Communication in Higher Education*, *16*(1), 7–22.

Giloi, S., & du Toit, P. (2013). Current approaches to the assessment of graphic design in a Higher Education context. *International Journal of Art and Design Education*, *32*(2), 256–268.

Gök, O., & Hacioglu, G. (2010). The organizational roles of marketing and marketing managers. *Marketing Intelligence and Planning*, *28*(3), 291–309.

Gorb, P. (1990). Design management : papers from the London Business School. In *Design Management Review* (Vol. 20, Issue 2). Architecture Design And Technology Press.

Gorichanaz, T., & Latham, K. F. (2016). Document phenomenology: a framework for holistic analysis. *Journal of Documentation*, *72*(6), 1114–1133.

Green, L. D., Smith, J., Bryant, G., Cooper, R., & Green, L. D. (2004). *Insightful Perspectives on experience designing design intuitive managers authority analytical empathetic*. 74–80.

Griffin, D. (2017). Survey Paper: Doctoral Education in (Graphic) Design. *Dialectic*, *I*(I), 135–153.

Harland, R. (2007). Redefining the plural domains of graphic design and orientating the subject towards a model that links practice, education and research. *Emerging Trends in Design Research: Proceedings of the International Association of Societies of Design Research (IASDR07)*.

Harland, R. (2011). The Dimensions of Graphic Design and Its Spheres of Influence. *Design Issues*, *27*(1), 21–34.

Harland, R. (2016). *Graphic Design in Urban Environments*. Bloomsbury Publishing.

Harland, R. (2017). Some important things to say about graphic design education. *Art, Design & Communication in Higher Education*, *16*(1), 3–6.

Harland, R. G. (2020). Graphic design studies: what can it be? Following in Victor Margolin's footsteps for possible answers. *DRS2020: Synergy*, *2*, 586–600.

Harland, R., James, C., Ian, G., Alison, H., & Steve, R. (2018). Graphic design research : a cause for the concerned. *DRS 2018*.

Harper, R. (2012). The collection and analysis of job advertisements: a review of research methodology. *Library & Information Research*, *36*(112), 29–54.

Hart, C. (1999). *Doing a literature review: Releasing the research imagination*. Sage.

Heller, S. (1998). *The Education of a Graphic Designer* (1st ed.). Allworth Press.

Heller, S. (1999). *Paul Rand*. Phaidon.

Heller, S. (2015). *The education of a graphic designer* (3rd ed.). Simon and Schuster.

Henderson, P. W., Cote, J. a, Leong, S. M., & Schmitt, B. (2003). Building strong brands in Asia: selecting the visual components of image to maximize brand strength. *International Journal of Research in Marketing*, *20*(4), 297–313.

Hollis, R. (2001). *Graphic design : a concise history*. Thames & Hudson.

Horváth, I. (2007). Comparison of three methodological approaches to design research. *International Conference on Engineering Design - ICED07*, 1–11.

Hox, J. J., & Boeije, H. R. (2005). Data Collection, Primary vs. Secondary. *Encyclopedia of Social Measurement*, *1*, 593–599.

Hummels, C., & Vinke, D. (2009). *Eindhoven designs. Volume two*. Retrieved from http://www.tue.nl/uploads/media/eindhoven_designs_volume2_24.pdf

Irons, J. G., & Buskist, W. (2008). Preparing the New Professoriate: What Courses Should They Be Ready to Teach? *Teaching of Psychology*, *35*(3), 201–204.

Jackson, M. (2007). How far merit selection? Social stratification and the labour market. *British Journal of Sociology*, *58*(3), 367–390.

Jacobs, J. (2017). Managing the Creative Process within Graphic Design Firms: A Literature Review. *Dialectic*, *1*(2).

Jones, E. A. ;, & Voorhees, R. A. (2002). *Defining and Assessing Learning: Exploring Competency-Based Initiatives.* ERIC.

Julier, G. (2010). Playing the system: Design consultancies, professionalization and value. In B. Townley & N. Beech (Eds.), *Managing Creativity: Exploring the Paradox* (pp. 237–259). Cambridge University Press.

Julier, G. (2013). *The culture of design* (3rd ed.). Sage.

Julier, G. (2017). *Economies of Design*. Sage.

Jung, H., Wiltse, H., Wiberg, M., & Stolterman, E. (2017). Metaphors, materialities, and affordances: Hybrid morphologies in the design of interactive artifacts. *Design Studies*, *53*, 24–46.

Karjalainen, T.-M., & Snelders, D. (2010). Designing Visual Recognition for the Brand. *Journal of Product Innovation Management*, *27*(1), 6–22.

Kaygan, P., Ilhan, A. O., & Oygür, I. (2020). Change in Industrial Designers' Jobs: The Case of Turkey, 1984-2018. *Design Journal*, *23*(6), 821–841.

Kennan, M. A., Cole, F., Willard, P., Wilson, C. C., Marion, L., Anne Kennan, M., Cole, F., Willard, P., Wilson, C. C., & Marion, L. (2006). Changing workplace demands: what job ads tell us. *Aslib Proceedings*, *58*(3), 179–196.

Kennan, M. A., Willard, P., & Wilson, C. S. C. S. (2006). What Do They Want?: A Study of Changing Employer Expectations of Information Professionals. *Australian Academic & Research Libraries*, *37*(1), 17–37.

Kiernan, L., & Ledwith, A. (2014). Is Design Education Preparing Product Designers for the Real World? A Study of Product Design Graduates in Ireland. *The Design Journal*, *17*(2), 218–237.

Kim, J., & Angnakoon, P. (2016). Research using job advertisements: A methodological assessment. *Library and Information Science Research*, *38*(4), 327–335.

Kim, J. Y., & Lee, C. K. (2016). An empirical analysis of requirements for data scientists using online job postings. *International Journal of Software Engineering and Its Applications*, *10*(4), 161–172.

Kinross, R. (2004). *Modern typography* (2nd ed.). Hyphen Press.

Korvenmaa, P. (2011). Rhetoric and Action : Design Policies in Finland at the Beginning of the third Millennia. *Scandinavian Journal of Design History*, *11*, 7–15.

Krippendorff, K. (2013). *Content Analysis: An Introduction to Its Methodology*. SAGE.

Kuokkanen, A., Varje, P., & Väänänen, A. (2013). Transformation of the Finnish employee ideal in job advertisements from 1944 to 2009. *Acta Sociologica 56*(3), 213–226.

Laing, S., Apperley, M., & Masoodian, M. (2017). Investigating the effects of client imagery on the ideation process of graphic design. *Design Studies*, *53*, 78–98.

Laing, S., & Masoodian, M. (2016). A study of the influence of visual imagery on graphic design ideation. *Design Studies*, *45*, 187–209.

Law, E. L.-C., Roto, V., Hassenzahl, M., Vermeeren, A. P. O. S., & Kort, J. (2009). Understanding, scoping and defining user experience: a survey approach. *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, 719–728.

Lawson, B. (2006). *How designers think: The design process demystified*. Routledge.

Lawson, B., & Dorst, K. (2015). *Design Expertise*. Architectural Press.

Lester, J. N., Muskett, T., & O'Reilly, M. (2017). Naturally Occurring Data Versus Researcher-Generated Data. In *A Practical Guide to Social Interaction Research in Autism Spectrum Disorders* (pp. 87–116). Springer.

Lewis, W. P., & Bonollo, E. (2002). An analysis of professional skills in design: implications for education and research. *Design Studies*, *23*(4), 385–406.

Liu, S. X., & de Bont, C. (2017). Barriers to Strategic Design: A Perspective from China. *She Ji: The Journal of Design, Economics, and Innovation*, *3*(2), 133–145.

Livingston, A., & Livingston, I. (2012). *The Thames & Hudson dictionary of graphic design and designers*. Thames & Hudson.

Lloyd, P., & Snelders, D. (2003). What was Philippe Starck thinking of? *Design Studies*, *24*(3), 237–253.

Lloyd, P. (2002). Making a drama out of a process: How television represents designing. *Design Studies*, *23*(2), 113–133.

Lupton, E., & Phillips, J. C. (2015). *Graphic Design: The New Basics: Revised and Expanded*. Chronicle Books.

Maeda, J. (2017). *In Tech Report 2017*. Retrieved from https://designintech.report/2017/03/11/design-in-tech-report-2017/

Marcusson, L., & Lundqvist, S. (2015). Why advertise the obvious? Learning outcomes from analyzing advertisements for recruitment of swedish IS/IT project managers. *International Journal of Information Systems and Project Management*, *3*(4), 39–56.

Margolin, V. (1994). Narrative Problems of Graphic Design History. *Visible Language*, *28*(3), 233–244.

Martin, D. (2012). The cooperative use of material resources and contextual features in graphic design work. *Design Studies*, *33*(6), 589–610.

McArthur, E., Kubacki, K., Pang, B., & Alcaraz, C. (2017). The Employers' View of "Work-Ready" Graduates: A Study of Advertisements for Marketing Jobs in Australia. *Journal of Marketing Education*, *39*(2), 82–93.

McDonald, A. (2006). In between: Challenging the role of graphic design by situating it in a collaborative, interdisciplinary class. *Design Studies: Theory and Research in Graphic Design*, 354–369.

McMahon, C. A. (2012). Reflections on diversity in design research. *Journal of Engineering Design*, *23*(8), 563–576.

Meggs, P. B., & Purvis, A. W. (2012). *Meggs' History of Graphic Design*. John Wiley & Sons.

Meggs, P. B., & Purvis, A. W. (2016). *Meggs' History of Graphic design*. John Wiley & Sons.

Meron, Y. (2021). Terminology and Design Capital: Examining the Pedagogic Status of Graphic Design through Its Practitioners' Perceptions of Their Job Titles. *International Journal of Art & Design Education*, *2021*.

Meyer, M. (2017). Qualifications and Competencies for Population Health Management Positions: A Content Analysis of Job Postings. *Population Health Management*, *20*(6), 475–485.

Micheli, P., Perks, H., & Beverland, M. B. (2018). Elevating Design in the Organization. *Journal of Product Innovation Management*, *35*(4), 629–651.

Moor, L., & Julier, G. (2009). Design and Creativity. In G. Julier & L. Moor (Eds.), *Design and creativity: Policy, Management and Practice*.

Moultrie, J., & Livesey, F. (2009). *International Design Scoreboard: Initial indicators of international design capabilities*. Retrieved from http://www.designcouncil.org.uk/Documents/Documents/Publications/Research/InternationalDesignScoreboard.pdf

Mutanen, U. M. (2008). Developing organisational design capability in a Finland-based engineering corporation: the case of Metso. *Design Studies*, *29*(5), 500–520.

Naveiro, R. M., & Pereira, R. C. de S. (2008). Design education in Brazil. *Design Studies*, *29*(3), 304–312.

Neuendorff, K. (2002). *The Content Analysis Guidebook*. SAGE Publications.

Newark, Q. (2008). What is Graphic Design Pub. *Rotovision SA, Switzerland*.

Nielsen, J. (1999). *Designing web usability: The practice of simplicity*. New Riders Publishing.

Nil G., M., Melioranski, R.-H., Er, O., & Fremantle, C. (2017). The Future of Design Support: What Can We Learn From Design Support Experience in the UK, Estonia and Turkey? *The Design Journal*, *20*(sup1), S669–S680.

Nini, P. (1996). What graphic designers say they do. *Information Design Journal*, *8*(2), 181–188.

Norman, D. (2010). *Why design education must change*. Core 77. Retrieved from https://www.researchgate.net/profile/Donald_Norman/publication/235700801_Wir_brauchen_neue_Designer_Why_Design_Education_Must_Change/links/54a2b47e0cf256bf8bb0d448.pdf

Oakley, M. (2011). Organising design activities. *The Handbook of Design Management*, 74–86.

Okamoto, K., & Polger, M. A. (2012). Off to market we go: A content analysis of marketing and promotion skills in academic librarian job ads. *Library Leadership and Management*, *26*(2).

Perks, H., Cooper, R., & Jones, C. (2005). Characterizing the role of design in new product development: An empirically derived taxonomy. *Journal of Product Innovation Management*, *22*(2), 111–127.

Person, O. (2011). *The strategic relevance of styling and the management of design styles* [Delft University of Technology]. Retrieved from http://resolver.tudelft.nl/uuid:19bd74bb-3770-4c53-af8b-cff6bad178f3

Poggenpohl, S., Chayutsahakij, P., & Jeamsinkul, C. (2004). Language definition and its role in developing a design discourse. *Design Studies*, *25*(6), 579–605.

Potter, J. (2002). Two kinds of natural. *Discourse Studies*, *4*(4), 539–542.

Potter, S. (2004). Undertaking a Literature Review. *Doing Postgraduate Research*, *4*(4), 411–429.

Preece, J., Rogers, Y., & Sharp, H. (2004). *Interaction design*. Apogeo Editore.

Rafaeli, A. (2006). Sense-making of employment: On whether and why people read employment advertising. *Journal of Organizational Behavior*, *27*(6), 747–770.

Rafaeli, A., & Oliver, A. L. (1998). Employment ads: A configurational research agenda. *Journal of Management Inquiry*, *7*(4), 342–358.

Ramírez, M. (2012). Employability Attributes for Industrial Design Graduates. *5th International Conference of Education, Research and Innovation*, 2462–2471.

Ramlau, U. H., & Melander, C. (2004). In Denmark, Design Tops the Agenda. *Design Management Review*, *15*(4), 48–54.

Rand, P. (1970). *Thoughts on design*. London : Studio Vista.

Rapley, T. (2007). Doing Conversation, Discourse and Document Analysis. In *Doing Conversation, Discourse and Document Analysis*.

Ratledge, D., & Sproles, C. (2017). An analysis of the changing role of systems librarians. *Library Hi Tech*, *35*(2), 303–311.

Raulik-Murphy, G. (2010). *A Comparative analysis of strategies for design promotion in different national contexts within the discipline of Design*. University of Wales.

Raulik-Murphy, G., Cawood, G., & Lewis, A. (2010). Design Policy: An Introduction to What Matters. *Design Management Review*, *21*(4), 52–59.

Raulik-Murphy, G. R., Murphy, D., Christiano, S. S. de, & Sanz, C. M. (2016). Revisão latino-americana do International Design Scoreboard: levantamento de dados para informação de políticas públicas de design. *Strategic Design Research Journal*, *9*(1), 2–13.

Ravasi, D., & Lojacono, G. (2005). Managing design and designers for strategic renewal. *Long Range Planning*, *38*(1), 51–77.

Ritchie, J., Lewis, J., Nicholls, C. M., & Ormston, R. (2013). *Qualitative research practice: A guide for social science students and researchers*. SAGE Publications, inc.

Roberts, L., & Mancuso, S. V. (2014). What kind of international school leaders are in demand around the world? A test of differences by region and stability over time. *Journal of Research in International Education*, *13*(2), 91–105.

Roberts, L., Wright, R., & Price, J. (2015). *Graphic Designers Surveyed*. GraphicDesign& Book.

Ross, R. (2018). Producing Knowledge with Billboards: Graphic Design and Research. *Design and Culture*, *10*(3), 299–321.

Ryynänen, T. (2009). Designed by the Media? *IASDR 2009 Rigor and Relevance in Design*, 2659–2669.

Schön, D. (1984). *The Reflectice Practitioner: How Professionals Think in Action*. Basic Books.

SEBRAE. (2014). *Design no Brasil*. Retrieved from https://www.sebrae.com.br/sites/PortalSebrae/artigos/o-design-no-brasil-relatorio-2014,6d818242d5e67410VgnVCM1000003b74010aRCRD

Shaughnessy, A. (2012). *How to be a graphic designer without losing your soul*. Chronicle Books.

Sheppard, B., Kouyoumjian, G., Sarrazin, H., Dore, F., & McKinsey & Co. (2018). *The Business Value of Design*. Retrieved from https://www.mckinsey.com/business-functions/mckinsey-design/our-insights/the-business-value-of-design

Shi, N., & Bennett, D. (2000). Information systems management positions - a market perspective. *Work Study*, *49*(7), 275–284.

Shou, Y., & Wang, W. (2017). Multidimensional competences of supply chain managers: an empirical study. *Enterprise Information Systems*, *11*(1), 58–74.

Silverman, D. (2013). *Doing qualitative research: A practical handbook*. Sage.

Silverman, D. (2015). *Interpreting qualitative data*. Sage.

Silverman, D. (2016). *Qualitative research*. Sage.

Singh, V., & Gu, N. (2012). Towards an integrated generative design framework. *Design Studies*, *33*(2), 185–207.

Snyman, R. M. M. (2001). Do employers really know what they want? An analysis of job advertisements for information and knowledge managers. *Aslib Proceedings*, *53*(7), 273–281.

Stamm, B. Von. (1998). Whose Design is it? The Use of External Designers. *The Design Journal*, *1*(1), 41–53.

Stones, C., & Cassidy, T. (2007). Comparing synthesis strategies of novice graphic designers using digital and traditional design tools. *Design Studies*, *28*(1), 59–72.

Stones, C., & Cassidy, T. (2010). Seeing and discovering: how do student designers reinterpret sketches and digital marks during graphic design ideation? *Design Studies*, *31*(5), 439–460.

Sun, Q, Williams, a, & Martyn, E. (2011). *A theoretical design management framework*. *14*(1), 112–132.

Sun, Qian. (2010). Design Industries and Policies in the UK and China: A Comparison. *Design Management Review*, *21*(4), 70–77.

Surry, D. W., & Robinson, M. A. (2001). A taxonomy of instructional technology service positions in higher education. *Innovations in Education and Teaching International*, *38*(3), 231–238.

Tan, S., & Melles, G. (2010). An activity theory focused case study of graphic designers' tool-mediated activities during the conceptual design phase. *Design Studies*, *31*(5), 461–478.

Todd, P., McKeen, J., & Gallupe, R. B. (1995). The Evolution of IS Job Skills: A Content Analysis of IS Job Advertisements from 1970 to 1990. *MIS Quarterly*, *19*(1), 1.

Tomes, A., Oates, C., & Armstrong, P. (1998). Talking design: Negotiating the verbal-visual translation. *Design Studies*, *19*(2), 127–142.

Townsend, S., & Armstrong, H. (2017). Introduction: The Value of Design in an Academic Context. *Design and Culture*, *9*(1), 67–68.

Triggs, T. (2011). Graphic Design History : Past , Present, and Future. *Design Issues2*, *27*(1).

Vaismoradi, M., Turunen, H., & Bondas, T. (2013). Content analysis and thematic analysis: Implications for conducting a qualitative descriptive study. *Nursing and Health Sciences*, *15*(3), 398–405.

Valencia, A., Person, O., & Snelders, D. (2013). An in-depth case study on the role of industrial design in a business-to-business company. *Journal of Engineering and Technology Management*, *30*(4), 363–383.

Valtonen, A. (2005). Six decades – and six different roles for the industrial designer . *Nordes Conference: In the Making*.

Valtonen, A. (2007). *Redefining industrial design: changes in the design practice in Finland*. University of Art and Design Helsinki.

Valtonen, A. (2016). Designing Universities of the Future. *Proceedings of DRS 2016, Design Research Society 50th Anniversary Conference*.

Van der Waarde, K. (2009). On graphic design: Listening to the reader? *Research Group Visual Rhetoric*.

Voorhees, R. (2001). Competency-Based Learning Models: A Necessary Future. In *New Directions for Institutional Research* (Vol. 2001, Issue 110).

Walker, H. J., & Hinojosa, A. S. (2013). *Recruitment: The Role of Job Advertisements*. *January*, 1–19.

Walker, S. (2017). Research in Graphic Design. *Design Journal*, *20*(5), 549–559.

Wheeler, A. (2009). *Designing brand identity*. John Wiley & Sons, Inc.

Xu, H., & Chen, H.-L. (2000). Whom do employers actually hire? the educating systems librarian research project report 2. *Electronic Library*, *18*(3), 171–182.

Yang, M. Y., You, M., & Chen, F. C. (2005). Competencies and qualifications for industrial design jobs: Implications for design practice, education, and student career guidance. *Design Studies*, *26*(2), 155–189.

Yeo, A., Legard, R., Keegan, J., Ward, K., McNaughton Nicholls, C., & Lewis, J. (2014). In-depth interviews. In *Qualitative research practice: A guide for social science students and researchers* (pp. 177–210). Sage Thousand Oaks, CA.

Yin, R. K. (2009). *Case Study Research: Design and Methods*. SAGE Publications.

# Appendix A. Thesis abstract in Portuguese


Resumo

A presente tese investiga a função e as qualificações de designers gráficos tais como são descritas por contratantes em anúncios de emprego. A literatura sobre design gráfico e sobre a área de design em geral sugere que a função dos designers está em mutação e que atualmente estes profissionais têm uma contribuição para as organizações que é mais estratégica do que já foi. Neste sentido, designers têm ampliado sua contribuição às organizações, de produções de orientação visual ao envolvimento determinante no desenvolvimento de bens e serviços. No caso do design gráfico, a literatura descreve que a função e as qualificações dos profissionais da área, tradicionalmente, estavam ligadas à execução de produtos visuais, tais como o projeto de cartazes, e ao conhecimento de tipografia e composição visual. Além disso, e em paralelo à expansão da função dos designers em geral, designers gráficos também atuam e têm competências em áreas como negócios, pesquisa e tecnologia (programação, por exemplo). A gama de funções e qualificações que se exige dos designers gráficos corresponde a desafios tanto para educadores quanto profissionais que precisam manter-se atualizados quanto aos cargos que os profissionais da área ocupam.

Ao fornecer uma perspectiva de base empírica sobre a função e as qualificações do designer gráfico, a tese contribui para a pesquisa nesta área. Até o momento, a literatura sobre o tema é predominantemente conceitual e baseia-se em preconcepções que não foram plenamente investigadas de modo empírico. O trabalho apresenta dados em profundidade a respeito de como empresas descrevem função e qualificações quando buscam profissionais de design gráfico,

com foco particular na expansão das competências. Em específico, a tese se soma a discussões a respeito do que é o design gráfico. Também soma informações específicas sobre a função e as qualificações dos designers a informes atuais sobre o mercado provenientes de instituições do Reino Unido (Design Council), Finlândia (Ornamo, Grafia) e Brasil (Sebrae). Além disso, contribui com pesquisas anteriores sobre gestão de design ao tratar dos primeiros passos na adoção do design em termos de recrutamento e construção da função do profissional da área inserido em uma organização. Por fim, a tese visa auxiliar educadores e profissionais da área do design com orientações para investigação do mercado de trabalho com o qual os profissionais do design se deparam.

Os estudos da presente tese apresentam a função e qualificações de designers gráficos tais como descritos em anúncios de emprego no Reino Unido (Estudo 1), na Finlândia (Estudo 2) e no Brasil (Estudo 3). Esta tese investiga o mercado de trabalho para designers gráficos nestes países e sobre suas diferenças conforme contexto geográficos, culturas de design, infraestruturas econômica e pedagógica, além de lançar luz a suas similaridades. Os estudos também abrangem como as competências de designers gráficos variam conforme senioridade (Estudo 1), trabalhar interna ou externamente às empresas (Estudo 1), transição para cargos com foco maior no digital (Estudo 2) e como as qualificações que as organizações exigem são similares ou diferentes das que as instituições de educação mobilizam (Estudo 3).

Os resultados destes estudos mostram que a função e as qualificações de designers gráficos são mais amplas do que se costuma caracterizar. Designers gráficos fornecem, acima de tudo, produções digitais e impressas. Para este tipo de trabalho, contudo, precisam não apenas de qualificações no design, mas também de qualificações comumente associadas a outros campos, tais como administração, gestão de projetos e pesquisa. Outro resultado dos estudos são definições quanto à descrição de qualificações quando designers gráficos passam (1) de cargos iniciais aos hierarquicamente superiores, (2) de departamentos internos à consultoria em design (ou vice-versa) e (3) de funções tradicionais às da gráfica digital (ou vice-versa).

Nesta tese, anúncios de emprego são a fonte primária dos dados. Este tipo de documento é usado de forma ampla em outros campos, tais como biblioteconomia e administração, para investigar as qualificações que

organizações buscam nos profissionais. No design, contudo, esta abordagem é rara. Enquanto muitos estudos sobre a profissão de design baseiam-se em entrevistas, anúncios de emprego são fontes de dados primários e naturais, o que possibilita análise detalhada dos documentos. Além disso, pesquisadores apontam a vantagem de manipular dados naturais (ou seja, não produzidos para fins de pesquisa) para abranger uma extensão maior do mercado de trabalho, dado que estes documentos estão à disposição na internet. No estudo 4, trato da relevância de anúncios de emprego para pesquisa, formação e prática em design. Embora fuja ao escopo da tese, com foco na profissão de designer gráfico, o estudo também pode contribuir de modo mais geral ao uso destes anúncios como método de pesquisa.

Em suma, os resultados que aqui se apresenta sugerem que anúncios de emprego refletem o passado e ao mesmo tempo orientam desenvolvimentos futuros no design gráfico. Para profissionais e educadores da área, os anúncios são um meio indireto de entender o mercado de trabalho, que pode moldar atividades pedagógicas e pensar sua evolução. Para pesquisadores, os anúncios de emprego fornecem informações a respeito das qualificações que as empresas buscam em profissionais de design e o quanto as organizações sabem (ou não) sobre a profissão. Por exemplo: organizações que acreditam que designers deveriam programar mencionam "programação" entre os requisitos para a vaga. Esta tese sugere, portanto, que pesquisadores da área deveriam tirar proveito do acesso e da abrangência que se encontra nos anúncios de emprego para investigar evoluções na profissão.

Paulo Roberto Nicoletti Dziobczenski is a graphic designer, researcher and educator. His interests are in the role of design in organizations, design adoption, design education, design management and how organizations recruit design professionals. Before starting his doctoral studies at Aalto University, Paulo worked as a graphic designer in design agencies and in-house teams in Brazil, where he also worked as a lecturer and thesis supervisor in a design bachelor programme. He has earlier published studies in how companies recruit designers in the International Journal of Design, The Design Journal and Design Research Society (DRS) Conferences.
For more information, visit paulodzi.com

Aalto-DD 131/2021



ISBN 978-952-64-0522-3 (printed)
ISBN 978-952-64-0523-0 (pdf)
ISSN 1799-4934 (printed)
ISSN 1799-4942 (pdf)

**Aalto University**
**School of Arts, Design and Architecture**
**Design**
**www.aalto.fi**

**BUSINESS + ECONOMY**

**ART + DESIGN + ARCHITECTURE**

**SCIENCE + TECHNOLOGY**

**CROSSOVER**

**DOCTORAL DISSERTATIONS**